

Post Mining Land Use | Rezoning | Macquarie Coal Complex Pilot Project

Traffic and Transport Report

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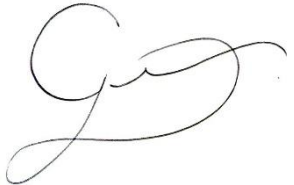
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Executive summary

This Traffic and Transport Report has been prepared to support the State led rezoning and associated master planning of the Macquarie Coal Complex (the Precinct) and Teralba South Gate sub-precinct (collectively referred to as the Site), a large post mining landholding west of Cockle Creek in the Lake Macquarie Local Government Area. The Site would be located between Teralba and Barnsley, framed by the M1 Pacific Motorway to the west, the B89 George Booth Drive and Main Road corridor to the north and the B53 arterial to the east. Wakefield Road is the key north-south spine through the area.

The Site would comprise approximately 211,100 m² of large format warehousing across the Cockle Creek and Killingworth employment cores, and a mixed-use core of around 18,100 m² at West Lake. At full development, the Site is anticipated to generate in the order of 12,470 daily trips of which 1,030 trips are expected to be heavy vehicles and 300 RAV 19 metre B-doubles. Peak hour volumes are expected to be in the order of 1,400 vehicles in the AM peak and 1,300 in the PM peak.

Midblock and intersection performance was reviewed across the base year and the short, medium and long term, with and without the Precinct and a new Cockle Creek crossing. In the base year the network is expected to operate at Level of Service (LoS) A to B, with only the Northville Drive corridor toward the B89 approaching LoS C to D.

Background growth is expected to drive the Northville Drive corridor to LoS E by the long term. The southern and western network would remain at LoS A to C with the Precinct throughout. The Northville Drive and The Weir Road intersections on the northern approach to the B89, would have adjoining links reaching LoS D in the short term and level of service F by the long term and consequently have been identified for potential upgrade, with proportionate contributions where they reach capacity under background growth.

Cockle Creek is a major line of severance for access in the wider area resulting in an already congested network to the southeast of the precinct. Reducing future vehicle demands on existing creek crossings through traffic management, mode shift, infrastructure upgrades and potential new infrastructure will be appropriately incorporated within the onward planning for the precinct.

Three structural findings drive the network strategy:

- The approved RAV B-double network does not extend into the Precinct, and heavy vehicle access would need to be newly established. Upgrading The Weir Road would provide the first compliant RAV route in the short term, the Wakefield Road route to the M1 would follow in the medium term as the principal freight route avoiding Barnsley, Holmesville and nearby schools. RAV access from the east through Teralba could be considered in the long term but would likely require a new road and creek crossing to bypass the grade constrained Rhondda Road
- Rhondda Road climbs features a sustained uphill climb of approximately 1 km for eastbound traffic with grades ranging from 5 to 10%, followed by a sustained downhill descent of over 1 km at similar grades with localised slopes over 12 per cent, beyond the Austroads limit for heavy vehicles, so the Precinct is not accessible for freight from Teralba by the existing Rhondda Road alignment. Alternative eastern access to the Precinct suitable for RAV would be a key part of the strategy, this could be a new Cockle Creek crossing or upgrades to other infrastructure that currently restricts RAV access.
- The Five Islands Road crossing of Cockle Creek is the only principal crossing and already operates at or near capacity. Therefore, the potential for a second crossing is a consideration in managing Precinct traffic in the long term.

A staged upgrade program has been identified:

- Short term (0 to 10 years). Establish The Weir Road as a RAV 19 metre B-double route to the northern access, deliver sub-precinct access intersections, and signalise and upgrade the Northville Drive intersections at The Weir Road and the B89 as well as the Northville Drive / Appletree Road / Wakefield Road intersection. Establish active transport links and extend public transport connections into the sub-precincts
- Medium term (10 to 20 years). Upgrade and approve Wakefield Road as the principal RAV 19 metre B-double route to the M1, upgrade the bridge over the railway in Teralba, deliver improved east-west

connectivity via a new internal road connecting to Wakefield Road, and provide internal spine duplication of the Northville Drive corridor.

- Long term (20 plus years). Protect and progress a direct M1 connection, potential reactivation of the colliery rail loop with a south facing junction connecting to the Main North Rail Line, railway overpass over Racecourse Road upgrade, and integration with the Lower Hunter Freight Corridor and High Speed Rail.

The assessment concludes that the proposed rezoning is not expected to be the sole contributor to future constraints on the surrounding road network, with background growth of local traffic being a key driver of the degradation in performance at the key intersections and corridors. Subject to the staged upgrade program identified in this report, there are no presently known transport issues that would make the rezoning inherently unfeasible.

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1 Introduction

1.1 Introduction

This Traffic and Transport Report (this Report) has been prepared by Aurecon on behalf of the Department Planning, Housing and Infrastructure (DPHI) to support the State-led rezoning of the Macquarie Coal Complex (the Precinct). This Report assesses the proposed rezoning and accompanying Master Plan in relation to traffic engineering and transport planning related matters.

This Report is structured as set out in Table 1-1.

Table 1-1: Report structure

Chapter	Description
Chapter 1 – Introduction	Explains the report's purpose and the methodology used to assess the rezoning. Covers the baseline and future horizons, trip generation and distribution, network performance approach, and data reliance.
Chapter 2 – Strategic Context	Sets out the federal, state and local policy and planning framework the report responds to. Includes freight, Movement and Place, regional plans and the local LEP, LSPS and DCP.
Chapter 3 – Baseline Analysis	Describes existing conditions across the road, active transport, bus, rail and freight networks. Covers crash history, traffic growth, trip patterns, existing midblock performance and the key existing traffic issues such as the Rhondda Road gradient and severance features for future traffic connectivity such as Cockle Creek and the Main Northern railway line.
Chapter 4 – Structure Plan	Presents the proposed structure plan, transport network and traffic generation, then assesses midblock and intersection performance across the horizons. Sets out the freight and RAV route strategy, the integration of transport opportunities and the staged required upgrades.
Chapter 5 – Conclusion	Summarises the baseline results against Precinct findings, a Cockle Creek crossing benefit and the staged upgrade program. Concludes the rezoning is feasible subject to staged delivery.
Appendices	Detailed midblock assessment results supporting the performance analysis.

1.2 Purpose of this report

This Report establishes an evidence based foundation to inform land use planning and infrastructure decision making for the Precinct to support the rezoning. The Report has been prepared to:

- Support the development of the Master Plan including the establishment of the movement network objectives, access arrangements and connectivity principles, and transport performance outcomes criteria aligned with intended land uses.
- Inform State-led rezoning by demonstrating that proposed land use allocations are supported by feasible and appropriate access arrangements, and that key transport opportunities and constraints are properly integrated into land use planning and staging.
- Identify and prioritise transport infrastructure requirements necessary to enable development, including key upgrades and new connections, and to establish indicative timing triggers across short, medium, and long-term horizons (0-10, 10-20, and 20+ years respectively), supporting an orderly and coordinated delivery of land use outcomes.

1.3 Methodology

The assessment follows the TfNSW *Guide to Transport Impact Assessment* (GTIA) (TS00085 v1.1, 2024), and the principles of the TfNSW *Transport Management and Accessibility Plan*, supported by the Austroads *Guide to Traffic Management* (Parts 3 and 12) and the Austroads *Guide to Road Design*. The methodology applied to each component of the assessment and is set out in the following sections.

1.3.1 Baseline and Future horizons

The baseline and future horizons adopted are as follows:

Horizon	Description
Base year (2026)	The existing network with current background traffic and no Precinct development. This establishes the baseline against which each future horizon is compared.
Short term (+5 years)	Background traffic is grown to 2031 and the Precinct is partially developed. Full build out is not expected within this horizon and a development reduction factor of 50 per cent is applied to the generated traffic. The key road network connectivity links, being a new Cockle Creek crossing, the railway overpass upgrade in Teralba and the proposed internal local road facilitating east-west access to the site, are assumed not yet to developed, as well as the formalisation of Wakefield Road, south of Rhondda Road as an approved RAV B-double route. This results in site access to Area 1 being primarily from the northern and southern access points via B89.
Medium term (+15 years)	Background traffic is further projected to 2041 and the Precinct is assumed to be at full build out, with full generation applied (no reduction factor). The connectivity links, including a new Cockle Creek crossing, the completion of upgrades to the railway overpass, development of the internal east-west local road and approval of Wakefield Road, south of Rhondda Road as a RAV B-double route, are assumed to have been established, activating the eastern access point to the Precinct.
Long term (+20 years)	Full build out and the connectivity links in place, with background traffic further projected to 2051.

1.3.2 Existing trip patterns

Existing travel behaviour is established from ABS Census data for 2016 and 2021. Vehicle ownership trends are assessed at the Bolton Point to Teralba SA2 level by comparing the distribution of motor vehicles per dwelling between the two census years and against the NSW average, indicating the degree of car dependency that new land uses must plan around.

Journey to work (JTW) patterns are assessed from ABS 2021 Census data tabulating place of usual residence by place of work at the SA2 level, for the suburbs in and around the master plan area (Teralba, West Wallsend, Barnsley, Holmesville, Edgeworth, Fassifern and Wakefield), identifying the dominant modes, the share of public and active transport and the principal origin and destination markets. These patterns inform both the mode share assumptions and the geographic distribution of light vehicle trips.

1.3.3 Background traffic growth

Background traffic data was drawn from several sources, each covering a different part of the network and period. The sources are as follows:

- Draft *West Wallsend Rezoning TIA* (GHD, 2025), providing 2024 survey volumes at a number of key intersections and the STFM (TZP22STMV3.93) growth rates.
- TfNSW permanent classifier stations on the corridors surrounding the study area. Annual Average Daily Traffic (AADT) is compiled for each station over the available data periods providing a long-term dataset suitable for understanding historic traffic patterns. with the 2020 and 2021 COVID-19 affected years

identified separately so that they do not distort the trend. Data from the following permanent TfNSW traffic counters were utilised:

- Lake Road, 130 metres north of T C Frith Avenue (Station ID: 05208), providing coverage from 2008 to 2023
- M1 Pacific Motorway northbound off-ramp to B89, 250 metres West of George Booth Drive (Station ID: 050995), providing coverage from 2007 to 2016
- Main Road on B89, 70 metres east of Fairleigh Street (Station ID: 05571), providing coverage from 2009 to 2020.

A global background growth rate of 1.2 per cent per annum, derived from the STFM approach rates, was adopted and applied linearly to forecast all source data to the common 2026 base year. The basis for this rate, and why the STFM data was preferred over the classifier trends for the lower order roads through the Precinct, is set out in the historical traffic growth section of the baseline analysis.

1.3.4 Trip generation calculations

Trip generation follows Chapter 5 of the TfNSW *GTIA*, applying daily, AM Peak and PM Peak vehicle trip rates to the structure plan land use schedule. The following vehicle trip rates have been adopted and are summarised in Table 1-2.

Site peak rates are adopted, being the peak traffic generating hour of the development itself, which is conservative for the network assessment as the full site peak is assigned to the network peak hour. The application of rates also does not consider any discounts to account for trip-chaining and/or self-containment within the Site given the early stage of planning. This further supports that the assessment is conservative.

Table 1-2: Traffic generation rates (GTIA, 2024)

Land use	Site AM Peak hour	Site PM Peak hour	Daily	Units	Source
Low density residential (Regional - Weekday)	0.83	0.84	7.53	Vehicle trips/dwelling	TfNSW <i>GTIA</i>
Large format warehousing (0 - 10,000 m ² GFA)	0.5	0.44*	4	Vehicle trips/100 m ² GFA	TfNSW <i>GTIA</i>
Large format warehousing (10,000 m ² - 100,000 m ² GFA)	0.26	0.23	2.83	Vehicle trips/100 m ² GFA	TfNSW <i>GTIA</i>
Commercial – Mixed use	4*		31*	Vehicle trips/100 m ² GLA	Derived. See below.

* Derived traffic generation rates. See below

The following derivations were adopted:

A trip generation rate is not published for large format warehousing (0 - 10,000 m² GFA) PM peak hour. The missing peak was derived using the published peak to peak ratio of large format warehousing (10,000 - 100,000 m² GFA) as similarities in trip generation patterns can be expected. Resultingly, a PM peak hour trip generation of 0.44 vehicle trips/100 m² GFA was adopted for large format warehousing (0 - 10,000 m² GFA).

- The TfNSW *GTIA* does not publish a traffic generation rate for the envisaged commercial mixed use, which is a composite of several retail and commercial formats rather than a single land use. A rate was therefore built up from the anticipated composition of the commercial mixed-use area, drawing each component rate from established published guidance, refer Table 1-3.

The daily and peak hour rates for the component land uses were then averaged to give a single blended rate for the commercial mixed use, resulting in 31 daily trips per 100m² and approximately 4 peak hour trips per 100 m² of gross leasable area (GLA). The averaging is appropriate at the rezoning stage as the precise tenancy mix is not yet fixed, and the rate is to be refined once the commercial composition is confirmed.

A GFA to GLA efficiency factor of 80 per cent was adopted, accounting for the common areas, circulation, amenities and back of house space within the gross floor area that are not leasable, so that trip generation is applied only to the tenanted floor area that actually drives demand.

Table 1-3: Component rates adopted for the commercial mixed use and the resulting average (per 100 m2 GLA)

Land use	Basis of rate	Daily rate (per 100 m2 GLA)	Peak hour rate (per 100 m2 GLA)
Large format retail (LFR)	TDT4a-2013	19	3
Retail	TDT4a-2013	39	5
Food and beverage (F&B)	RTA Guide NSW	60	5
Showroom	TDT4a-2013	15	4
Leisure	ITE (USA) PM peak rates, mix of uses	20	2
Cinema	ITE (USA) PM peak rates, multiplex cinema	49	5
Office / commercial	RTA Guide NSW	15	6
Adopted average		31 daily trips per 100 m2 GLA	4.29 per 100 m2 GLA

The rates above give the trip generation at full development. A development reduction factor is also adopted to represent the short term, when the Precinct is only partially built out during the short term:

- A factor of 50 per cent is applied for the short term, reducing all generated trips to 50 per cent of the full development total
- This recognises that the rezoned land uses will not be established or fully tenanted in the short term, and that demand accrues progressively as parcels are released and occupied
- The factor is applied uniformly across all land uses and vehicle classes and flows through to the distributions and the network assessment. As such, both the residential and the employment demand reduce proportionately
- The factor is adjustable and is to be refined as the staging plan and release sequence are confirmed.

1.3.5 Trip distributions

Directional splits consistent with the surveyed entry and exit splits in the TfNSW *GTIA* were adopted to determine trip entry and exit movements throughout the Precinct and are summarised in Table 1-4. The residential entry and exit distributions are not provided in the *GTIA*, the adopted values align with:

- The commuter based nature of residential travel, where trips are predominantly exiting movements in the AM peak (journey to work or education) and entering in the PM peak
- Standard practice in NSW traffic and transport assessments, where residential areas exhibit a strong directional bias during peak periods.

Table 1-4: Vehicle trip entry and exit distribution (GTIA, 2024)

Lane use	Site AM Peak	Site AM Peak	Site PM Peak	Site PM Peak
	Entry	Exit	Entry	Exit
Residential	20%	80%	80%	20%
Large format warehousing	62%	38%	34.5%	66.5%
Mixed use commercial	50%	50%	50%	50%

Light vehicle trip distribution

Light vehicle trips are distributed using ABS 2021 Census JTW desire lines (place of usual residence by place of work at SA2 level) for the suburbs in and around the Precinct, with each area assigned to its logical corridor of approach. The tabulation of around 1,600 resident workers shows that for the nearer destinations:

- North (38 per cent) to the Edgeworth and Cameron Park SA2
- South (24 per cent combined) to the Toronto, Morisset and western lake SA2s
- Local west (23 per cent) within the West Wallsend, Barnsley and Killingworth area
- East (16 per cent) to the Bolton Point and Teralba SA2, across Cockle Creek via Five Islands Road.

The adopted shares combine this local pattern with the longer distance commute to central Newcastle and the Central Coast recorded in the wider Census data, which adds weight to the B53 and B89 corridors and the M1.

Heavy vehicle trip distribution

Heavy vehicle trips are distributed by freight market analysis, identifying the long haul, port, regional industrial and local markets the employment land would serve and assigning each to the route it would use under the heavy vehicle access strategy. The market identification draws on:

- *NSW Freight and Ports Plan 2018 to 2023* (TfNSW, 2018), forecasting the NSW freight task to grow 28 per cent to 618 million tonnes by 2036 and identifying the M1 as the principal road freight corridor between Greater Sydney, the Hunter and Queensland
- *Lower Hunter Freight Corridor Strategic Environmental Assessment* (TfNSW, 2021), describing the Hunter freight task as movements within the region and between Greater Sydney, northern NSW and Queensland
- Port of Newcastle trade data, recording more than 160 million tonnes of total trade and a record 11.1 million tonnes of diversified non-coal cargo across 25 cargo types in 2025
- Greater Newcastle Metropolitan Plan (DPE, 2018), assigning the strategic freight and logistics role to this catalyst area.

The heavy vehicle share of employment trips and the B-double share of heavy vehicles are benchmarked against the vehicle classification splits in the TfNSW *Large Format Warehousing – Data and Analysis note* (2024). The combined light and heavy vehicle distributions are then assigned to the external network to produce corridor volumes for the midblock and intersection assessments.

1.3.6 Road network performance

An assessment of the performance of the road network under the future traffic demands has been undertaken. This includes evaluating the quantum of increase in average daily traffic on relevant roads and a Level of Service (LoS) review at key intersections or midblock sections and comparing these to established thresholds for LoS. The assessment also considered critical volumes as the maximum experienced flows in

AM and PM peaks for each midblock or intersection location. A midblock refers to a section of road located between two intersections.

Given the expedited nature of the assessment required to support the state-led rezoning, this analysis has been undertaken using available information and in line with methodologies adopted for other rapid precinct rezoning projects.

Midblock assessment

Midblock performance is assessed using volume to capacity ratios (VCR) consistent with the Austroads *Guide to Traffic Management Part 3*. One-way lane capacities at level of service E are adopted by functional class, being 1,400 vehicles per hour per lane for arterials, 1,200 for sub arterials, 1,000 for collectors and 900 for local streets. This methodology and the associated capacity parameters are consistent with those adopted for the adjacent West Wallsend rezoning, ensuring a uniform assessment criterion. The LoS traffic flow descriptions are provided in Table 1-5.

Table 1-5: LoS traffic flow descriptions

LOS	Traffic flow description
A	Free flow conditions with unimpeded manoeuvrability and minimal delay
B	Stable flow with some restriction on manoeuvrability and acceptable delay
C	Stable but constrained flow, speeds and lane changing more restricted
D	Approaching unstable flow, limited gaps, speeds declining with little headroom
E	Operating at capacity, unstable flow with queues forming on any disruption
F	Demand exceeds capacity, forced flow, breakdown conditions and persistent queuing

Conversion between VCR and LoS is speed dependent and based on methodologies outlined in Austroads guidelines. With regards to lane capacity, the following assumptions have been made based on the functional hierarchy of road as per Austroads *Guide to Traffic Management Part 3: Transport Study and Analysis Methods*:

- Arterial road (typically regional roads or state roads in urban areas) represents interrupted road conditions with access to some properties and streets with operation speed of 70 km/hr to 90 km/hr
- Collector roads (typically regional roads and some Local roads) are similar to arterial roads with speed limit less than or equal to 60 km/hr
- Local roads represent interrupted conditions with access to properties, parking lanes and posted speed of 50 km/hr.

Table 1-6 provides the reference for conversion of VCR and LoS and lists the lane capacity based on functional classification of roads based on Austroads (2020) *Guide to Traffic Management Part 3: Transport Study and Analysis Methods*.

Table 1-6: Level of Service description for road performance

LOS	Arterial		Sub-arterial		Collector		Local	
	VCR	Capacity	VCR	Capacity	VCR	Capacity	VCR	Capacity
A	0.26	360	0.26	310	0.24	240	0.24	220
B	0.41	570	0.41	490	0.38	380	0.38	340
C	0.59	830	0.59	710	0.54	540	0.54	490
D	0.81	1,130	0.81	970	0.78	780	0.78	700

LOS	Arterial		Sub-arterial		Collector		Local	
	VCR	Capacity	VCR	Capacity	VCR	Capacity	VCR	Capacity
E	1.00	1,400	1.00	1,200	1.00	1,000	1.00	900
F	>1		>1		>1		>1	

Passenger car equivalent

For the purpose of the midblock assessment, all vehicles travelling in each direction were also converted to the Passenger Car Unit (PCU). As per TfNSW's recommended use of the Passenger Car Equivalent (PCE) table from *Transport for NSW Economic Parameter Values (2025)*, passenger and light vehicles are considered as 1.0 PCU. Heavy vehicles are converted to passenger car units using a PCE of 8 for general B-double volumes, and 3.0 for general access heavy vehicles. The PCE factors per vehicle type are provided in Table 1-7.

Table 1-7: PCE factors per vehicle type (TfNSW Economic Parameter Values, 2025)

Vehicle type	PCE factors
Passenger vehicles & LCVs	1.00
Rigid trucks	3.00
Trailers	6.00
Articulated trucks	5.00
B-doubles	8.00
Double road trains	8.00
Triple road train	10.00
2 axle buses	2.00
3 axle buses	3.00

Intersection performance

Intersection modelling was not undertaken for this assessment as base traffic volumes and performance was not available, therefore only qualitative assessment was undertaken. For the qualitative assessment, approaching midblock LoS were adopted as a guide to suggest intersection performance, as follows:

- Intersections between midblock segments with reserve capacity (LoS C or better) would generally operate with reasonable levels of services
- Intersections with any adjoining approach midblock capacity of LoS D or above may be subject to poorer LoS D and therefore may require upgrades

1.3.7 Traffic Assessment Assumptions

The following assumptions were adopted for this traffic assessment:

- A gross floor area to gross leasable area efficiency factor of 80 per cent is adopted for the commercial floorspace, accounting for common areas, circulation, amenities and back of house space that are not leasable
- The commercial mixed-use rate of 31 daily and 4.29 peak hour trips per 100 m² GLA is a blended average of component retail and commercial rates, as the GTIA does not publish a rate for this composite use
- A 50 per cent development reduction factor represents partial build out and occupancy in the short term (0 to 10 years). The factor is applied uniformly across all land uses and vehicle classes, a simplification to be refined against the staging plan
- AM and PM inbound and outbound splits adopted from GTIA guidance (employment 62 per cent inbound AM, 34.5 per cent inbound PM, residential 20 per cent inbound AM, 80 per cent inbound PM)
- PM trips assumed to be the reverse leg of AM trips, with origin and destination reversed
- Site peak hour rates are adopted and the full site peak is assigned to the network peak hour, which is conservative
- Highest order vehicle assumed to be the 19 metre B-double over 50 tonnes
- Freight market shares are professional judgement informed by freight strategy, not a surveyed origin destination matrix, supporting the market hierarchy rather than precise percentages
- A new Cockle Creek crossing is assumed, for the purposes of the with-Crossing scenario only and is assumed to be delivered during the medium term horizon
- A heavy vehicle share of 15 per cent of employment trips and a B-double share of 3 per cent of heavy vehicles are adopted, benchmarked against the TfNSW September 2024 large format warehousing surveys
- Light vehicle trips are distributed on ABS 2021 Census journey to work patterns and heavy vehicle trips on a freight market analysis, with the same corridor and market shares applied in every scenario
- The M1 (Central Coast and Sydney) and Teralba local shares are professional judgement, not direct Census outputs, pending an extended journey to work extraction
- Industrial light vehicle distribution assumed to mirror the residential corridor split
- Distribution assumed constant across all horizons, with only volume scaling
- Background traffic is grown from a 2026 base year at a single global linear rate of 1.2 per cent per annum, derived from the Draft *West Wallsend Rezoning Traffic and Transport Assessment* (GHD, 2025), STFM data and applied uniformly to all corridors
- Survey volumes and turning counts adopted from the Draft *West Wallsend Rezoning Traffic and Transport Assessment* (GHD, 2025), pending updated surveys for the Precinct
- Passenger car equivalents of 8.0 for B-doubles and 3.0 on general heavy vehicles
- Intersection performance is not assessed quantitatively at the rezoning stage. It is inferred qualitatively from the adjoining midblock level of service
- Any intersection whose adjoining midblock approaches reach level of service D or worse is flagged for mitigation consideration and detailed SIDRA modelling at the development application stage
- The assessment is a strategic, rezoning stage analysis to establish structure plan feasibility and the order of upgrades, not design stage performance.

1.4 Reliance on data and information

Advice and recommendations in this report rely on the accuracy of the existing input documentation and advice provided to date, including:

- Post Mining Land Use – Deliverability of Pilot Projects – Due Diligence Report – Macquarie Coal Complex (Aurecon, February 2026)
- Preliminary statutory planning due diligence (Patch Planning)
- *Draft West Wallsend Rezoning Traffic and Transport Assessment* (GHD, August 2025)
- *West Wallsend Colliery, Killingworth Continued Operations Project – Traffic Impact Assessment* (Stapleton Transportation & Planning Pty Ltd, 2010)
- *Teralba Sustainable Resource Centre Traffic Assessment* (AECOM, 2010)
- Insights from internal workshops with DPHI, specialists and agencies
- Updated traffic surveys have not been undertaken for this assessment. Survey data has been adopted from the West Wallsend assessment (November 2024 counts) and is to be refreshed before exhibition.

2 Strategic Context

The planning policy documents in the following sections establish the strategic context that the transport response considers.

2.1 Regional Context and Connectivity

The Precinct is located between Teralba and Barnsley, framed by the M1 Pacific Motorway to the west, the B89 George Booth Drive and Main Road corridor to the north, and the B53 arterial route to the east. Wakefield Road forms the primary north–south spine through the area, providing key internal connectivity.

At a regional scale, the Precinct is well positioned to leverage established transport and infrastructure networks that have historically supported mining operations. These include direct access to the Hunter Expressway, the M1 Pacific Motorway, and broader freight corridors connecting to the Port of Newcastle. This strategic connectivity underpins the Precinct's capacity to accommodate a diverse range of employment uses, including advanced manufacturing, logistics, and emerging industries, aligning with broader regional economic transition objectives. The regional context and connectivity are shown in Figure 2-1.



Figure 2-1 Regional context and connectivity (PMLU Master Plan, 2026)

2.2 Federal

2.2.1 National Freight and Supply Chain Strategy

The National Freight and Supply Chain Strategy is the Commonwealth framework for improving the efficiency, safety and resilience of the national freight task, supported by national action plans for heavy vehicle productivity and access. It is relevant as the employment cores are positioned as freight and logistics land served by the M1 national corridor and the Port of Newcastle, and the strategy direction toward high productivity vehicle access and harmonised heavy vehicle regulation under the Heavy Vehicle National Law underpins the RAV route approach taken in this report.

2.2.2 Sydney to Newcastle High Speed Rail

The Sydney to Newcastle High Speed Rail program, progressed by the Commonwealth High Speed Rail Authority, is developing a preferred scheme with stations proposed at Broadmeadow and at Morisset in the Lake Macquarie local government area, with the alignment still being finalised. It is relevant as development within the Precinct must not prejudice the corridor or preclude any future desire to incorporate a tie into its infrastructure, and as even without a local stop the project would materially improve regional accessibility through interchange via the Northern line and could release capacity on the existing line for additional local passenger and freight services.

2.3 State

2.3.1 Future Transport Strategy

The Future Transport Strategy is the NSW Government framework for transport planning and investment, directing a multi modal and place-based approach, successful places, net zero emissions and resilient networks. It is relevant as the Precinct creates an entirely new movement network, so every new road can be designed from the outset as a complete street with walking, cycling and public transport provision, and due to the strategy elevates network resilience.

2.3.2 NSW Movement and Place Framework

The Movement and Place Framework is the NSW Government approach to planning, designing and operating roads and streets, prepared jointly by TfNSW and the Government Architect NSW and embedded in the Future Transport Strategy, supported by the Practitioners Guide to Movement and Place and the Design of Roads and Streets guideline. It classifies every road by two, its movement function for people and goods and its place intensity as a destination in its own right, producing street types that range from movement corridors with high movement and low place significance through main streets where both are high to local streets and places for people where place dominates. It is relevant for two reasons:

- The Precinct creates a new network on a largely blank canvas, so every street can be assigned its movement and place street type from the outset and designed to it, rather than retrofitting place outcomes onto roads built only for movement
- The framework gives the master plan its sharpest tool for resolving the central tension in this Precinct, being a substantial freight task passing close to the high place value Teralba centre, by concentrating freight on movement corridors that bypass the centre and protecting the centre streets as places for people.

2.3.3 Hunter Regional Plan 2041

The Hunter Regional Plan 2041 sets the land use vision for the region, promoting sustainable growth, 15-minute neighbourhoods and the global gateway roles of the Port of Newcastle and Newcastle Airport. It is relevant as the Precinct contributes regionally significant employment land within the fastest growing corridor

in the State, and the plan requires everyday needs to be met close to home, which shapes the residential components at West Lake, Teralba and Barnsley West and their walking and cycling connections to the Teralba centre and the rail stations.

2.3.4 Greater Newcastle Metropolitan Plan

The Greater Newcastle Metropolitan Plan identifies North West Lake Macquarie as a gateway to Greater Newcastle with a strategic freight and rail position. It is relevant as the employment cores at Cockle Creek and Killingworth give effect to that gateway role, and the plan therefore requires the Precinct to be served by a purpose built restricted access vehicle network, protection of the West Wallsend Colliery rail loop and consistency with the Lower Hunter Freight Corridor alignment.

2.3.5 NSW Freight Policy Reform and the Lower Hunter Freight Corridor

The NSW freight policy reform program and the protected Lower Hunter Freight Corridor between Fassifern and Hexham preserve a future dedicated freight rail bypass of the Newcastle passenger network, with no committed delivery timeframe. They are relevant as the corridor passes the western side of the study area and strengthens the long term case for freight oriented land uses, while the absence of a committed timeframe means land uses relying on the corridor must be treated as a longer term proposition and local freight planning must remain consistent with the corridor in the interim.

2.3.6 Speers Point Transport Improvements (TfNSW)

The Speers Point Transport Improvements are committed TfNSW works on the B53 at the Five Islands Road roundabout, forecast to reduce travel times by up to 50 per cent. They are relevant as the assessment assumes this short-term relief at the only existing Cockle Creek crossing, while identifying that residual capacity works will be required in the medium term as background growth and development demand continue.

2.3.7 West Wallsend Rezoning

The Hunter and Central Coast Development Corporation (HCCDC) owns more than 1,500 hectares of former BHP Steelworks land at West Wallsend, immediately adjoining the Precinct, which planning strategies including the Hunter Regional Plan 2041 have identified as a potential site for environmental, economic and community benefits.

HCCDC is undertaking technical investigations to inform a future rezoning expected to deliver a mix of conservation, residential, employment and recreation land uses. Of particular relevance to this assessment, HCCDC will work with TNSW to preserve land for the future Lower Hunter Freight Corridor in West Wallsend. Given its scale, adjacency and shared freight corridor, the West Wallsend rezoning is a key cumulative development context for the Precinct, requiring coordination of access, network upgrades and staging as both progress.

2.4 Local

2.4.1 Lake Macquarie Local Strategic Planning Statement and Imagine Lake Mac Community Strategy

The Local Strategic Planning Statement (LSPS) and the Imagine Lake Mac community strategy set Council direction for jobs growth and 15-minute neighbourhoods. They are relevant as the structure plan gives effect to that direction, delivering regionally significant employment land and walkable residential neighbourhoods served by an integrated movement network.

2.4.2 Walking, Cycling and Better Streets Strategy

The Walking, Cycling and Better Streets Strategy sets Council direction for active transport across the local government area and identifies a rail trail from Cockle Creek to West Wallsend along the colliery rail corridor crossing the creek in the north east of the precinct and running along the northern boundary to Barnsley. This is a key link to Cockle Creek Station and also would be a safe attractive link between the precinct and key areas of population that may well work in the precinct. It is also relevant as the rail loop corridor is protected for the rail trail should freight reactivation not proceed, and the active transport network within the Precinct is designed to connect to this strategic route.

2.4.3 North West Lake Macquarie Catalyst Area Place Strategy

The Place Strategy is the most location specific document, setting place outcomes for Teralba, Boolaroo and the surrounding centres. It identifies a walkable Teralba centre, improved station access at Teralba and Booragul (items S13 and S16), an active transport crossing of Five Islands Road from Lake Crescent (items S12 and S15), the Pitt Street intersection upgrade (item L19) and the Munibung Road multi modal freight corridor treatment in the Cardiff Advanced Industry Precinct. The North West Catalyst Area is shown in Figure 2-2.

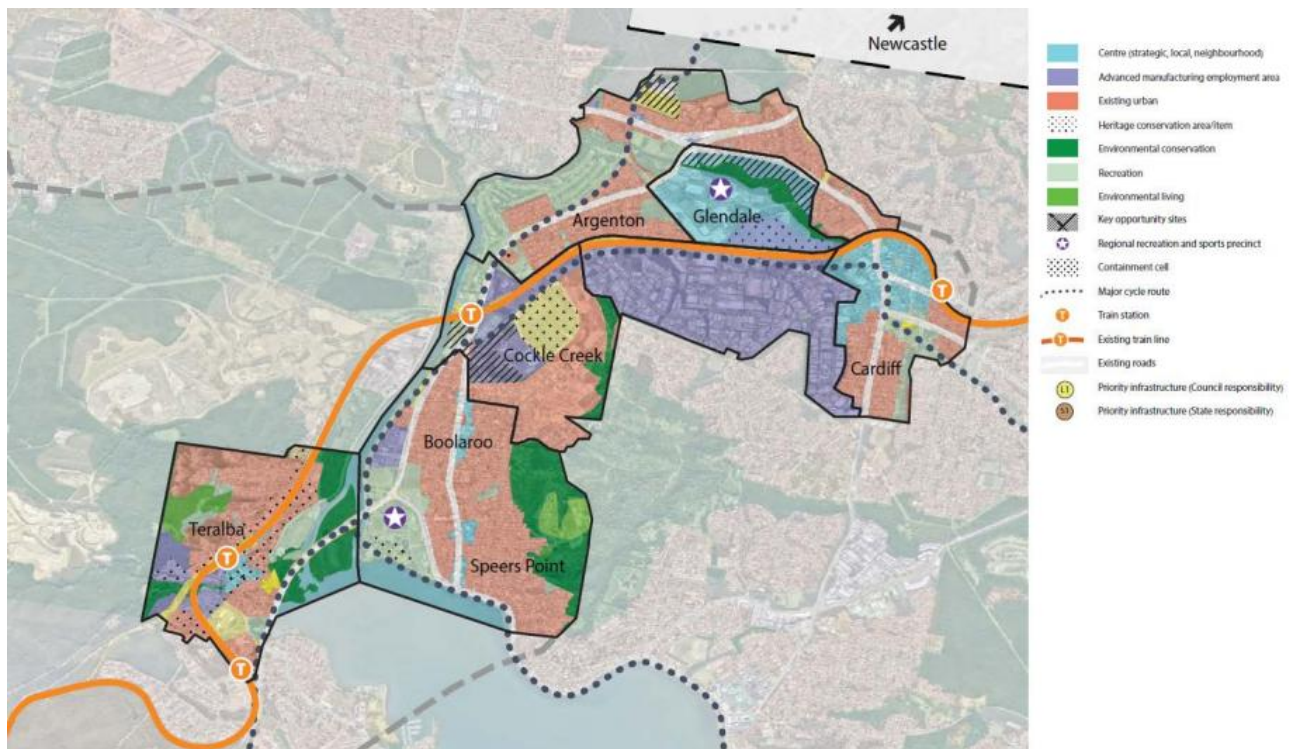


Figure 2-2 North West Catalyst Area (NWCA Place Strategy, LCC 2025)

3 Baseline Analysis

This section provides a strategic, desktop-based assessment of existing traffic and transport conditions, constraints and opportunities of the Precinct. It also provides recommended actions for the post-exhibition phase. Table 3-1 provides details of the following:

- The performance of existing multimodal transport networks surrounding the site, with a particular focus on the areas likely to present capacity constraints to development if not addressed
- Identifies any traffic and transport projects (including both service and capital improvements) currently proposed, in planning or under development for the surrounds
- Provides a high-level indication of the quantum of traffic demand from future land-use options based on the benchmark of IPG Badgerys Creek Road Master Plan
- Provides guidance on prioritisation of transport improvements to ensure the surrounding network can support planned growth and improved precinct access can be established
- Identifies opportunities to incorporate active and public transport to support proposed land uses.

Table 3-1 Traffic and transport analysis summary

Traffic and Transport Attributes	Description
Current conditions	■ The site has good proximity to the M1 Pacific Motorway (3.5 to 6 km), B53 and B89 arterial routes, existing rail infrastructure (Cockle Creek and Teralba stations), and a degraded internal rail loop. ■ Coal was exported by rail and when Westside mine was operational coal was hauled via internal haul road to Eraring power station. Traffic at this time was therefore generated by employees and deliveries, and likely included a low volume of heavy vehicles. ■ As at May 2024 (post active mining operations), the average daily traffic on B53 arterial was as follows: – The Esplanade between T C Frith Avenue and Park Road is 26,000 vehicles – Five Islands Road south of The Esplanade is 40,000 vehicles – T C Frith Avenue between Munibung Road and Seventh Street is 22,000 vehicles.

Traffic and Transport Attributes	Description
Opportunities	■ Existing rail loop and train load-out facilities could present a long-term opportunity for freight-based uses, noting that infrastructure is degraded. There is also an opportunity to reuse the rail loop creek crossing to provide a rail trail connecting Cockle Creek to West Wallsend as identified in Lake Macquarie City Council planning documents <u>Council Walking and Cycling Strategy</u> ■ Proximity to proposed Lower Hunter Freight Corridor strengthens freight-based use potential depending on timing and commitment to delivering the corridor ■ Booragul, Teralba and Cockle Creek stations provide local options for public transport connectivity, subject to the delivery of strong active transport or road based public transport connections linking the Precinct to each station. Planned upgrades to Cockle Creek station by TfNSW to meet accessibility requirements will further strengthen connectivity <u>Cockle Creek Station Upgrade</u> ■ Upgrades to the Northville Drive corridor are the primary means of addressing access to and from the Precinct, providing the capacity for employment cores as background growth and Precinct demand increase. A potential Cockle Creek crossing connecting to the B53 is identified as a longer term opportunity rather than a near term requirement, which could improve directness and reliability of access for high productivity vehicles and present an opportunity for active transport connectivity to Cockle Creek Station. Its need, timing and form would be subject to further investigation as the Precinct develops, and in the longer term such a link could potentially connect through to the M1 Motorway ■ The Speers Point Transport Improvements project is planned to address existing congestion and delays along B53 and is forecast to reduce travel times by up to 50% <u>Speers Point Transport Improvements</u> ■ If M1 Motorway access was available, there are limited existing capacity issues in the section between Awaba Road and B89 with issues mainly confined to further north and being addressed through the M1 Raymond Terrace project ■ There are limited current public transport and active transport infrastructure with no significant current plans given the Precinct's future land use plans are yet to be defined. This provides a somewhat 'blank canvas' to ensure public and active transport is incorporated from day 1 ■ The existing rail loop may require a significant rebuild and would need to be considered in conjunction with the future Lower Hunter Freight Corridor and need to reduce freight on the northern line to increase passenger services ■ Given the uncertainty of the timing for delivery of the Lower Hunter Freight Corridor, land uses that rely on this should be considered as a longer-term proposition.

Traffic and Transport Attributes	Description																
Constraints and risk	- There are current capacity issues at Five Islands Road / The Esplanade roundabout on B53 causing delays. Short to medium term this will be addressed through Speers Point Transport Improvements but may still be a longer-term capacity constraint with growth in the area. Currently significant queuing and delays are experienced in AM, PM and Saturday peak hours - Direct site road access is limited, with challenging terrain on local routes such as Rhondda Road and Wakefield Road and low-lying flood prone routes along Racecourse Road and The Weir Road - Cockle Creek and the railway line are two corridors that sever the precinct from populations to the east and north, with limited crossing opportunities to access Booragul, Teralba and Cockle Creek stations - Internal rail loop requires significant rebuild and new signalling infrastructure - Limited direct connections to M1 Motorway with access via Wakefield Road with steep grades or Appletree Rd / Northville Dr through residential areas to B89. B89 also only provides south facing ramp access to M1 Motorway - Depending on the selected land use, the Precinct could generate volumes of peak traffic that need to be assessed on the wider network. As a guide for large scale industrial precincts, the below table outlines the forecast peak traffic generation for the IPG Badgerys Creek Road Master Plan (Badgerys Creek TMAP). The site area was 184 hectares which is comparable to the Precinct. The Badgerys Creek TMAP provides a useful benchmark for forecast peak traffic generation. As shown in Table 3-2, the Precinct comprises approximately 236,000 m² of GFA, around 38 per cent of the Badgerys Creek master plan's 625,467 m². Notwithstanding, this substantially smaller floor area, the Precinct is forecast to generate approximately 30% less peak hour trips of that master plan in both the AM and PM peaks. This reflects the land use mix and development intensity proposed for the Precinct, which generates a higher rate of trips per square metre than the predominantly industrial Badgerys Creek development. Table 3-2: Daily trip generation comparison against Badgerys Creek TMAP (2036) <table><tr><th>Metric</th><th>Badgerys Creek Road Master Plan</th><th>Proposed Precinct</th><th>Precinct as proportion</th></tr><tr><td>Total GFA (m²)</td><td>625,467</td><td>~236,000</td><td>~38%</td></tr><tr><td>AM peak trips (vph)</td><td>2,080</td><td>1,427</td><td>~69%</td></tr><tr><td>PM peak trips (vph)</td><td>1,988</td><td>1,344</td><td>~68%</td></tr></table>	Metric	Badgerys Creek Road Master Plan	Proposed Precinct	Precinct as proportion	Total GFA (m ²)	625,467	~236,000	~38%	AM peak trips (vph)	2,080	1,427	~69%	PM peak trips (vph)	1,988	1,344	~68%
Metric	Badgerys Creek Road Master Plan	Proposed Precinct	Precinct as proportion														
Total GFA (m ²)	625,467	~236,000	~38%														
AM peak trips (vph)	2,080	1,427	~69%														
PM peak trips (vph)	1,988	1,344	~68%														
Recommendations																	
Post-exhibition	- Staged public transport and active transport connectivity plan to support employment and residential uses - Network capacity assessment to quantify demand from future land uses and identify required upgrades - Detailed assessment of rail loop rebuild requirements, costs, and staging (if required) - Staged road network upgrade strategy, including M1 connection options and local arterial improvements.																

3.1 Road Network

In NSW, roads are formally classified under the *Roads Act 1993*, which defines responsibilities for ownership, funding and management. There are three key road classifications being:

- State roads, which form the primary routes for the movement of people and goods within and between major urban centres. This includes roads classified as freeways, State highways and main roads under the *Roads Act 1993*. State roads are managed by TfNSW
- Regional roads, which provide for travel between smaller towns and districts, as well as perform a sub-arterial function within major urban centres. These roads are managed by local councils but often receive funding from the State Government due to their importance to the road network in NSW
- Local roads, which provide for local circulation and access to properties. Local roads also include collector and local access roads and are managed by local councils.

The surrounding roads and their road classification are shown in Figure 3-1.

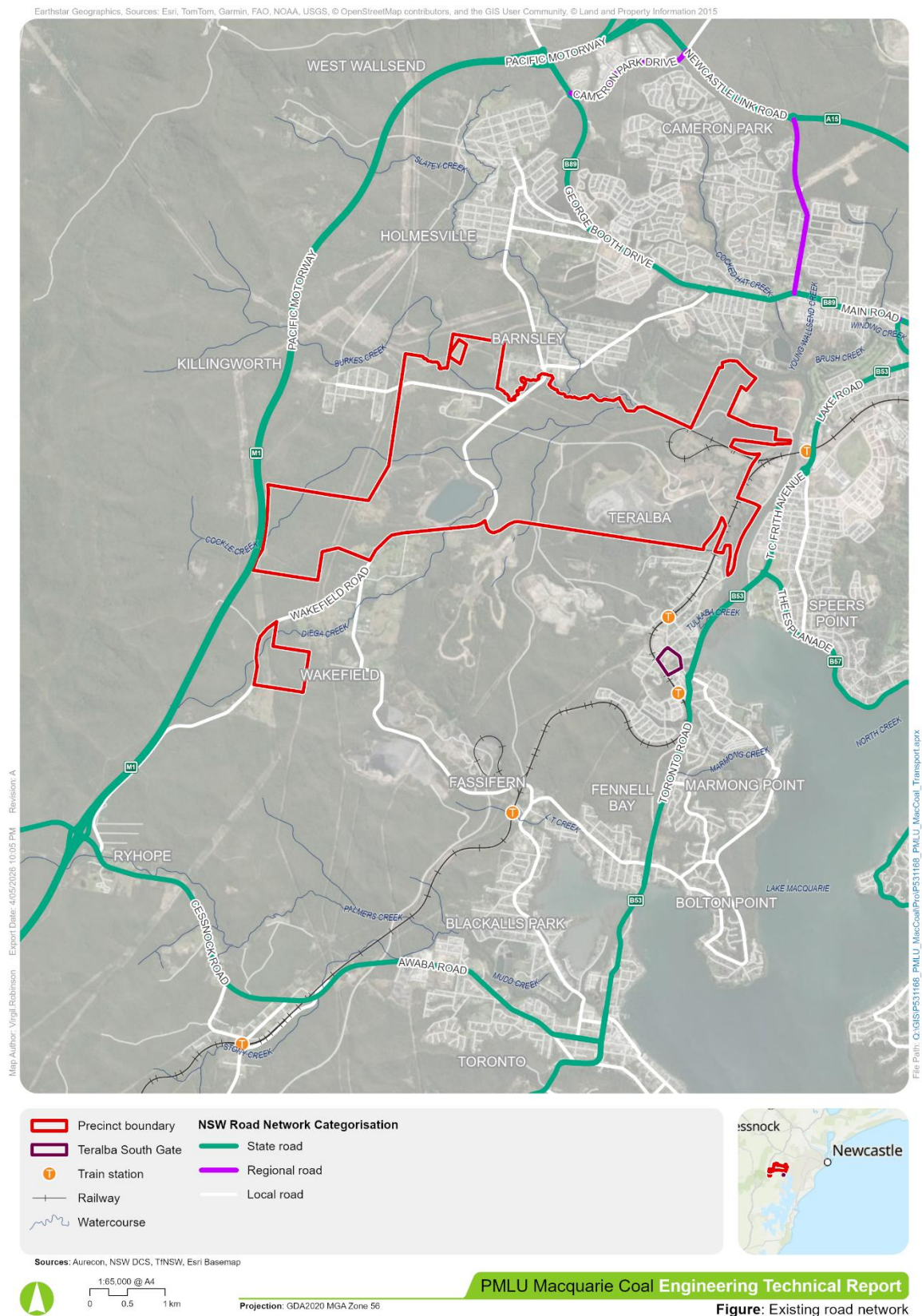


Figure 3-1 Existing road network

3.1.1 State Roads

M1 Pacific Motorway

The M1 Pacific Motorway is a major motorway running north to south that forms part of the Sydney to Brisbane transport corridor. The motorway provides a key connection between Sydney, the Central Coast, Newcastle and the hunter region. Near the study area, the motorway is a divided multi-laned road with a posted speed limit of 100km/hr or 110km/hr. Access between the M1 and the study area is provided via the Newcastle Link (for south bound) or the B89 Route (for north bound). From this point the study area can be accessed via regional roads such as Northville Drive. The route accommodates significant commuter, freight, and tourist traffic volumes and is approved for use by heavy vehicles, including B-doubles and carries approximately 80,000 vehicles per day.

B89 Route – George Booth Drive and Main Road

The B89 route of George Booth Drive and Main Road form a regional arterial connection linking the north bound M1 Pacific Motorway with suburbs surrounding Lake Macquarie and western Newcastle. George Booth Drive serves as a commuter and freight movements between the industrial precinct and motorway network whilst Main Road provides access to urban centres. West of Edgeworth the B89 typically operates as a two-lane undivided road with wider carriageways at key intersections. East of Edgeworth the route B89 has a shoulder lane mostly used for unrestricted parking with a cycle lane. The B89 route connects with Northville Drive via a two-laned roundabout then a single-laned roundabout to Wakefield Road into the study area. The route is also connected to the B53 via a multilane signalised intersection. The route is approved for use by heavy vehicles and accommodates regular bus services and has a posted speed limit between 50km/hr and 80km/hr depending on the surrounding land use.

B57 Route – The Esplanade

The Esplanade (B57 route) is an arterial road providing access along Lake Macquarie and connecting residential and commercial areas. The route comprises a single laned urban road with a cycle lane and signalised intersections. It provides direct property access and accommodates local traffic movements, transport services and cyclists' movements. The route to the west connects to the B53 at Speers point at a two-laned roundabout and the connects to the A37 east of Hillsborough. Parking is prohibited along its length. The posted speed limit ranges between 50km/hr and 60km/hr. Although not a primary freight corridor the route is approved for heavy vehicle use and accommodates approximately 22,000 vehicles per day.

B53 Route – Lake Road | T C Frith | Five Islands Road | Toronto Road

The B53 route comprises of Lake Road, T C Frith Avenue, Five Islands Road and Toronto Road. The route is a major arterial corridor connecting western Lake Macquarie with the M1 Pacific Motorway (via the B89) and surrounding regional road network. The route consists of a two-lane urban road with additional turning lanes at major intersections including the B89 signalised intersection and the two-laned roundabout connecting to the B57. The B53 does not have any existing links over Cockle Creek to access the study area except using single laned regional roads through Teralba. Posted speed limits vary between 50km/hr and 80km/hr depending on the surrounding land use and road environment. The corridor supports a mix of commuter, freight, commercial, and public transport movements and is approved for heavy vehicle movements.

3.1.2 Regional Roads

Cameron Park Drive

Cameron Park Drive is a regional collector road allowing south bound vehicles from the M1 to access the B89 Route from the Newcastle Link. The road generally operates as a two-lane urban carriageway with signalised intersections and turning lanes provided at key intersections. The road connects to the B89 at a two-laned roundabout and the Newcastle link at a signalised intersection. The road has a posted speed limit of 80 km/hr and parking is not permitted. The route accommodates regular heavy vehicle and service vehicle movements associated with residential construction, commercial activity, and regional freight access. Due to significant residential growth in the western Newcastle corridor, the surrounding road network experiences notable peak period congestion, particularly near the Newcastle Link intersection.

3.1.3 Local Roads

Carrington Street

Carrington Street is a sealed, two-way local road that connects to the B89. It carries a posted speed limit of 50 km/hr and traverses a residential area, with on-street parking lanes provided on both sides. The road intersects Withers Street at an unsignalised junction. Carrington Street forms part of the local school bus network.

Withers Street

Withers Street is a sealed, two-way local road with a posted speed limit of 50 km/hr through a residential area. On-street parking lanes are provided on both sides. The road meets Appletree Road at an unsignalised T-junction and forms part of the local school bus network.

Appletree Road

Appletree Road is a sealed, two-way local road with on-street parking lanes on both sides. It passes through a residential area and a school zone associated with West Wallsend High School, with a posted speed limit of 60 km/hr which decreases to 50 km/hr on the approach to Holmesville. The road crosses Slaty Creek via a small bridge. Appletree Road terminates at a single-lane roundabout in Barnsley at its intersection with Northville Drive and Wakefield Road, and forms part of the local school bus network.

Northville Drive

Northville Drive is a sealed, two-way local road connecting to the B89. It passes through a school zone associated with Edgeworth Heights Public School and provides on-street parking lanes within residential areas, with a posted speed limit of 50 km/hr. The road crosses Slaty Creek via a small bridge and meets Wakefield Road at a single-lane roundabout. Northville Drive forms part of the local school bus network.

Wakefield Road

Wakefield Road is a sealed, two-way local road with a posted speed limit of 80 km/hr, reflecting its function as a connecting route between the Barnsley roundabout and the M1 Pacific Motorway at its southern extent.

Rhondda Road

Rhondda Road is a sealed, two-way local road serving an area of mixed residential and industrial land use in Teralba, including established industrial operations. The posted speed limit is 80 km/hr which decreases to 50 km/hr on the approach to Teralba. The road runs broadly parallel to the rail corridor and the Cockle Creek floodplain. The steep descent into Teralba at its eastern end is a significant constraint on heavy vehicle

movement, so in practice Rhondda Road is only readily traversable by larger vehicles toward its western end, with the grade limiting its suitability as a heavy vehicle route into Teralba. Further detail on the Rhondda Road gradient is provided in Section 3.10.5.

Railway Street

Railway Street is a sealed, two-way local road within a predominantly residential area of Teralba, with a posted speed limit of 50 km/hr. The road aligns with the rail corridor and provides local access to adjoining properties, with on-street parking available along the frontages. The road connects to the surrounding local road network at each end. The connecting roads and their intersection control types should be confirmed against the design base.

3.2 Crash statistics

A review of the crash history within the study area was undertaken utilising the publicly available data from the TfNSW Centre for Road Safety. This tool captures recorded crash history in the 5 year period between 2020 and 2024. The analysis breaks down the crash history to outline the trends in the crashes, accident prone areas, the type, severity, and number of crashes, categorised by natural lighting conditions. The crash degree definitions adopted for this analysis are outlined in Table 3-3 below.

Table 3-3: Crash degree definitions

Term	Definition
Fatal	A crash in which a person dies within 30 days form injuries received in a road traffic crash.
Serious	A crash in which a person is admitted to hospital as a result of injuries sustained in a road traffic crash.
Moderate injury crash	A crash in which a person attends an emergency department following a crash but is not admitted to hospital.
Minor / other injury crash	A crash in which a person identified as injured in police crash data who did not attend an emergency department and was not admitted to hospital.
Non-casualty (towaway)	A crash in which at least one vehicle is towed away where there's no death and no person injured.

The crash history of the Precinct road network and its surrounding corridors recorded 235 crashes. The records confirm that the existing network already carries a measurable safety burden, concentrated on the higher order corridors that the Precinct traffic would use. Accordingly, upgrades proposed as part of the Precinct may help with addressing the safety burdens currently present. The crashes are summarised below by severity, by the most frequent Road User Movement (RUM) types and by location. The locations of the recorded crashes are shown in Figure 3-2. Summaries of the recorded crash data along the Precinct road network are provided in Table 3-4 to Table 3-7.

The records establish three points relevant to the rezoning being:

- The existing network already carries a measurable safety burden, with around 29 per cent of all crashes resulting in death or serious injury
- The crashes are concentrated on the higher order corridors and at a small number of intersections rather than spread evenly across the network, which means the safety risk is location specific and can be targeted by treatments at defined points
- The dominant crash types, being rear-end, run-off-road movements and cross traffic, would be the most responsive to the proposed intersection upgrades proposed in this report.

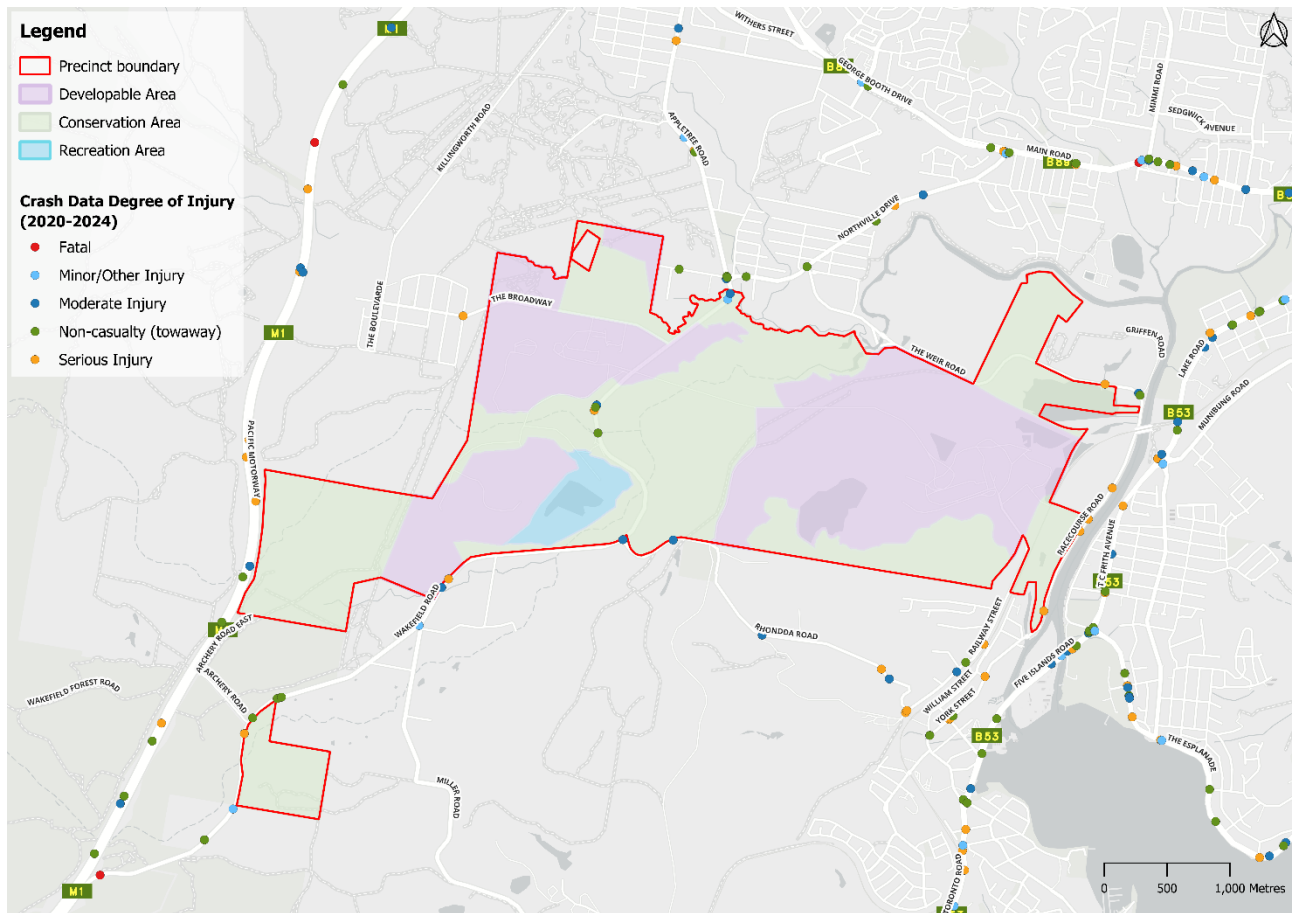


Figure 3-2: Recorded crashes (2020-2024) (TfNSW, 2026)

Table 3-4: Reported crashes by severity, Precinct road network and surrounding corridors

Degree of crash	Number
Fatal	3
Serious injury	65
Moderate injury	69
Minor or other injury	30
Non-casualty (tow-away)	68
Total	235

Table 3-5: Most frequent crash types by Road User Movement (RUM) code

RUM code	RUM description	Number
30	Rear end	51
71	Off road to the left into object	22
10	Cross traffic	14
21	Right through	13
13	Right near	13
20	Head on	11

Table 3-6: Locations with the highest reported crash numbers

Location	Number
Cessnock Road at Wakefield Road	12
Five Islands Road at The Esplanade	10
Newcastle Expressway at Palmers Road	9
The Esplanade at Park Street	8
Main Road at Minmi Road	7
Main Road at Frederick Street	6
Lake Road at Main Road	6

Fatal and serious injury crashes

Three crashes over the period were fatal and a further 65 resulted in serious injury, so killed and seriously injured crashes account for 68 of the 235 records, or around 29 per cent. The serious injury crashes are distributed evenly across the period, at 12 to 14 per year, indicating a persistent rather than improving trend.

The three fatal crashes each involved different conditions and movements and are summarised in Table 3-7.

Table 3-7: Fatal crashes recorded along Precinct road network (2020-2024)

Year	Location	Vehicles involved	Weather	Lighting	RUM code
2020	Wakefield Road at Cessnock Road, Freemans Waterhole	Car and car	Overcast	Daylight	20 - Head on
2022	Newcastle Expressway at George Booth Drive, Holmesville	Car and large rigid	Raining	Daylight	33 - Lane sideswipe
2024	Main Road at Park Street, Edgeworth	Single car	Fine	Darkness	71 - Off road left into object

Across the fatal crashes the common factors are higher speed environments and a mix of run-off-road, head-on and sideswipe movements rather than intersection conflicts, with two of the three occurring in adverse conditions (overcast or raining) or in darkness.

The serious injury crashes follow a broader pattern, dominated by intersection movements (right near, cross traffic and right through together accounting for around a third) and run-off-road events. Most serious crashes occurred in fine weather (~68 per cent) and in daylight (~57 per cent), as such the safety burden reflects the volume and geometry of the network rather than being driven by weather or lighting.

Heavy vehicles were involved in a small number of serious crashes, including rear-end and sideswipe events on Rhondda Road, Cessnock Road and the Newcastle Expressway, and a notable share of serious crashes involved vulnerable road users, with pedal cyclists, motorcyclists and pedestrians appearing across Main Road, Lake Road, The Esplanade, T C Frith Avenue, Appletree Road, Racecourse Road and Railway Street.

3.3 Traffic Growth

Historical traffic growth on the higher-order corridors framing the Precinct (the M1, the B53 and the B89) shows modest long term background growth, broadly in the order of 0.5 to 1 per cent per annum, with further detail provided in Appendix A.

A global background traffic growth rate of 1.2 per cent per annum has been adopted for the assessment, based on the STFM growth rates documented in the Draft West Wallsend Rezoning TIA (GHD, 2025). The available traffic count stations are located on surrounding arterial roads rather than lower-order local roads within the Precinct; therefore, the STFM growth profile was considered a suitable basis for applying a consistent background growth allowance.

Review of the STFM data indicates that the highest forecast growth occurred during the 2021–2026 period, with annual growth rates of approximately 1.9 to 2.0 per cent. This period also aligns with substantial recent development activity evident from historical satellite imagery, with much of the surrounding area now appearing largely built out. For the subsequent five-year periods from 2026 to 2041, the STFM indicates lower annual growth rates, generally ranging from approximately 1.0 to 1.59 per cent. As most base traffic volumes used in the assessment are from 2024, an adopted growth rate of 1.2 per cent per annum, rounded for assessment and reporting purposes, is considered to appropriately reflect the forecast growth profile beyond the initial high-growth period.

3.4 Existing Trip Patterns

Existing trip patterns confirm a strongly car dependent baseline, with private vehicles accounting for around 63 to 66 per cent of journeys to work and only 0 to 1 per cent by public transport and 1 to 2 per cent by active transport.

This reflects the current regional, lower density context and limited public and active transport connectivity rather than a desired end state. Consistent with the mode shift sought by NSW Future Transport, the Movement and Place framework and Council's active transport strategies, the Precinct is planned to improve on this baseline rather than perpetuate it, through land use, density and connectivity that support higher walking, cycling and public transport mode share over time. As such the car-based trip generation adopted in this assessment represents a conservative approach pending the delivery of those improvements.

3.5 Existing Midblock Performance Results

The 2026 baseline AM and PM Peak hour midblock performance assessment was undertaken for locations illustrated in Figure 3-3. The 2026 baseline indicates how the network performs in its present state, before any Precinct demand, and identifies where the spare capacity sits and where it does not. LoS A to C indicates free to stable flow with little delay, LoS D indicates the onset of congestion where flow becomes sensitive to small increases in demand, level of service E indicates operation at or near capacity, and level of service F indicates demand exceeding capacity with queuing and breakdown.

- The Wakefield Road spine operates at LoS A to B in both directions and both peaks, the strongest result in the network. The northbound (inbound to the employment cores) direction carries marginally more than the southbound, but both sit well within capacity, confirming the spine has the potential to absorb some level of additional freight tasks prior to any upgrades.
- Rhondda Road, The Broadway, Appletree Road, Racecourse Road and Charlton Street all operate at LoS A to B in both directions, reflecting their current role as lightly utilised local and collector roads.
- Northville Drive is the exception. The segment north of The Weir Road already runs at LoS C to D and the segment between Wakefield Road and The Weir Road at LoS C during AM peak and LoS D during PM peak. The eastbound and northbound toward the B89 direction is the more loaded in the AM and the pattern reverses in the PM, consistent with a commuter tidal flow to and from Newcastle.

The 2026 baseline therefore frames the assessment. The Wakefield Road, Rhondda Road and western local corridors have ample capacity, while the Northville Drive corridor toward the B89 is the part of the network closest to its limit based on current traffic volumes and patterns, making it the corridor to monitor as demand grows. The 2026 baseline AM and PM Peak hour midblock performance are summarised in Table 3-8.

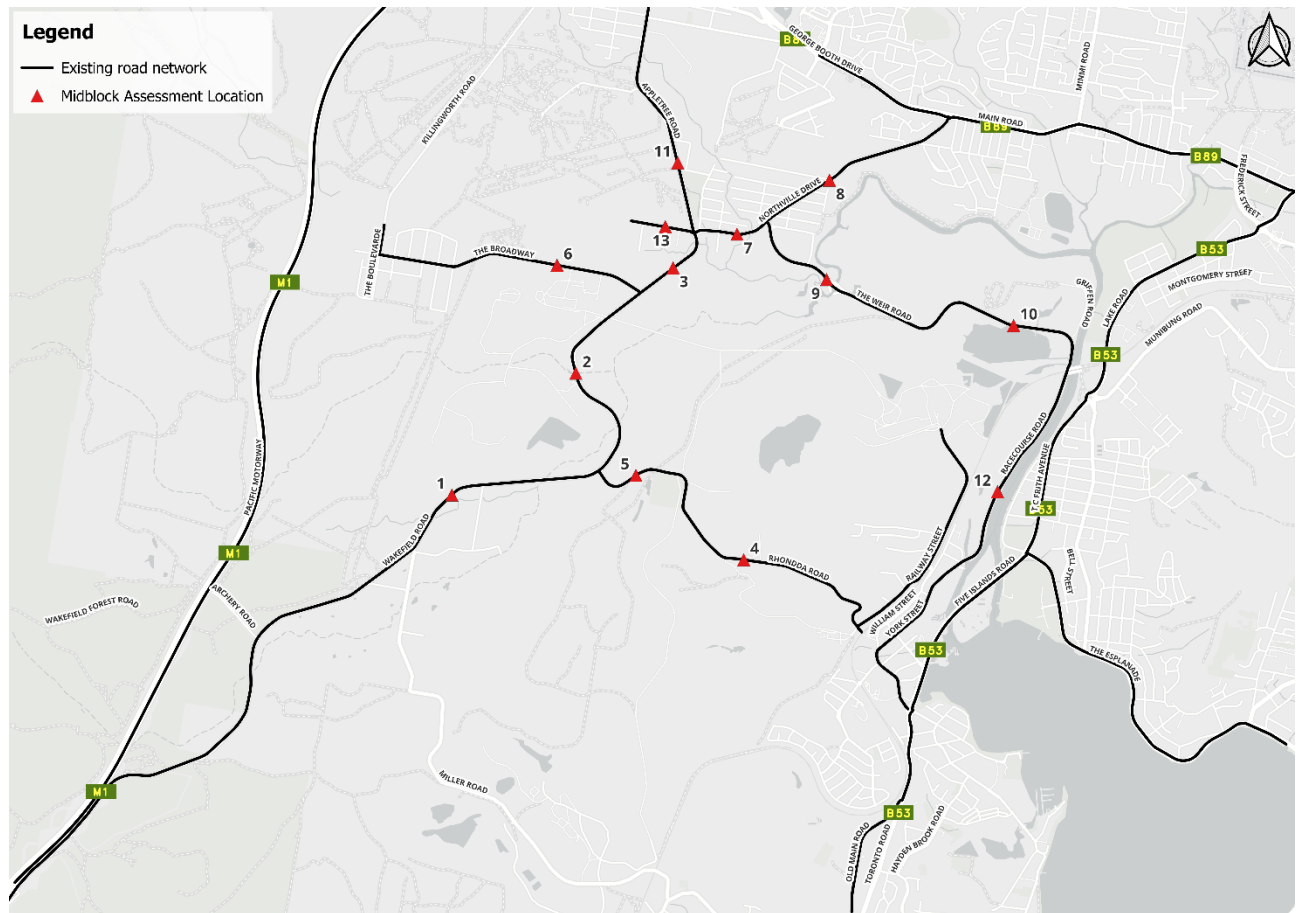


Figure 3-3: 2026 baseline midblock assessment locations

Table 3-8: 2026 baseline midblock performance

ID	Segment	Direction	AM Peak		PM Peak	
			VCR	LoS	VCR	LoS
1	Wakefield Road, south of Rhondda Road	NB	0.16	A	0.14	A
		SB	0.15	A	0.18	A
2	Wakefield Road, between Rhondda Road and The Broadway	NB	0.22	A	0.19	A
		SB	0.21	A	0.23	A
3	Wakefield Road, between The Broadway and Northville Drive	NB	0.32	B	0.23	A
		SB	0.23	A	0.31	B
4	Rhondda Road, east of Area 1 southern Access Point	EB	0.03	A	0.03	A
		WB	0.04	A	0.02	A
5	Rhondda Road, west of Area 1 southern Access Point	EB	0.04	A	0.05	A
		WB	0.06	A	0.03	A
6	The Broadway	EB	0.15	A	0.11	A
		WB	0.07	A	0.16	A
7	Northville Drive, between Wakefield Road and The Weir Road	EB	0.51	C	0.40	C
		WB	0.34	B	0.55	D

ID	Segment	Direction	AM Peak		PM Peak	
			VCR	LoS	VCR	LoS
8	Northville Drive, north of The Weir Road	EB	0.54	D	0.52	C
		WB	0.48	C	0.55	D
9	The Weir Road, west of Area 1 northern Access Point	EB	0.35	B	0.32	B
		WB	0.25	B	0.45	C
10	The Weir Road, east of Area 1 northern Access Point	EB	0.35	B	0.32	B
		WB	0.25	B	0.45	C
11	Appletree Road	NB	0.27	B	0.31	B
		SB	0.25	B	0.32	B
12	Racecourse Road	NB	0.10	A	0.14	A
		SB	0.14	A	0.10	A
13	Charlton Street	EB	0.05	A	0.03	A
		WB	0.02	A	0.07	A

3.6 Active Transport Network

The existing active transport network within and surrounding the Precinct is limited and fragmented, reflecting the area's historical industrial and mining land uses. Within the study area itself, there is currently no formal active transport infrastructure no dedicated pedestrian paths, shared use paths, or cycling facilities exist internal to the site.

In the broader locality, active transport infrastructure is generally concentrated along the surrounding arterial road network as show in Figure 3-4. The most significant facility in the immediate area is a shared use path running along The Esplanade and T C Frith Avenue on the eastern fringe of the study area, providing a north-south connection toward Speers Point. Beyond this, cycling is largely accommodated via mixed traffic arrangements on local roads, with a dedicated bicycle path present along the M1 Pacific Motorway corridor further to the west. There are no dedicated separated cycling facilities directly serving the study area.

Connectivity across Cockle Creek is a significant constraint. Five Islands Road provides the only crossing point available to pedestrians and cyclists within the immediate area, limiting direct active transport access between the study area, Cockle Creek Station, and the residential areas to the east.

Pedestrian and cycling access to the nearest rail stations is provided via the local road and footpath network. A shared use path runs along the rail corridor in the vicinity of both Teralba and Booragul stations, providing a degree of active transport connectivity between the two stations and the study area boundaries. Cockle Creek Station is accessible primarily via footpaths and the shared use path along Five Islands Road, though no dedicated cycling infrastructure connects directly to the station from the study area.

Overall, the existing active transport network provides limited connectivity to surrounding public transport services, with significant gaps in infrastructure continuity both within the study area and across the broader locality. These key gaps are constraints to achieving the shift to more sustainable travel modes sought by the strategy for the Precinct. Addressing these gaps through a connected walking and cycling network linking the developable areas to the nearby train stations, local centres and bus services will be necessary to support higher active and public transport mode share over time.

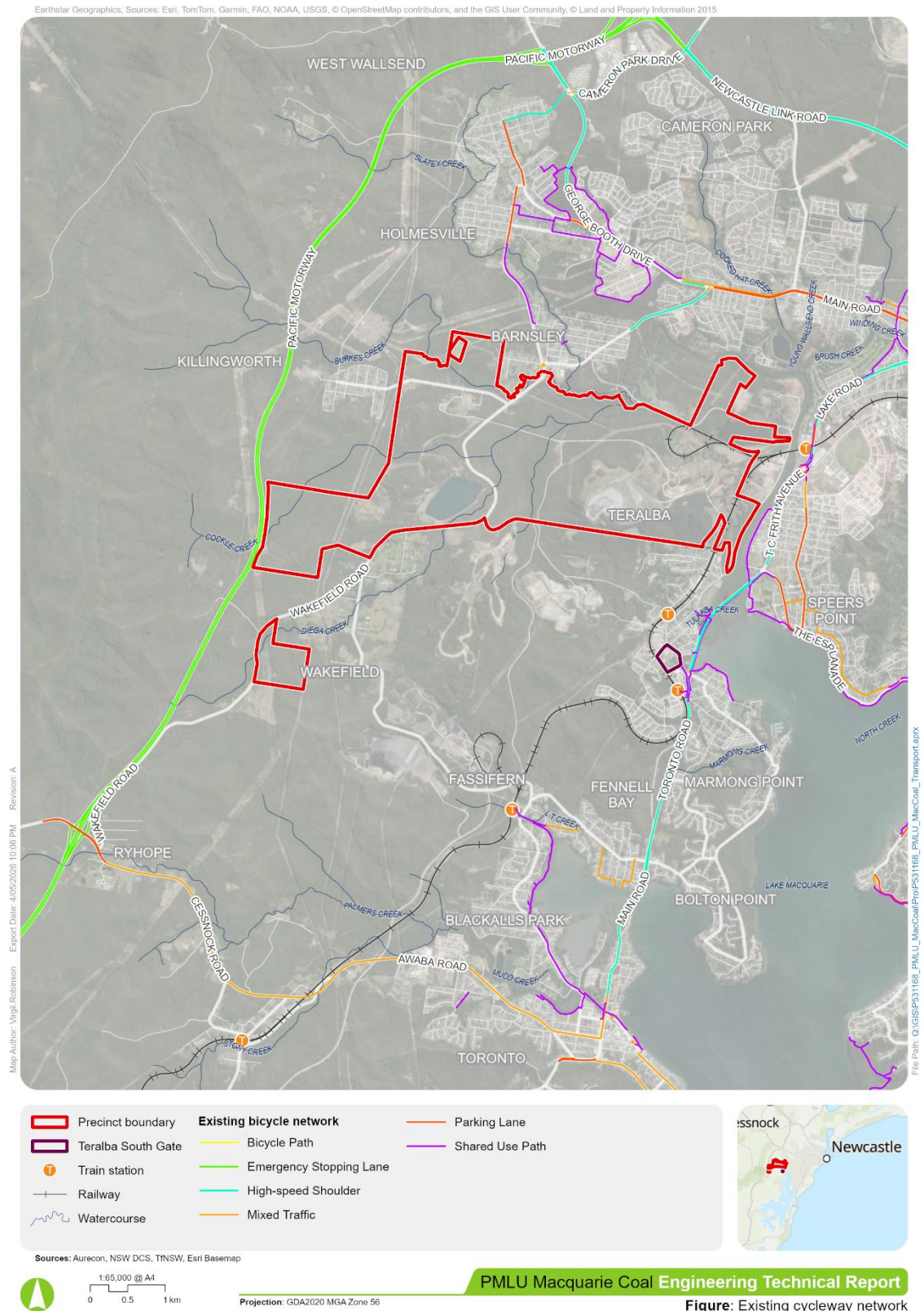


Figure 3-4 Existing cycleway network

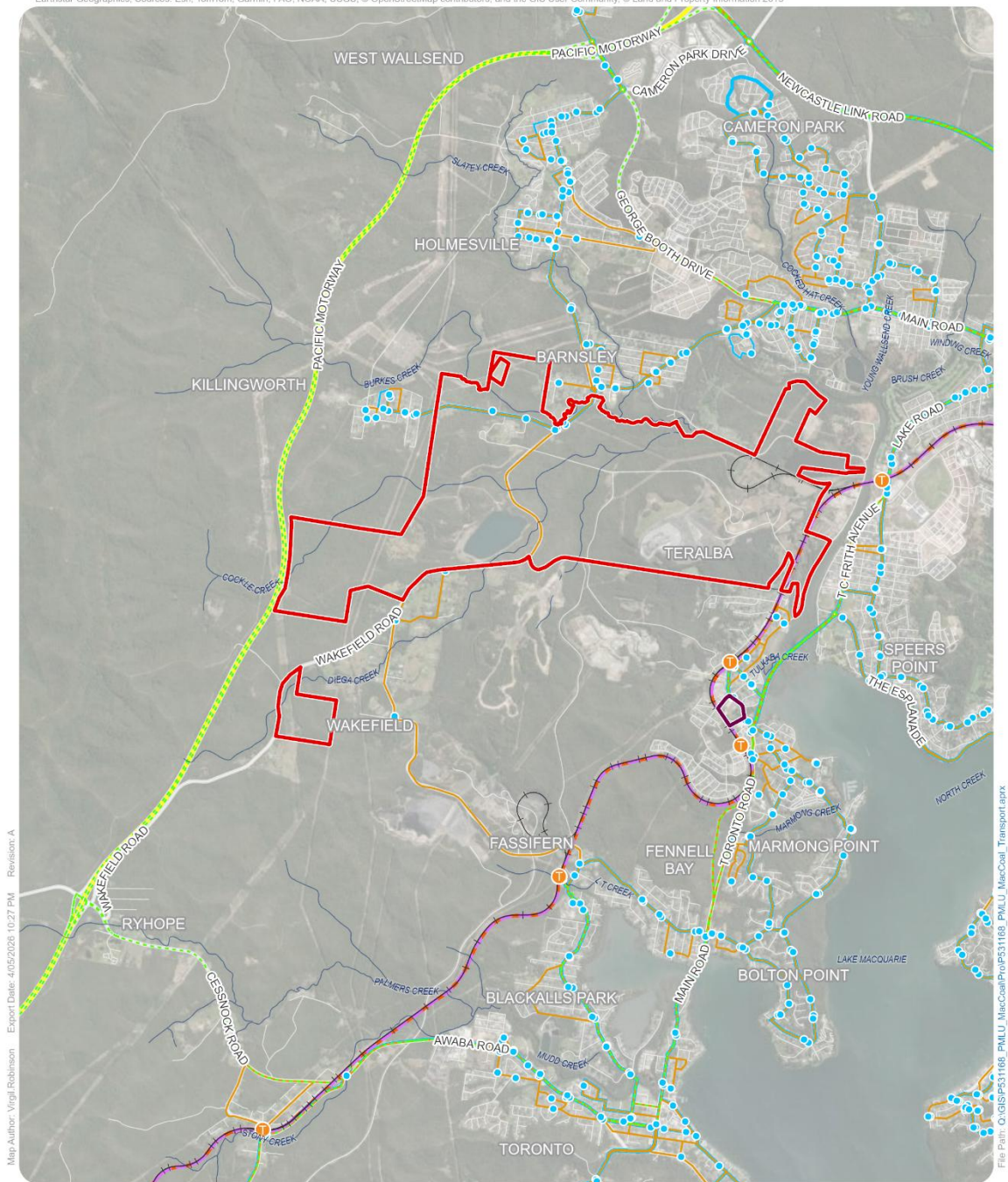
3.7 Bus Network

The Precinct has limited access to the general public bus network, with the Hunter Buses Network providing service along the northern edge of the study area. Bus stops are present at or near the northern boundary, associated with services connecting Killingworth and Glendale via the local road network. Most of the study area, including its central and southern portions, remains beyond walkable distance of any regular route bus service, as shown in Figure 3-5.

More broadly, bus services are concentrated along the Five Islands Road, Seventh Street, Lake Road and Main Road corridor to the east, forming the primary bus route serving the immediate locality under the Hunter Buses Network, operated under the TfNSW public transport network. Services provide connections to surrounding suburban centres including Speers Point and broader regional destinations including Newcastle CBD.

School bus services operate through the Precinct along Wakefield Road, routing between Wakefield and Barnsley Public School via Killingworth during the morning and afternoon school peaks. These services do not form part of the general public bus network and are not available for general passenger use. However, they are recognised as it could be expected that bus stops on these routes would likely attract concentrations of vulnerable users, particularly children walking during the school start and finish periods coinciding with peak times, which represents a road safety consideration where these routes coincide with proposed heavy vehicle and RAV movements. This reinforces the strategy of routing heavy vehicle movements away from sensitive areas where possible, and of providing safe crossing points and separation between heavy vehicles and pedestrians at the respective intersections.

In terms of network capacity, bus services along the Five Islands Road, Seventh Street, Lake Road and Main Road corridor operate at a combined peak frequency of approximately 4 services during the AM peak hour (8:00 am to 9:00 am) and approximately two services per hour during the PM peak (4:00 pm to 6:00 pm). Based on an assumed seated capacity of approximately 45 passengers per service, this equates to a theoretical capacity of approximately 180 passengers per hour in the AM peak and 90 passengers per hour in the PM peak direction.



Map Author: Virgil Robinson Export Date: 4/05/2020 10:27 PM Revision: A

File Path: O:\GIS\531168_PMLU_MacCoal\Prop531168_PMLU_MacCoal_Transport.aprx



Sources: Aurecon, NSW DCS, TNSW, Esri Basemap



1:65,000 @ A4
0 0.5 1 km

Projection: GDA2020 MGA Zone 56



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Figure: Existing public transport network

Figure 3-5 Existing public transport network

3.8 Rail Network

The Precinct is served by the Central Coast & Newcastle Line (CCN), which provides intercity rail connections between Newcastle Interchange and Sydney Central via the Central Coast. The line serves as the primary heavy rail corridor for the region, providing connections to major destinations including Newcastle CBD, Gosford, and Sydney, with broader interchange opportunities available at key nodes along the corridor.

Four stations are located in the surrounding area particularly on the eastern side, as shown in Figure 3-5. Teralba Station is located east of the Precinct providing the most direct rail access. The southern parcel of the Precinct sits between Teralba Station to the north and Booragul Station to the south, affording this portion of the site particularly strong access to the rail network. Cockle Creek Station, located to the north, serves as a broader interchange station and key public transport node for the locality. Fassifern Station, located further south beyond the study area boundary, serves as the most significant station in the broader locality, accommodating CCN services as well as intercity and regional services to Brisbane, Casino, Grafton and Armidale.

CCN services at Teralba, Booragul and Cockle Creek stations operate at a frequency of approximately 2 services per hour in each direction during the AM peak (7:00 am 8:00 am) and PM peak (5:00 pm 6:00 pm) periods, resulting in a combined frequency of approximately 4 services per hour across both directions. Services are operated by the Mariyung (D-Set) fleet, which operates in configurations of 4 to 10 cars with a seated capacity ranging from 325 passengers (4-car) to 1,312 passengers (10-car) per service. Based on a conservative assumed 4-car consist, this equates to a theoretical capacity of approximately 650 passengers per hour in the peak direction.

3.9 Freight Network

3.9.1 Road freight network

The road freight network in the vicinity of the Precinct is served by multiple approved heavy vehicle routes. B-double approved routes, as identified on the National Heavy Vehicle Regulator (NHVR) network, in the immediate area include the Pacific Motorway (M1), the B53 arterial corridor (Five Islands Road and The Esplanade), the B89 arterial route, Northville Drive and TC Frith Avenue. These routes are approved for B-double vehicles up to 19m, 23m and 25/26m configurations.

Within and immediately adjacent to the Precinct, Rhonda Road along the southern boundary, a section of Wakefield Road within the western portion of the study area, and York Street east of the southern parcel are also B-double approved for the same vehicle configurations. These routes provide connectivity between the study area and the broader regional road freight network, including access to the M1 Pacific Motorway.

3.9.2 Rail freight network

The existing and planned freight network in the surrounding area is shown in Figure 3-6. The Precinct currently has no active rail freight operations. The only rail freight infrastructure within the Precinct is the disused internal rail loop associated with the former Macquarie Coal Complex, which is currently in a dilapidated and unusable state. The disused West Wallsend Colliery rail loop formerly carried coal to the Port of Newcastle but is dilapidated, and its junction with the Main Northern line near Cockle Creek faces north only, therefore, trains cannot turn directly south toward Teralba without a reversing movement via Port Waratah.

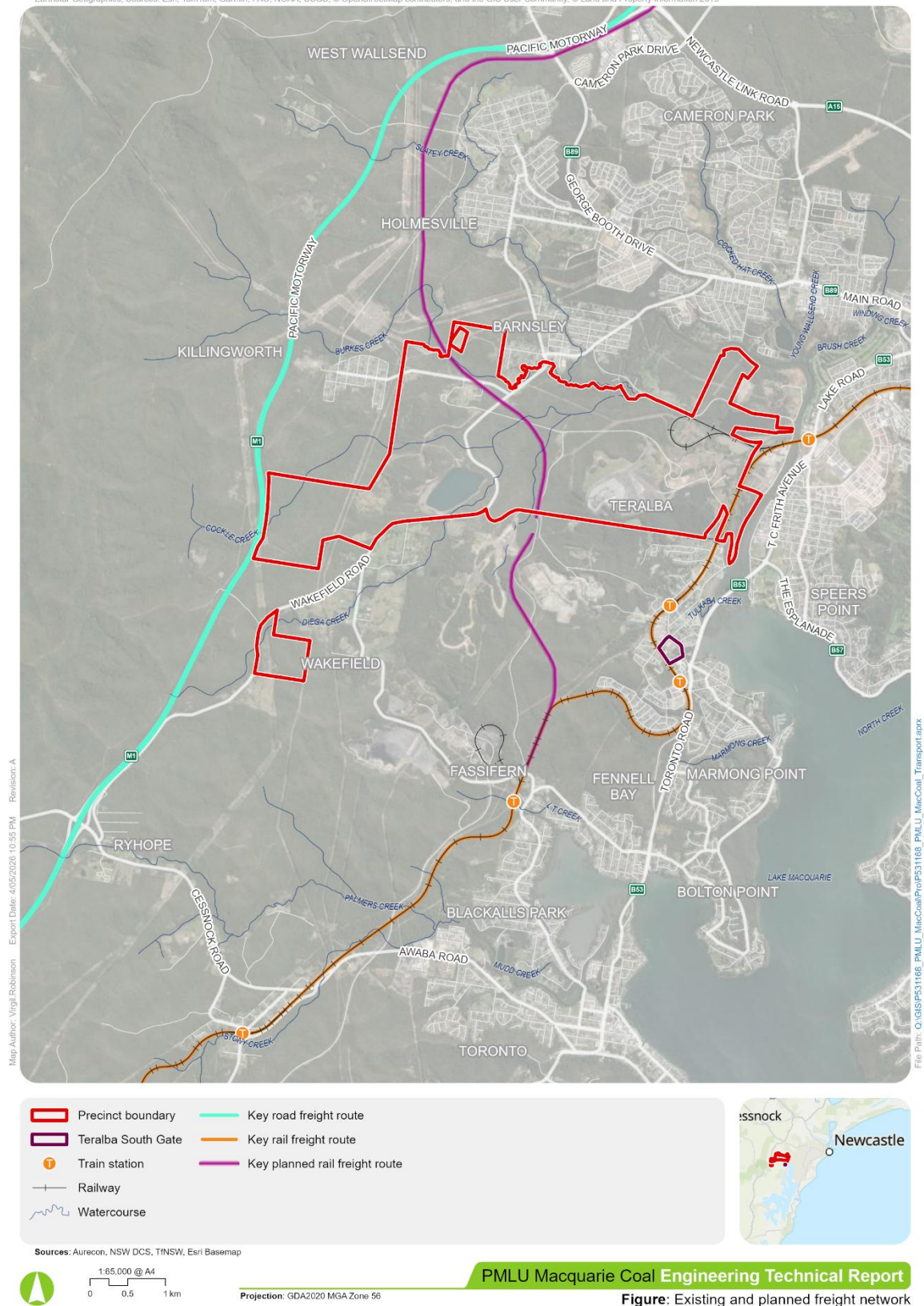
This loop historically connected the site to the broader rail network and the Port of Newcastle for coal export purposes but no longer functions in this capacity. The disused rail loop also presents a separate long-term opportunity for adaptive reuse as a rail trail connecting Cockle Creek to West Wallsend, as identified in the Lake Macquarie City Council Walking and Cycling Strategy.

The Main Northern Line passes adjacent to the study area via Teralba and Cockle Creek stations, managed by the Australian Rail Track Corporation (ARTC) and carrying both passenger and freight services connecting the Hunter Valley to the Port of Newcastle. While this provides a degree of proximity to the active

freight network, there is currently no operational connection between the study area's internal rail infrastructure and the Main Northern Line.

The proposed Lower Hunter Freight Corridor (LHFC), shown as a key planned rail freight route in Figure 3-6, is a TfNSW initiative to provide a dedicated freight rail line between Fassifern and Hexham station (located further up north of the Precinct). The corridor alignment was protected in 2022, though no construction timeline has been committed to. Subject to its delivery, the LHFC may present a long-term freight connectivity opportunity that could strengthen the case for freight-based land uses on the site, and this potential should be protected through the structure plan.

Earthstar Geographics, Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community, © Land and Property Information 2015



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Figure: Existing and planned freight network

Figure 3-6 Existing and planned freight network

3.10 Existing Traffic Issues

The existing road network in the vicinity of the study area is subject to areas of known performance and connectivity constraints, which are summarised below.

3.10.1 Roundabout Congestion (Five Islands Road / The Esplanade / TC Frith Avenue)

The roundabout at the intersection of Five Islands Road, The Esplanade and TC Frith Avenue on the B53 arterial corridor is a known congested location, experiencing significant queuing and delays during AM, PM and Saturday peak periods. As at May 2024, the average daily traffic volumes of approximately 40,000 vehicles on Five Islands Road south of The Esplanade, 26,000 vehicles on The Esplanade between TC Frith Avenue and Park Road, and 22,000 vehicles on TC Frith Avenue between Munibung Road and Seventh Street reflect the considerable demand pressure on this intersection.

Live traffic conditions illustrating stop-and-go and slow traffic on the Five Islands Road approach are shown in Figure 3-7. The Speers Point Transport Improvements project is planned to address this existing congestion and delays along the B53 corridor and is forecast to reduce travel times by up to 50% (TfNSW, Speers Point Transport Improvements).



Figure 3-7 Five Islands Road / The Esplanade / TC Frith Avenue Roundabout traffic conditions (Source: ArcGIS Online Live Traffic Services, 9:30 am to 10:00 am)

3.10.2 Connectivity over Cockle Creek

East West connectivity across Cockle Creek is limited to the B53 Five Islands Road corridor, with no alternative crossing available in the immediate area. This constraint concentrates traffic demand onto a single crossing point, contributing to and exacerbating the congestion issues at the Five Islands Road / The Esplanade / TC Frith Avenue roundabout described above.

3.10.3 Connectivity to the M1 Pacific Motorway

Direct connectivity between the study area and the M1 Pacific Motorway is limited. The absence of a direct arterial connection requires traffic to navigate the local road network, reducing the efficiency and reliability of freight and general vehicle movements to and from the motorway network.

3.10.4 Limited road connectivity - Teralba

Teralba has poor connectivity to the broader road network, with traffic concentrated through the William Street bridge over the rail line north of Teralba Station, as shown in Figure 3-8. This bridge represents a significant pinch point, limiting access from the western portions of the study area to the broader Teralba road network via a single narrow crossing.



Figure 3-8 William Street Bridge - Single Rail Crossing Point, Teralba

3.10.5 Rhondda Road Gradient

Connectivity between Wakefield Road and Teralba is limited to Rhondda Road. Rhondda Road presents significant challenges for heavy vehicle movements due to its vertical alignment. As illustrated in Figure 3-9, the road features a sustained uphill climb of approximately 1 km for eastbound traffic with grades ranging from 5 to 10%, followed by a sustained downhill descent of over 1 km at similar grades.



Figure 3-9 Rhondda Road Elevation Profile (Source: Google Earth)

Assessed against Table 8.2 of the Austroads *Guide to Road Design Part 3*, reproduced in Table 3-9, the sustained grades sit in the 9 to 12 per cent band, where heavy vehicles are significantly slower uphill with substantial downhill reductions on winding alignments, while the localised slopes approaching over 12 per cent which Austroads classifies as not negotiable by heavy vehicles. Rhondda Road therefore is not recommended for heavy vehicle access from Teralba.

Table 3-9: Effect of grade on heavy vehicles (Austroads Guide to Road Design Part 3, Table 8.2)

Grade	Heavy vehicle effect uphill	Heavy vehicle effect downhill	Road type suitability
0 to 3%	Minimal	Minimal	All roads
3 to 6%	Some reduction on high speed roads	Minimal	Low to moderate speed roads including high volume roads
6 to 9%	Significantly slower	Minimal on straight alignment, substantial on winding alignment	Mountainous terrain, auxiliary lanes usually needed at high volumes
9 to 12%	Much slower	Significantly slower on straight alignment, much slower on winding alignment	Auxiliary lanes needed at moderate to high volumes, runaway vehicle facilities considered where commercial vehicle proportion is high
12 to 15%	15% is the maximum negotiable	Extremely slow	Satisfactory only on low volume roads with very few or no commercial vehicles
15 to 33%	Not negotiable	Not negotiable	Extreme cases only, short lengths, no commercial vehicles

Rhondda Road is also the main sealed connection between Teralba and the interior of the Precinct. The only other sealed connecting, Powerhouse Road, has access through Teralba as it is geometrically constrained by a narrow bridge over the railway that limits its suitability for larger vehicles. With both eastern approaches constrained, the issue is not an inconvenience of one of severable viable routes, but a structural barrier on the eastern side of the Precinct. Access to the Precinct from Teralba is accordingly difficult for all vehicles and unsuitable for heavy vehicles, which underpins strategy adopted in this report.

Freight access is established from the south via Wakefield Road and the north via Northville Drive rather than from the east. The east access through Teralba is limited to general heavy vehicles for local servicing and is not available for the RAV access strategy in the short term.

A potential future crossing of Cockle Creek linking The Weir Road and Racecourse Road could improve connectivity between Teralba, the B53 corridor and the interior of the Precinct, offering a more direct eastern connection than Rhondda Road. If delivered, such a connection has the potential to support heavy vehicle access via the B53, which could help divert freight away from residential catchments such as Barnsley and reduce demand on Northville Drive. The crossing is identified as an opportunity to enhance connectivity rather than a prerequisite for the Precinct, and its delivery, timing and form would be subject to further investigation.

3.10.6 Racecourse Road and The Weir Road

Travel on this route is currently limited to smaller vehicles due to a rail overpass which restricts vertical clearance to 4.2 metres. This overpass supporting structure also introduces width constraints with a structural wall dividing the traffic lanes. More broadly, the active rail line presents three compounding constraints:

- The limited clearance beneath the existing road-under-rail bridge
- The restricted capacity and geometry of the existing road-over-rail bridge
- The significant cost and complexity of crossing an active rail line to establish any additional crossing point.

Together these limit the practical options for improving eastern access and are a key consideration in the connectivity and freight routing strategy for the Precinct.

4 Structure Plan

4.1 Proposed Structure Plan

The Structure Plan (GHD, 2026) organises the Macquarie Coal Complex into a series of sub-precincts set within a framework of conservation lands, creek corridors and the existing rail and road network. The developable areas are concentrated in the centre and east of the site and are separated from one another by extensive conservation areas, such that the transport network must connect a set of discrete parcels rather than a single contiguous landholding. The Structure Plan distinguishes developable areas, conservation areas, recreation areas, and further investigation areas, and identifies the indicative road hierarchy and connectivity links that the transport assessment tests.

The principal sub precincts and their transport roles are:

- Areas 1 forms the central and eastern employment core around the former Cockle Creek workings, the largest developable holdings, accessed from The Weir Road, Northville Drive with a potential future Cockle Creek crossing to the B53 identified as an opportunity to improve eastern connectivity
- Area 2 is the northern developable area between The Broadway and Wakefield Road, accessed from the Wakefield Road spine and The Broadway
- Area 4 is the western developable area south of Wakefield Road around the water body, accessed from Wakefield Road and Rhondda Road
- Area 5 is a small developable and consideration areas on the northern and eastern fringes, accessed from Appletree Road, Northville Drive and the local road network
- Area 6 is largely conservation and recreation areas to the west and south-west, framed by the Sugarloaf Conservation Area, with limited or no developable yield.

Wakefield Road, collector roads distributing through the developable areas, and a network of existing and proposed local roads. The Structure Plan is shown below Figure 4-1.

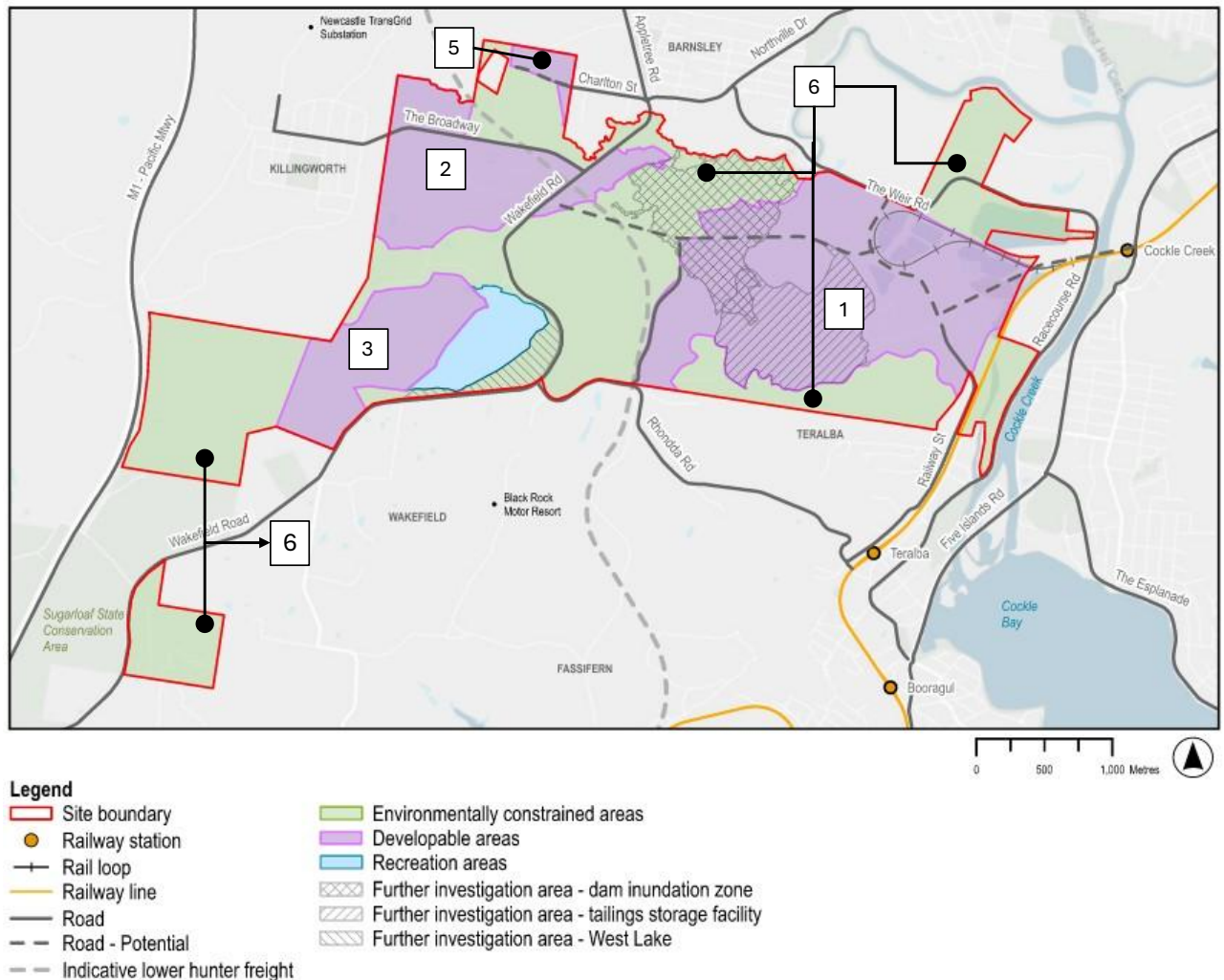


Figure 4-1: Structure Plan (GHD, 2026)

4.2 Transport Network

4.2.1 Overview

The road network is organised as a clear hierarchy connecting the developable land to the regional network, consistent with the master plan. Three corridors frame the area, being the B53 along the eastern edge, the B89 to the north and the M1 to the west. A sub arterial road enters off Wakefield Road and the western connections to carry the main internal movement, while collector and local roads distribute traffic to the developable parcels.

Access to the developable parcels is taken from defined points on Rhondda Road, Wakefield Road, The Broadway, Charlton Street, The Weir Road and Powerhouse Road. A new local road connection across the centre of the area is contingent on declassification of the tailings storage facility and an interim connection arrangement is to be provided. The short-term and long-term road networks are shown below in Figure 4-2 and Figure 4-3 respectively.

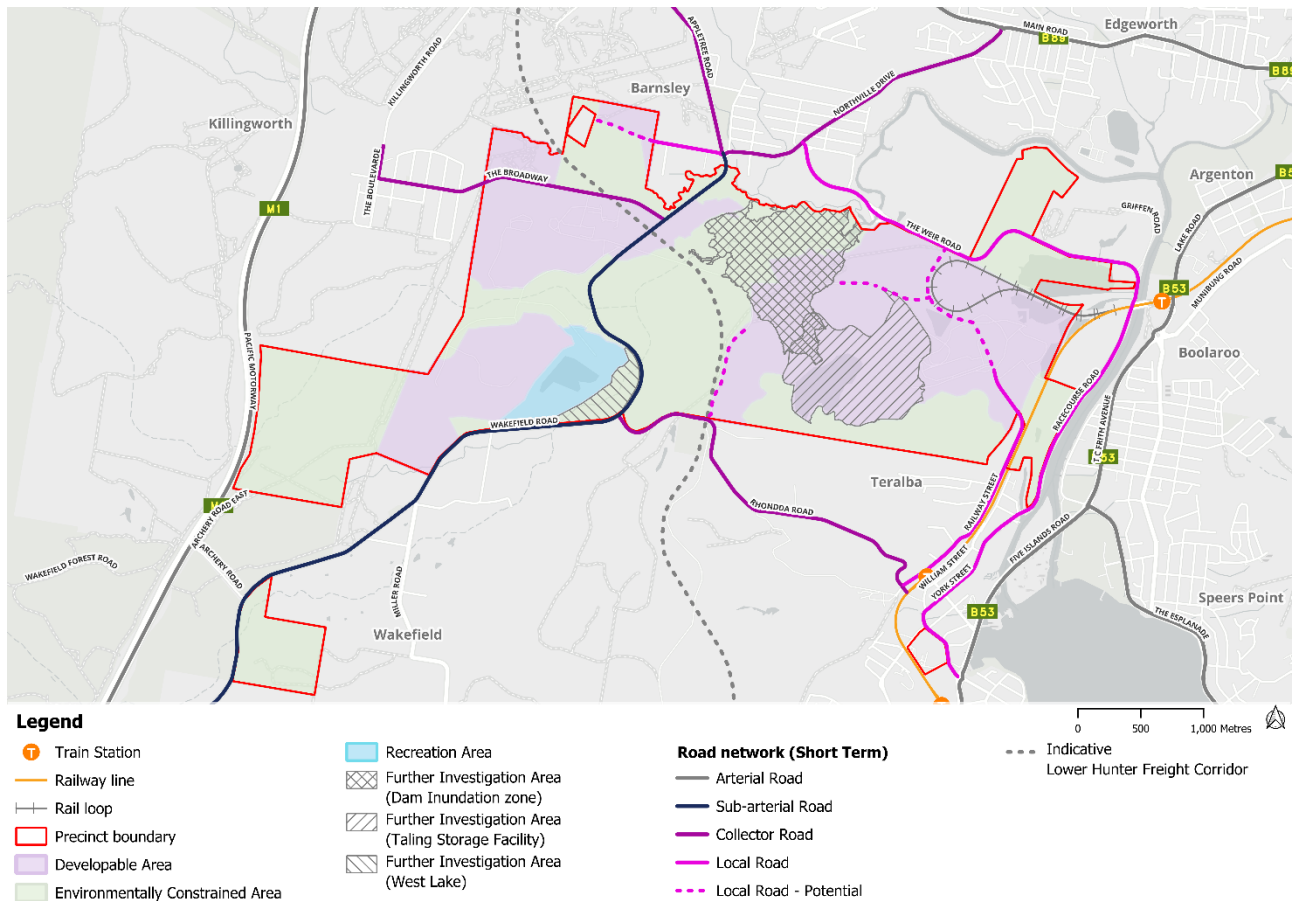


Figure 4-2: Future road network - short term

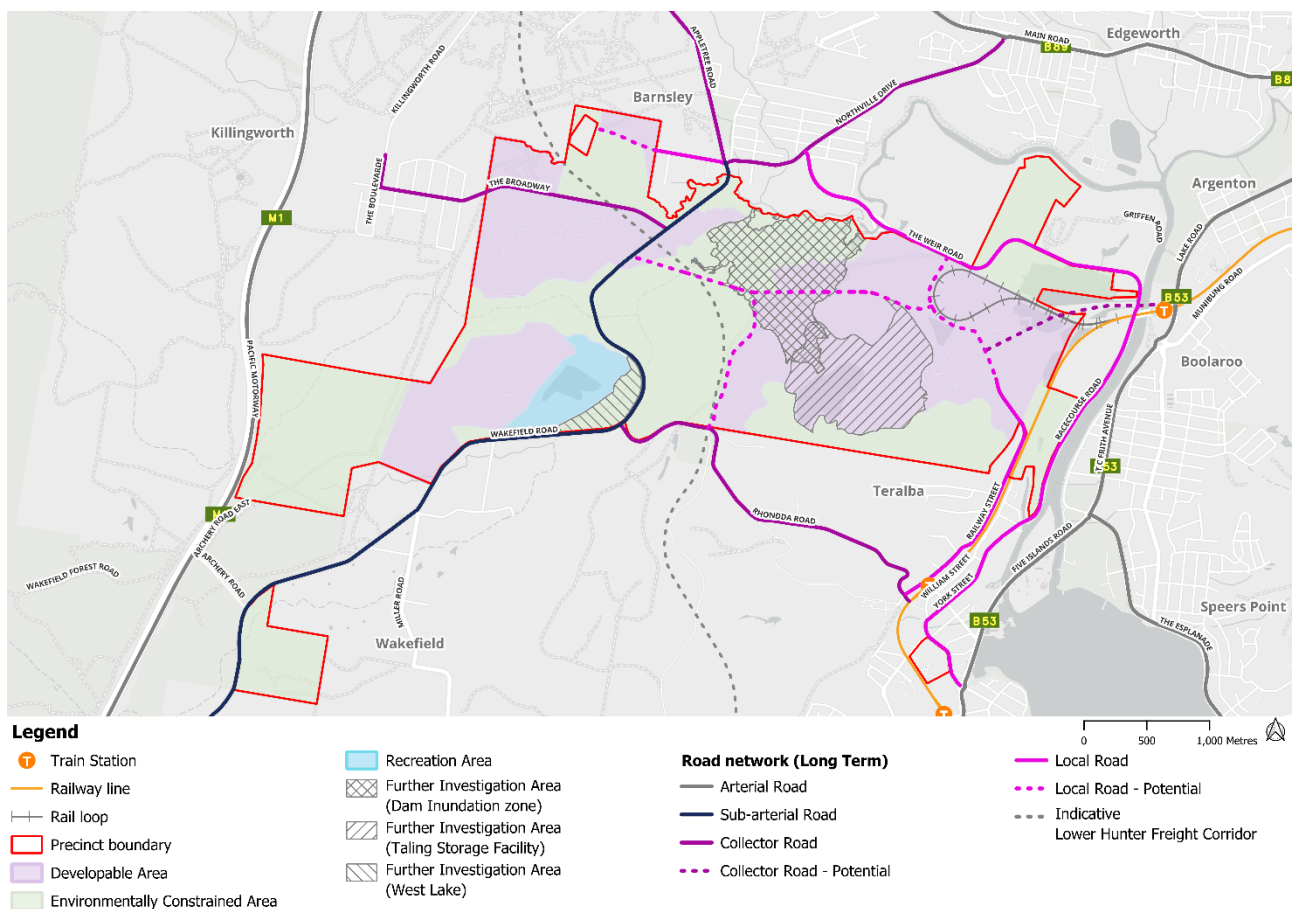


Figure 4-3: Future road network - long term

4.2.2 Benefits of the proposed road network

For general vehicles the proposed network delivers three principal benefits:

- Demand is distributed across three to four directions of approach rather than loading a single access, keeping every access within a manageable range so all internal corridors are expected to operate at LoS C or better at full development once appropriate upgrades to the surrounding network have been delivered
- The hierarchy is legible, with a single sub arterial spine collecting from collectors and locals, so through movements and freight stay on the roads designed for them and the residential streets are protected from intrusion
- Resilience is built in, with multiple accesses and internal connections forming a connected network that offers alternative paths when an access is closed by incident, flood or maintenance. Future connectivity improvements, including a potential Cockle Creek crossing, would further strengthen this network over time.

The Wakefield Road spine gives the employment cores a direct, sensitive receiver free run to the M1 Pacific Motorway, the highest capability freight corridor in the region, while Northville Drive provides a second classified route to the B89 for the port and Newcastle industrial markets.

Significant or major freight generating lots front the spine and the principal collectors, allowing heavy vehicles to navigate without traversing local streets. The separation of the residential sub-precincts from the employment cores by the conservation lands and the creek corridors allows freight amenity impacts to be managed by geography as much as by regulation. The deferred central link, contingent on declassification of the tailings storage facility, will shorten internal freight paths between Cockle Creek and Killingworth and remove circuitous routing via the external network.

4.2.3 Freight access points

The Precinct has three defined freight access points, each with a distinct role:

- South Access from Wakefield Road. The primary RAV gateway to the M1 Pacific Motorway, carrying the Sydney, Central Coast long haul markets. Its upgrade and NHVR approval are the critical enabling aspect
- North Access from Northville Drive and the B89. Serves the Port of Newcastle, Newcastle industrial and Hunter Valley markets and long haul routes on the M1 Pacific Motorway to northern regions
- East Access from Powerhouse Road, Teralba. Limited to general heavy vehicles for local servicing in the short term and enabled for RAV access in the long term. A potential new Cockle Creek crossing is one option that could alleviate heavy vehicle traffic through Teralba and improve connection to the eastern regions, though it may not be required if other upgrades deliver improved conditions through to the east.

4.2.4 Transport Authority Engagement

Engagement with TfNSW, Lake Macquarie City Council, the NHVR and ARTC is to continue through the rezoning. The traffic modelling methodology, including assessment intersections, horizons and assumptions, are to be confirmed with TfNSW.

4.3 Traffic generation and distribution

4.3.1 Trip generation results

Applying the methodology and rates set out in the Section 1.3.4 to the structure plan land use schedule results in the trip generation summarised in Table 4-1. The schedule derives GFA at 1,000 m² per hectare of land use area per unit of floor space ratio, a convention to be confirmed as trip generation scales directly with it, and all values are able to be refined as the schedule is finalised.

Table 4-1: Indicative Precinct trip generation

No.	Sub precinct	Land use	Yield	AM peak	PM peak	Daily
1	Cockle Creek - Employment Core	Large format warehousing, (10,000 to 100,000 m ²)	139,121 m ² GFA	362	320	3,937
2	Killingworth - Employment Core	Large format warehousing, (0 to 10,000 m ²)	72,027 m ² GFA	360	319	2,881
3	West Lake - Mixed Use	Commercial – mixed use	18,082 m ² GLA (18,082 m ² of 22,602 m ² GFA at 0.8 efficiency)	697	697	5,605
5	Barnsley West - Residential	Low density residential (regional)	10 dwellings	8	8	49
Total				1,427 vehicles per hour	1,344 vehicles per hour	12,472 vehicles per day

The heavy vehicle task is derived from the employment land component. The heavy vehicle share is taken as 15 per cent of both daily and peak hour employment trips, being the average of the medium (class 2 to 5, 21 per cent) and long (class 6 to 9, 7 per cent) class shares observed in the TfNSW September 2024 surveys, and the B-double share is taken as 3 per cent of heavy vehicles consistent with the double combination class observed at the surveyed estates.

On this basis the Precinct is expected to generate approximately 1,030 heavy vehicles per day (two way), around 108 in the AM peak hour and 96 in the PM peak hour, of which approximately 30 daily movements are RAV 19 metre B-doubles (over 50 tonnes). The Precinct is anticipated to generate approximately 11,450 daily light vehicle trips, comprising around 1,320 in the AM peak hour and 1,250 in the PM peak hour.

4.3.2 Trip distribution

Light vehicle distribution

Light vehicle trips are distributed using ABS 2021 Census journey to work desire lines for the suburbs in and around the Precinct. The observed place of work pattern for the around 1,600 resident workers tabulated from these suburbs is dominated by the Edgeworth and Cameron Park SA2 to the north at 38 per cent, the Toronto, Morisset and western lake SA2s to the south at 24 per cent combined, work within the West Wallsend, Barnsley and Killingworth area itself at 23 per cent, and the Bolton Point and Teralba SA2 to the east at 16 per cent. Based broadly on this distribution the anticipated light vehicle traffic volumes per approach are summarised in Table 4-2

Table 4-2: Indicative light vehicle distribution at full development

Origin and destination	Share	Daily LV	AM LV (in / out)	PM LV (in / out)
Newcastle centres	30%	3,435	396 (219/176)	374 (163/212)
Newcastle West centres	25%	2,862	330 (183/147)	312 (136/177)
Toronto centres	15%	1,717	198 (110/88)	187 (81/106)
Barnsley centres	12%	1,374	158 (88/71)	150 (65/85)
Sydney	10%	1,145	132 (73/59)	125 (54/71)

Origin and destination	Share	Daily LV	AM LV (in / out)	PM LV (in / out)
Teralba centre	8%	916	106 (58/47)	100 (43/56)

Heavy vehicle distribution

Heavy vehicle trips are distributed by freight market. Long haul movements are routed from the local network to the M1 Pacific Motorway, predominantly via Wakefield Road to the south, while Port of Newcastle and Newcastle industrial movements are routed via Northville Drive to the B89. Local Lake Macquarie movements south of the area are routed to the B53 and could benefit from a new Cockle Creek crossing which would enhance connectivity.

The market shares are assumptions informed by the published freight evidence for the corridor rather than a surveyed origin and destination matrix, which does not yet exist for land that is not yet developed, and can be revised as occupier developments emerge. The evidence basis for each market is set out in Table 4-3.

Table 4-3: Evidence basis for the adopted heavy vehicle market shares

Market and share	Share	Evidence basis	Source
Sydney and Central Coast long haul	30%	Greater Sydney is the dominant freight market in NSW and the destination of the largest share of the regional task. The NSW freight task is forecast to grow 28 per cent to 618 million tonnes by 2036 with the largest growth in Greater Sydney, and large format warehousing is a metropolitan distribution land use whose primary exchange is with the Sydney market via the M1	NSW Freight and Ports Plan 2018 to 2023 (TfNSW, 2018). TfNSW Large Format Warehousing Note (2024)
Port of Newcastle	20%	The port is the regional import and export gateway 25 km from the area, handling more than 160 million tonnes in 2025 including a record 11.1 million tonnes of diversified non coal cargo across 25 cargo types, with machinery and project cargo growing strongly. Circular economy, energy transition and freight and logistics occupiers in the employment cores would exchange directly with the port	Port of Newcastle trade results 2025. Port of Newcastle Master Plan 2040
Newcastle west industrial	20%	Established industrial precincts at Cardiff, Glendale, Beresfield, Tomago and Mayfield West form the regional supply chain the employment cores would trade with, reached via the B89 and B53. The GNMP identifies these precincts and assigns the freight and logistics role to the catalyst area	Greater Newcastle Metropolitan Plan (DPE, 2018)
Brisbane and northern long haul	10%	The M1 and Pacific Highway carry the Sydney to Brisbane interstate road freight task through the lower Hunter, a smaller market for metropolitan distribution floorspace than Sydney, reflected in the lower share	NSW Freight and Ports Plan 2018 to 2023 (TfNSW, 2018). Lower Hunter Freight Corridor SEA (TfNSW, 2021)
Hunter Valley	10%	Mining services, agribusiness and energy markets of the Hunter Valley are reached directly via George Booth Drive and the Hunter Expressway, a natural market for circular economy and industrial occupiers	Hunter Regional Plan 2041 (DPE, 2022)

The resulting aggregate orientation, being 50 per cent of the heavy vehicle task to the M1, 40 per cent to the B89 and Newcastle and 10 per cent local south, is consistent with the Lower Hunter Freight Corridor Strategic Environmental Assessment description of the Hunter freight task as movements within the region and between Greater Sydney, northern NSW and Queensland, and with the heavy vehicle access strategy of this report, which provides compliant RAV routes only to the M1 (south via Wakefield Road) and the B89 (north via Northville Drive). The establishment of a new Cockle Creek crossing as a RAV route would also alleviate RAV movements through Northville Drive and offer safer amenities.

Table 4-4: Indicative heavy vehicle origin and destination distribution (daily two-way trips)

Market	Share	Daily HV	Daily RAV B-double	AM HV (in / out)	PM HV (in / out)
Sydney and Central Coast long haul	30%	307	9	32 (20/12)	29 (10/19)
Port of Newcastle	20%	205	6	22 (13/8)	19 (7/13)
Newcastle west industrial	20%	205	6	22 (13/8)	19 (7/13)
Brisbane and northern long haul	10%	102	3	11 (7/4)	10 (3/6)
Hunter Valley	10%	102	3	11 (7/4)	10 (3/6)
Local Lake Macquarie south (Toronto, Morisset)	10%	102	3	11 (7/4)	10 (3/6)

The M1 Pacific Motorway captures 50 per cent of the heavy vehicle task (Sydney, Brisbane and Hunter markets), the B89 toward the Port of Newcastle and Newcastle industrial areas capture 40 per cent and the B53 south captures 10 per cent.

Heavy vehicle peak flows follow the warehousing entry and exit profile of the GTIA, with around 67 of the 109 AM peak movements inbound (around 62 per cent entry) and 63 of the 97 PM peak movements outbound (around 65 per cent exit), reflecting the morning delivery and dispatch cycle and the afternoon outbound distribution run.

At approximately 30 daily B-double movements, vehicles average roughly one to three per hour across the business day on the combined route set, with the Wakefield Road routes to the M1 Pacific Motorway carrying around 12 per day, RAV movements typically occur throughout the outside of the AM and PM peak periods. The broader heavy vehicle task on Wakefield Road of around 410 movements per day equates to roughly 28 per hour two way across the business day.

4.3.3 Distribution summary by scenario

The same corridor and market shares are applied in every scenario, therefore, the difference between the short term and the medium and long term distributions is due to volume rather than direction. The short term applies the 50 per cent development reduction factor, while the medium and long term apply full development generation.

Table 4-5 and Table 4-6 summarises the two distributions for light vehicle and heavy vehicles respectively and the share applied to each location. Light vehicle volumes combine the West Lake, Teralba and Barnsley West residential trips with the employment light vehicles, and heavy vehicle volumes are the employment freight task. As noted in the previous section, RAV movements typically occur throughout the outside of the AM and PM peak periods and therefore are not considered in the assessment of the AM and PM peak periods.

Table 4-5: Light vehicle distribution summary by corridor (daily two-way trips)

Market	Share	Short term daily	Medium / long term daily
Newcastle centres	30%	1,720	3,440
Newcastle West centres	25%	1,430	2,860
Toronto centres	15%	860	1,720
Barnsley centres	12%	690	1,370
Sydney	10%	570	1,150
Teralba centre	8%	460	920
Total light vehicles	100%	5,730 vehicles daily	11,460 vehicles daily

Table 4-6: Heavy vehicle distribution summary by market (daily two-way trips)

Market	Share	Short term HV	Medium and long term HV
Sydney and Central Coast long haul	30%	150	310
Port of Newcastle	20%	100	210
Newcastle west industrial	20%	100	210
Brisbane and northern long haul	10%	50	100
Hunter Valley	10%	50	100
Local Lake Macquarie south	10%	50	100
Total heavy vehicles	100%	500 HV vehicles daily	1,030 HV vehicles daily

4.4 Transport Performance Analysis

4.4.1 Midblock Performance

Midblock performance is assessed using volume to capacity ratios (VCR) against lane capacities as set out in Section 1.3.6. Given the uncertainty in both the existing traffic baseline and the ultimate development detail, this assessment adopts a conservative LoS D as the criterion for identifying where upgrades may be warranted, with confirmed upgrade requirements to be assessed through more detailed modelling as more definitive information becomes available.

The assessment is structured around the staging of the Precinct rather than a single full development case, so that the network response can be tied to demand staging and to the potential upgrades that unlock the network to support this demand. Three development horizons are tested, being:

- Short term 2031 - 5 years
- Medium term 2041 - 15 years
- Long term 2051 - 25 years.

Background traffic is grown linearly at 1.2 per cent per annum. Within each horizon the network is tested without the Precinct (background growth only) and with the Precinct. The medium and long term the with-Precinct case is run both with and without a potential Cockle Creek crossing in place. This allows for the impacts to be identified from background growth, the Precinct traffic generation and the relief provided by a new Cockle Creek crossing, allowing each upgrade trigger to be attributed targeted issues.

To represent the short term, when the Precinct is only partially developed and occupied, a development reduction factor of 50 per cent is applied to all generated trips, reducing them to 50 per cent of the full development total for the short-term scenario.

This recognises that the rezoned land uses will not be established or fully tenanted in the short-term horizon, that demand accrues progressively as parcels are released and occupied, and that the warehousing and commercial floorspace in particular will fill over a period of years rather than on opening. The factor is applied uniformly across all land uses and vehicle classes and flows through to the distributions and the network assessment. Therefore, short term scenario reflects a proportionate reduction in both the residential and the employment generated demand. The factor is adjustable and is to be refined as the staging plan and release sequence are confirmed.

The medium and long term results show the with-Precinct cases at full generation, with and without a potential Cockle Creek crossing. A potential Cockle Creek crossing (Corridor 14) is only present in the medium and long term scenarios, accordingly, no results are presented for this corridor in the base year or short term. The AM and PM Peak hour midblock performance assessment for development scenarios was undertaken for the locations illustrated in Figure 4-4.

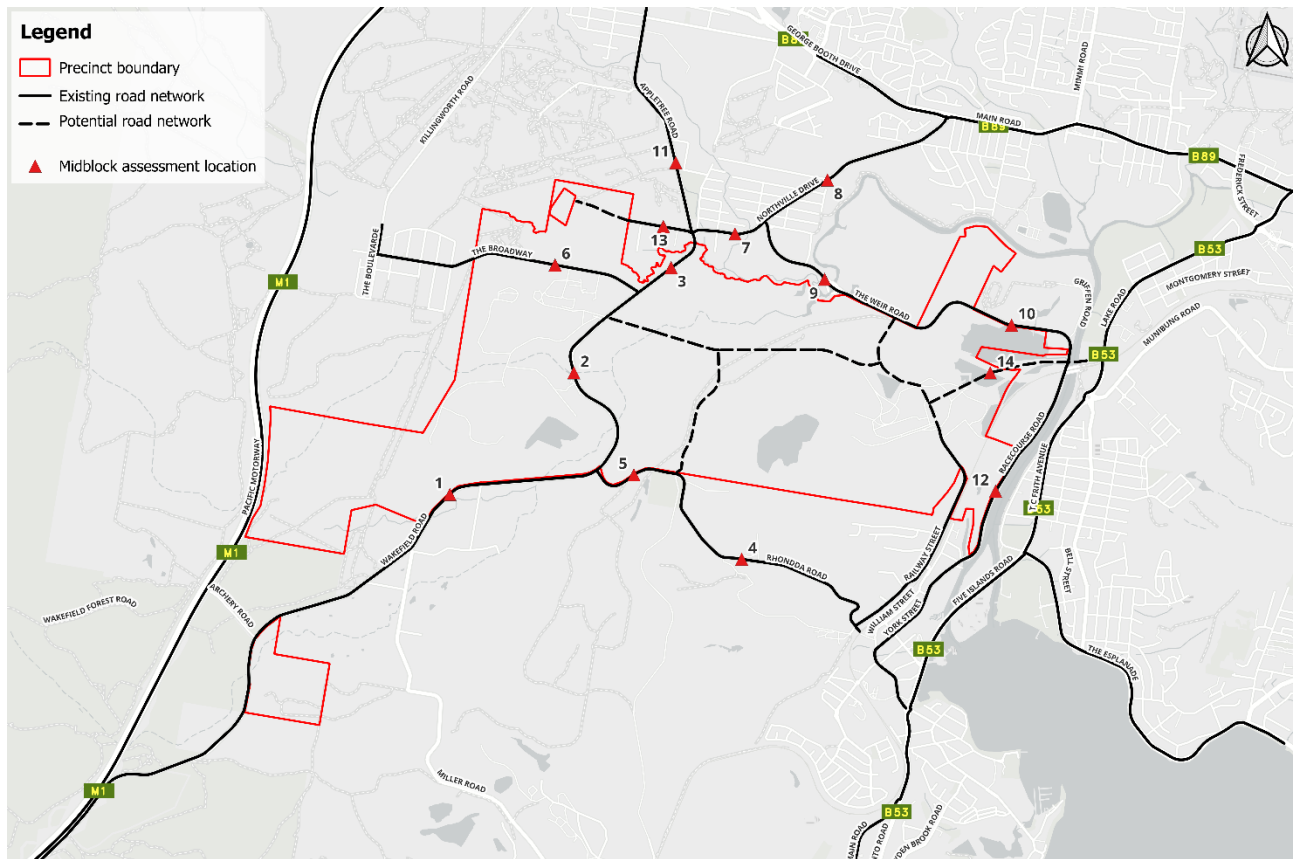


Figure 4-4: Development scenarios - midblock assessment locations

Midblock Performance Results – Short Term

The short term compares the 2031 network with and without the partially developed Precinct (80 per cent generation), which isolates the traffic the Precinct itself adds in this period. A new Cockle Creek crossing and the Teralba rail overpass upgrade are assumed not to be in place, therefore, all Precinct traffic enters and leaves through the northern and southern access points for Area 1, concentrating the load on The Weir Road, Northville Drive and Rhondda Road.

- Without the Precinct, every corridor stays at or close to its base year LoS except for Northville Drive north of The Weir Road where LoS in base year changes from LoS C to D, with the increase in background traffic.
- Adding the Precinct is clearly visible on the northern access corridors.
- During AM peak, eastbound section of Northville Drive, between Wakefield Road and The Weir Road is expected to operate at LoS D with precinct in comparison to LoS C without precinct and Westbound section is expected to operate at LoS C with precinct in comparison to LoS B without precinct. During PM peak the midblock performance of this section in eastbound is expected at LoS C and in westbound section at LoS D in both with precinct and without precinct scenarios.
- The eastbound section of Weir Road west of the northern access is expected to operate at LoS B during both the AM and PM peak without precinct, with precinct this section is expected to operate at LoS D in the AM peak and LoS C in the PM peak. The performance of westbound section is expected to change from LoS B to LoS C in the AM peak and from LoS C to LoS D in the PM peak under with precinct scenario when compared to without precinct.
- The southern access performs better than the northern access. Rhondda Road east and west of the southern access stays at LoS A in both directions even with the Precinct, confirming the southern connections have substantial spare capacity to carry early employment demand and that the constraint is specific to the northern corridors, not to Precinct access generally.

- The Wakefield Road spine and the western local roads is expected to continue to operate at level of service A to B throughout. Short term Precinct traffic is expected to result in only minor changes to VCR values, with no change to level of service across the western or southern network.

In performance terms the short term shows that the Precinct could be established and partially built without triggering a duplication anywhere, however, that the northern access corridors including The Weir Road and Northville Drive would be the corridors most affected by Precinct related traffic as all site traffic is funnelled north and south while the eastern access is absent.

The short-term midblock performance results compared against the 2026 baseline is summarised in Table 4-7.

Table 4-7: Short term - 2031 midblock performance

ID	Segment	Direction	AM Peak						PM Peak					
			2026 Base		2031 without Precinct		2031 with Precinct		2026 Base		2031 without Precinct		2031 with Precinct	
			VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS
1	Wakefield Road, south of Rhondda Road	NB	0.16	A	0.17	A	0.23	A	0.14	A	0.15	A	0.19	A
		SB	0.15	A	0.16	A	0.20	A	0.18	A	0.19	A	0.24	A
2	Wakefield Road, between Rhondda Road and The Broadway	NB	0.22	A	0.24	A	0.24	A	0.19	A	0.21	A	0.21	A
		SB	0.21	A	0.22	A	0.22	A	0.23	A	0.25	A	0.25	A
3	Wakefield Road, between The Broadway and Northville Drive	NB	0.32	B	0.34	B	0.34	B	0.23	A	0.25	A	0.25	A
		SB	0.23	A	0.25	A	0.25	A	0.31	B	0.33	B	0.33	B
4	Rhondda Road, east of Area 1 southern Access Point	EB	0.03	A	0.03	A	0.09	A	0.03	A	0.04	A	0.11	A
		WB	0.04	A	0.05	A	0.12	A	0.02	A	0.03	A	0.07	A
5	Rhondda Road, west of Area 1 southern Access Point	EB	0.04	A	0.05	A	0.12	A	0.05	A	0.05	A	0.10	A
		WB	0.06	A	0.07	A	0.12	A	0.03	A	0.04	A	0.10	A
6	The Broadway	EB	0.15	A	0.16	A	0.16	A	0.11	A	0.12	A	0.12	A
		WB	0.07	A	0.07	A	0.07	A	0.16	A	0.17	A	0.17	A
7	Northville Drive, between Wakefield Road and The Weir Road	EB	0.51	C	0.54	C	0.56	D	0.40	C	0.43	C	0.45	C
		WB	0.34	B	0.36	B	0.38	C	0.55	D	0.58	D	0.60	D
8	Northville Drive, north of The Weir Road	EB	0.54	D	0.58	D	0.69	D	0.52	C	0.56	D	0.72	D
		WB	0.48	C	0.51	C	0.69	D	0.55	D	0.58	D	0.69	D
9	The Weir Road, west of Area 1 northern Access Point	EB	0.35	B	0.37	B	0.58	D	0.32	B	0.34	B	0.47	C
		WB	0.25	B	0.26	B	0.40	C	0.45	C	0.48	C	0.66	D
10		EB	0.35	B	0.37	B	0.39	C	0.32	B	0.34	B	0.37	B

ID	Segment	Direction	AM Peak						PM Peak					
			2026 Base		2031 without Precinct		2031 with Precinct		2026 Base		2031 without Precinct		2031 with Precinct	
			VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS
	The Weir Road, east of Area 1 northern Access Point	WB	0.25	B	0.26	B	0.28	B	0.45	C	0.48	C	0.50	C
11	Appletree Road	NB	0.27	B	0.29	B	0.29	B	0.31	B	0.33	B	0.33	B
		SB	0.25	B	0.27	B	0.27	B	0.32	B	0.34	B	0.34	B
12	Racecourse Road	NB	0.10	A	0.11	A	0.13	A	0.14	A	0.15	A	0.17	A
		SB	0.14	A	0.15	A	0.17	A	0.10	A	0.11	A	0.13	A
13	Charlton Street	EB	0.05	A	0.06	A	0.06	A	0.03	A	0.03	A	0.03	A
		WB	0.02	A	0.03	A	0.03	A	0.07	A	0.07	A	0.07	A

Midblock Performance Results – Medium Term

The medium term applies full Precinct generation against background growth to 2041 and is the first horizon to test the network both with and without a potential Cockle Creek crossing. Comparing these two cases isolates the benefit the crossing provides.

- Without the Precinct, the background growth alone pushes Northville Drive and The Weir Road to level of service D, confirming these corridors are on a congestion path independent of the Precinct.
- With the Precinct but without the crossing, the northern corridors are expected to deteriorate further. Northville Drive north of The Weir Road is expected to reach a VCR of around 0.96 (LoS E) during the AM peak and PM peak in both directions of travel. The westbound section of Weir Road west of the northern access is expected to reach LoS E at a VCR of 0.89 during PM peak, as the full eastern employment demand would still need to access and exit to the north to and from these roads.
- Under the with-precinct scenario, provision of the crossing is expected to improve operations on the same corridors by reducing the VCR and improving performance from LoS E to LoS D. This indicates the potential benefit of the crossing, with an improvement of one LoS band on the constrained corridor under the same precinct traffic demand.
- A new Cockle Creek crossing is expected to operate at level of service A in both directions and both peaks, with a very low VCR.
- Charlton Street, Racecourse Road and Appletree Road all remain at LoS A, and the Wakefield Road spine and southern corridors operate at LoS B or better.

The medium term therefore shows that full build out is expected to be achievable without the northern corridors failing, and that some alleviated corridor performance could be achieved if a Cockle Creek crossing or other improvements to the east were delivered. Improving connections to the east would have dual benefits, providing direct eastern access to the site and relieving the northern collectors by giving the B53 bound traffic a route that would not pass through the Northville Drive and The Weir Road intersections. Further investigation is needed into wider network distribution impacts of a potential crossing, including the potential for rat-running to avoid congestion on the B53 corridor.

The medium term midblock performance results without the Precinct, with the Precinct and without a new Cockle Creek crossing, and with the Precinct and new Cockle Creek crossing during the AM and PM peak are summarised in Table 4-8.

Table 4-8: Medium term - 2041 midblock performance

ID	Segment	Direction	AM Peak						PM Peak					
			2041 without Precinct		2041 with Precinct without new Cockle Creek crossing		2041 with Precinct and new Cockle Creek crossing		2041 without Precinct		2041 with Precinct without new Cockle Creek crossing		2041 with Precinct and new Cockle Creek crossing	
			VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS
1	Wakefield Road, south of Rhondda Road	NB	0.20	A	0.31	B	0.31	B	0.18	A	0.25	A	0.25	A
		SB	0.20	A	0.27	B	0.27	B	0.23	A	0.33	B	0.33	B
2	Wakefield Road, between Rhondda Road and The Broadway	NB	0.28	B	0.28	B	0.28	B	0.24	A	0.25	A	0.25	A
		SB	0.26	B	0.27	B	0.27	B	0.29	B	0.29	B	0.29	B
3	Wakefield Road, between The Broadway and Northville Drive	NB	0.40	B	0.40	B	0.40	B	0.30	B	0.30	B	0.30	B
		SB	0.30	B	0.30	B	0.30	B	0.39	B	0.39	B	0.39	B
4	Rhondda Road, east of Area 1 southern Access Point	EB	0.04	A	0.10	A	0.08	A	0.05	A	0.11	A	0.08	A
		WB	0.06	A	0.12	A	0.09	A	0.03	A	0.08	A	0.07	A
5	Rhondda Road, west of Area 1 southern Access Point	EB	0.06	A	0.19	A	0.19	A	0.06	A	0.15	A	0.15	A
		WB	0.08	A	0.18	A	0.18	A	0.05	A	0.18	A	0.18	A
6	The Broadway	EB	0.19	A	0.19	A	0.19	A	0.14	A	0.14	A	0.14	A
		WB	0.09	A	0.09	A	0.09	A	0.20	A	0.20	A	0.20	A
7	Northville Drive, between Wakefield Road and The Weir Road	EB	0.64	D	0.68	D	0.68	D	0.51	C	0.55	D	0.55	D
		WB	0.43	C	0.48	C	0.48	C	0.69	D	0.74	D	0.74	D
8	Northville Drive, north of The Weir Road	EB	0.68	D	0.94	E	0.76	D	0.66	D	1.00	E	0.74	D
		WB	0.60	D	0.98	E	0.69	D	0.69	D	0.92	E	0.76	D
9	The Weir Road, west of Area 1 northern Access Point	EB	0.43	C	0.77	D	0.56	D	0.41	C	0.59	D	0.49	C
		WB	0.31	B	0.53	C	0.41	C	0.56	D	0.89	E	0.69	D
10		EB	0.43	C	0.48	C	0.48	C	0.41	C	0.45	C	0.45	C

ID	Segment	Direction	AM Peak						PM Peak					
			2041 without Precinct		2041 with Precinct without new Cockle Creek crossing		2041 with Precinct and new Cockle Creek crossing		2041 without Precinct		2041 with Precinct without new Cockle Creek crossing		2041 with Precinct and new Cockle Creek crossing	
			VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS
	The Weir Road, east of Area 1 northern Access Point	WB	0.31	B	0.35	B	0.35	B	0.56	D	0.61	D	0.61	D
11	Appletree Road	NB	0.34	B	0.34	B	0.34	B	0.39	C	0.39	C	0.39	C
		SB	0.32	B	0.32	B	0.32	B	0.40	C	0.40	C	0.40	C
12	Racecourse Road	NB	0.13	A	0.18	A	0.18	A	0.18	A	0.22	A	0.22	A
		SB	0.18	A	0.22	A	0.22	A	0.13	A	0.18	A	0.18	A
13	Charlton Street	EB	0.07	A	0.07	A	0.07	A	0.04	A	0.04	A	0.04	A
		WB	0.03	A	0.03	A	0.03	A	0.09	A	0.09	A	0.09	A
14	New Cockle Creek crossing	EB					0.13	A					0.20	A
		WB					0.21	A					0.10	A

Midblock Performance Results – Long Term

The long term applies full Precinct generation against background growth to 2051. With background growth dominating by this horizon, the with and without crossing comparison shows that a Cockle Creek crossing does not fully address the constraints on the northern corridors and is not the full solution to the longer term access constraints to the north, which would require separate upgrades to the Northville Drive corridor toward the B89.

- Northville Drive reaches LoS E in both segments and both directions under background growth without the Precinct. The segment north of The Weir Road is expected to operate at LoS E both directions of travel during the AM and PM peak, meaning operation is at capacity and unstable flow and queuing is expected without duplication.
- With precinct and without crossing, the performance is expected to deteriorate with the addition of precinct traffic demand, is expected to lead to LoS F for eastbound and westbound section of Northville Drive, north of The Weir Road during AM and PM peak.
- The crossing is expected to provide some relief to the northern corridor and result in a level of service broadly comparable to the background-growth scenario without precinct traffic. With the crossing, Northville Drive north of The Weir Road is expected to operate at LoS E, compared with LoS F without the crossing. Similarly, The Weir Road west of the northern access is expected to operate at LoS E or better in both directions with the crossing, compared with LoS F (Westbound PM peak) without the crossing.
- Charlton Street and Racecourse Road remain at LoS A or better, and Appletree Road is expected to operate at LoS C and the Wakefield Road segments at LoS C or better, therefore, these collector and local roads are identified for monitoring as background growth accumulates even though they sit below a duplication trigger.
- The crossing operates at level of service A with very low VCR ratios in both directions, which suggests it could carry additional diverted demand if required, though it would have limited ability to provide long term relief to the northern corridor on its own.

In performance terms the long term confirms that the primary constraint would be the Northville Drive corridor to the B89, which is expected to reach capacity under cumulative background growth regardless of the Precinct. The Precinct would contribute to the timing, and as such duplication of Northville Drive may need to be considered and could be progressed alongside eastern connectivity improvements such as a Cockle Creek crossing, with the development contributing proportionately to works that would serve both the Precinct and the wider catchment.

Across the horizons, the results show consistent trend. The Wakefield Road spine, the southern Rhondda Road access and the western local roads carry the Precinct comfortably in every scenario, and the south and west corridors are not constraints. The northern access corridors, being Northville Drive and The Weir Road, are the corridors that carry the Precinct traffic toward the B89 and they begin the assessment at LoS C to D but are expected to reach a duplication trigger under cumulative background growth in the long term rather than purely due to the traffic generated by the Precinct.

Improving connectivity to the east to reduce reliance on Northville Drive would be important for the precinct, and the wider community however whether this is via a new crossing of the creek or improvements to the existing network will require further investigation. Whether the Precinct significantly uses the northern corridor or not, it is apparent that this corridor may struggle in the long term without some forms of intervention to either reduce demands or increase capacity.

The long term midblock performance results without the Precinct, with the Precinct and without the new Cockle Creek crossing, and with the Precinct and new Cockle Creek crossing during the AM and PM peak are summarised in Table 4-9.

Table 4-9: Long term - 2041 midblock performance

ID	Segment	Direction	AM Peak						PM Peak					
			2051 without Precinct		2051 with Precinct without new Cockle Creek crossing		2051 with Precinct and new Cockle Creek crossing		2051 without Precinct		2051 with Precinct without new Cockle Creek crossing		2051 with Precinct and new Cockle Creek crossing	
			VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS
1	Wakefield Road, south of Rhondda Road	NB	0.26	B	0.38	B	0.38	B	0.24	A	0.31	B	0.31	B
		SB	0.26	A	0.34	B	0.34	B	0.29	B	0.40	B	0.40	B
2	Wakefield Road, between Rhondda Road and The Broadway	NB	0.37	B	0.37	B	0.37	B	0.32	B	0.32	B	0.32	B
		SB	0.34	B	0.35	B	0.35	B	0.38	B	0.38	B	0.38	B
3	Wakefield Road, between The Broadway and Northville Drive	NB	0.52	C	0.52	C	0.52	C	0.39	B	0.39	B	0.39	B
		SB	0.39	B	0.39	B	0.39	B	0.50	C	0.50	C	0.50	C
4	Rhondda Road, east of Area 1 southern Access Point	EB	0.06	A	0.11	A	0.09	A	0.06	A	0.12	A	0.10	A
		WB	0.08	A	0.14	A	0.11	A	0.05	A	0.10	A	0.08	A
5	Rhondda Road, west of Area 1 southern Access Point	EB	0.08	A	0.21	A	0.21	A	0.08	A	0.17	A	0.17	A
		WB	0.11	A	0.20	A	0.20	A	0.06	A	0.19	A	0.19	A
6	The Broadway	EB	0.25	B	0.25	B	0.25	B	0.18	A	0.18	A	0.18	A
		WB	0.11	A	0.11	A	0.11	A	0.26	B	0.26	B	0.26	B
7	Northville Drive, between Wakefield Road and The Weir Road	EB	0.83	E	0.88	E	0.88	E	0.66	D	0.70	D	0.70	D
		WB	0.56	D	0.61	D	0.61	D	0.90	E	0.94	E	0.94	E
8	Northville Drive, north of The Weir Road	EB	0.89	E	1.15	F	0.97	E	0.86	E	1.20	F	0.94	E
		WB	0.78	E	1.16	F	0.88	E	0.90	E	1.13	F	0.97	E
9	The Weir Road, west of Area 1 northern Access Point	EB	0.56	D	0.90	E	0.69	D	0.53	C	0.72	D	0.61	D
		WB	0.41	C	0.63	D	0.50	C	0.74	D	1.06	F	0.86	E
10		EB	0.56	D	0.61	D	0.61	D	0.53	C	0.57	D	0.57	D

ID	Segment	Direction	AM Peak						PM Peak					
			2051 without Precinct		2051 with Precinct without new Cockle Creek crossing		2051 with Precinct and new Cockle Creek crossing		2051 without Precinct		2051 with Precinct without new Cockle Creek crossing		2051 with Precinct and new Cockle Creek crossing	
			VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS	VCR	LoS
	The Weir Road, east of Area 1 northern Access Point	WB	0.41	C	0.45	C	0.45	C	0.74	D	0.78	D	0.78	D
11	Appletree Road	NB	0.44	C	0.44	C	0.44	C	0.51	C	0.51	C	0.51	C
		SB	0.41	C	0.41	C	0.41	C	0.52	C	0.52	C	0.52	C
12	Racecourse Road	NB	0.17	A	0.22	A	0.22	A	0.23	A	0.28	B	0.28	B
		SB	0.23	A	0.28	B	0.28	B	0.17	A	0.22	A	0.22	A
13	Charlton Street	EB	0.09	A	0.09	A	0.09	A	0.06	A	0.06	A	0.06	A
		WB	0.04	A	0.04	A	0.04	A	0.12	A	0.12	A	0.12	A
14	New Cockle Creek crossing	EB					0.13	A					0.20	A
		WB					0.21	A					0.10	A

4.4.2 Intersection Performance

As detailed in Section 1.3.6, intersection performance is not assessed quantitatively at this stage but is inferred from the LoS of the adjoining midblock segments. Any intersection whose approaches reach LoS D or worse is identified as a potential intersection that may require improvements subject to further investigation as development is defined and progresses.

In the short term the intersections that may require early consideration and potential mitigation works are:

- Northville Drive / Appletree Road / Wakefield Road intersection
- Northville Drive intersections at The Weir Road and at George Booth Drive and Main Road (the B89)
- Northern access on The Weir Road

The Northville Drive / The Broadway and Rhondda Road accesses are not considered candidates for early capacity upgrade considerations as their adjoining links remain at level of service C however spatial and geometrical works may be required in the form of access treatments, turn lanes to accommodate RAV swept paths.

By the medium term the same Northville Drive and The Weir Road intersections remain the binding constraints, now reaching level of service E on several approaches assuming no short term works are undertaken, and The Weir Road at Racecourse Road is added as the east-west movements grow.

By the long term the most of Northville Drive approaches reach level of service E (assuming works have not been completed) irrespective of whether the Precinct is developed or not, confirming Northville Drive, and the Appletree Road approaches to the Northville Drive / Appletree Road / Wakefield Road intersection will require considerations in relation to increasing capacity.

Detailed SIDRA network modelling will be required to confirm capacity constraints and potential opportunities or requirements for specific treatment for each identified intersection at the development application or subdivision stage, consistent with the master plan performance criteria.

The indicative intersection performance based on the LoS of adjoining midblock segments are summarised in Table 4-10 to Table 4-12 for short, medium and long term horizons respectively.

Table 4-10: Indicative short term intersection performance

Intersection	Period	Adjoining link	Direction	Base 2026 LoS (VCR)	Without Precinct LoS (VCR)	With Precinct, no crossing LoS (VCR)
Northville Drive / Appletree Road / Wakefield Road	AM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.51)	C (0.54)	D (0.56)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	D (0.55)	D (0.58)	D (0.6)
Northville Drive / The Weir Road	AM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.51)	C (0.54)	D (0.56)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	D (0.55)	D (0.58)	D (0.6)
Northville Drive / George Booth Drive / Main Road (B89)	AM peak	Northville Drive, Wakefield Road to Weir Rd	Eastbound	D (0.54)	D (0.58)	D (0.69)
	AM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	C (0.48)	C (0.51)	D (0.69)
	PM peak	Northville Drive, Wakefield Road to Weir Rd	Eastbound	C (0.52)	D (0.56)	D (0.72)

Intersection	Period	Adjoining link	Direction	Base 2026 LoS (VCR)	Without Precinct LoS (VCR)	With Precinct, no crossing LoS (VCR)
The Weir Road / Area 1 northern access	PM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	D (0.55)	D (0.58)	D (0.69)
	AM peak	The Weir Road, west of Area 1 north access	Eastbound	B (0.35)	B (0.37)	D (0.58)
	PM peak	The Weir Road, west of Area 1 north access	Westbound	C (0.45)	C (0.48)	D (0.66)

Table 4-11: Indicative medium term intersection performance

Intersection	Period	Adjoining link	Direction	Base 2026 LoS (VCR)	Without Precinct LoS (VCR)	With Precinct, no crossing LoS (VCR)	With Precinct, with crossing LoS (VCR)
Northville Drive / Appletree Road / Wakefield Road	AM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.51)	D (0.64)	D (0.68)	D (0.68)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.40)	C (0.51)	D (0.55)	D (0.55)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	D (0.55)	D (0.69)	D (0.74)	D (0.74)
Northville Drive / The Weir Road	AM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.51)	D (0.64)	D (0.68)	D (0.68)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.40)	C (0.51)	D (0.55)	D (0.55)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	D (0.55)	D (0.69)	D (0.74)	D (0.74)
Northville Drive / George Booth Drive / Main Road (B89)	AM peak	Northville Drive, north of The Weir Road	Eastbound	D (0.54)	D (0.68)	E (0.94)	D (0.76)
	AM peak	Northville Drive, north of The Weir Road	Westbound	C (0.48)	D (0.6)	E (0.98)	D (0.69)
	PM peak	Northville Drive, north of The Weir Road	Eastbound	C (0.52)	C (0.41)	D (0.59)	C (0.49)
	PM peak	Northville Drive, north of The Weir Road	Westbound	D (0.55)	D (0.56)	E (0.89)	D (0.69)
The Weir Road / Area 1 northern access	AM peak	The Weir Road, west of Area 1 north access	Eastbound	B (0.35)	C (0.43)	D (0.77)	D (0.56)
	PM peak	The Weir Road, west of Area 1 north access	Eastbound	B (0.32)	C (0.41)	D (0.59)	C (0.49)

Intersection	Period	Adjoining link	Direction	Base 2026 LoS (VCR)	Without Precinct LoS (VCR)	With Precinct, no crossing LoS (VCR)	With Precinct, with crossing LoS (VCR)
	PM peak	The Weir Road, west of Area 1 north access	Westbound	C (0.45)	D (0.56)	E (0.89)	D (0.69)

Table 4-12: Indicative long term intersection performance

Intersection	Period	Adjoining link	Direction	Base 2026 LoS (VCR)	Without Precinct LoS (VCR)	With Precinct, no crossing LoS (VCR)	With Precinct, with crossing LoS (VCR)
Northville Drive / Appletree Road / Wakefield Road	AM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.51)	E (0.83)	E (0.88)	E (0.88)
	AM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	B (0.34)	D (0.56)	D (0.61)	D (0.61)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.40)	D (0.66)	D (0.7)	D (0.7)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	D (0.55)	E (0.9)	E (0.94)	E (0.94)
Northville Drive / The Weir Road	AM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	C (0.40)	D (0.66)	D (0.7)	D (0.7)
	AM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	D (0.55)	E (0.9)	E (0.94)	E (0.94)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Eastbound	D (0.54)	E (0.89)	F (1.15)	E (0.97)
	PM peak	Northville Drive, Wakefield Road to Weir Road	Westbound	C (0.48)	E (0.78)	F (1.16)	E (0.88)
Northville Drive / George Booth Drive / Main Road (B89)	AM peak	Northville Drive, north of The Weir Road	Eastbound	C (0.52)	E (0.86)	F (1.2)	E (0.94)
	AM peak	Northville Drive, north of The Weir Road	Westbound	D (0.55)	E (0.9)	F (1.13)	E (0.97)
	PM peak	Northville Drive, north of The Weir Road	Eastbound	B (0.35)	D (0.56)	E (0.9)	D (0.69)
	PM peak	Northville Drive, north of The Weir Road	Westbound	B (0.25)	C (0.41)	D (0.63)	C (0.5)
The Weir Road / Area 1	AM peak	The Weir Road, west of Area 1 north access	Eastbound	B (0.32)	C (0.53)	D (0.72)	D (0.61)

Intersection	Period	Adjoining link	Direction	Base 2026 LoS (VCR)	Without Precinct LoS (VCR)	With Precinct, no crossing LoS (VCR)	With Precinct, with crossing LoS (VCR)
northern access	AM peak	The Weir Road, west of Area 1 north access	Westbound	C (0.45)	D (0.74)	F (1.06)	E (0.86)
	PM peak	The Weir Road, west of Area 1 north access	Eastbound	C (0.51)	E (0.83)	E (0.88)	E (0.88)
	PM peak	The Weir Road, west of Area 1 north access	Westbound	B (0.34)	D (0.56)	D (0.61)	D (0.61)

4.5 Transport Considerations for Structure Plan

The Precinct has a strategically valuable but presently underused relationship to the rail network. The Central Coast and Newcastle line runs along the eastern edge with stations at Teralba, Booragul and Cockle Creek, each within 1 to 3 kilometres of the developable land.

The committed Cockle Creek Station accessibility upgrade by TfNSW will strengthen its role as the main public transport hub for the area. Cockle Creek Station is the closest station (within 400m of the site) to the precinct but is not currently accessible other than by vehicle. Walking catchments to the stations are presently constrained to the eastern fringe of the Precinct, and most of the future employment land sits beyond a comfortable 800 metre walk, making private vehicle, feeder bus services and cycling connections the realistic public transport access modes for the employment cores.

4.5.1 Active Transport

Connectivity for pedestrians and cyclists must largely be built rather than upgraded. The active transport network is to connect homes, the local centre and the stations with safe continuous routes for all ages and abilities, delivered as early as practicable in each stage, with end of trip facilities in new development and all road reserves designed to for the inclusion of active transport infrastructure. Priority links are the off-road shared cycleway from Pitt Street to Anzac Parade (Place Strategy item L19), the active transport crossing of Five Islands Road from Lake Crescent (item S12/S15), and the new active transport links shown on the structure plan toward Cockle Creek Station, potentially via the Racecourse Road bridge over Cockle Creek, which is proposed to be replaced by Council. The existing and future active transport network is shown in Figure 4-5.

The disused colliery rail corridor presents a strategic opportunity as the alignment for a rail trail from Cockle Creek to West Wallsend identified in the Council Walking, Cycling and Better Streets Strategy, providing a flat, direct and largely grade separated spine through the area.

Active transport constraints are topographical and structural, being severance by Cockle Creek, the rail line and the steep terrain on the western half of Rhondda Road, distances between the employment cores and the stations at the upper end of cycling catchments, and an arterial environment on Five Islands Road that is hostile to walking and cycling without the Lake Crescent crossing identified in the Place Strategy (items S12 and S15).

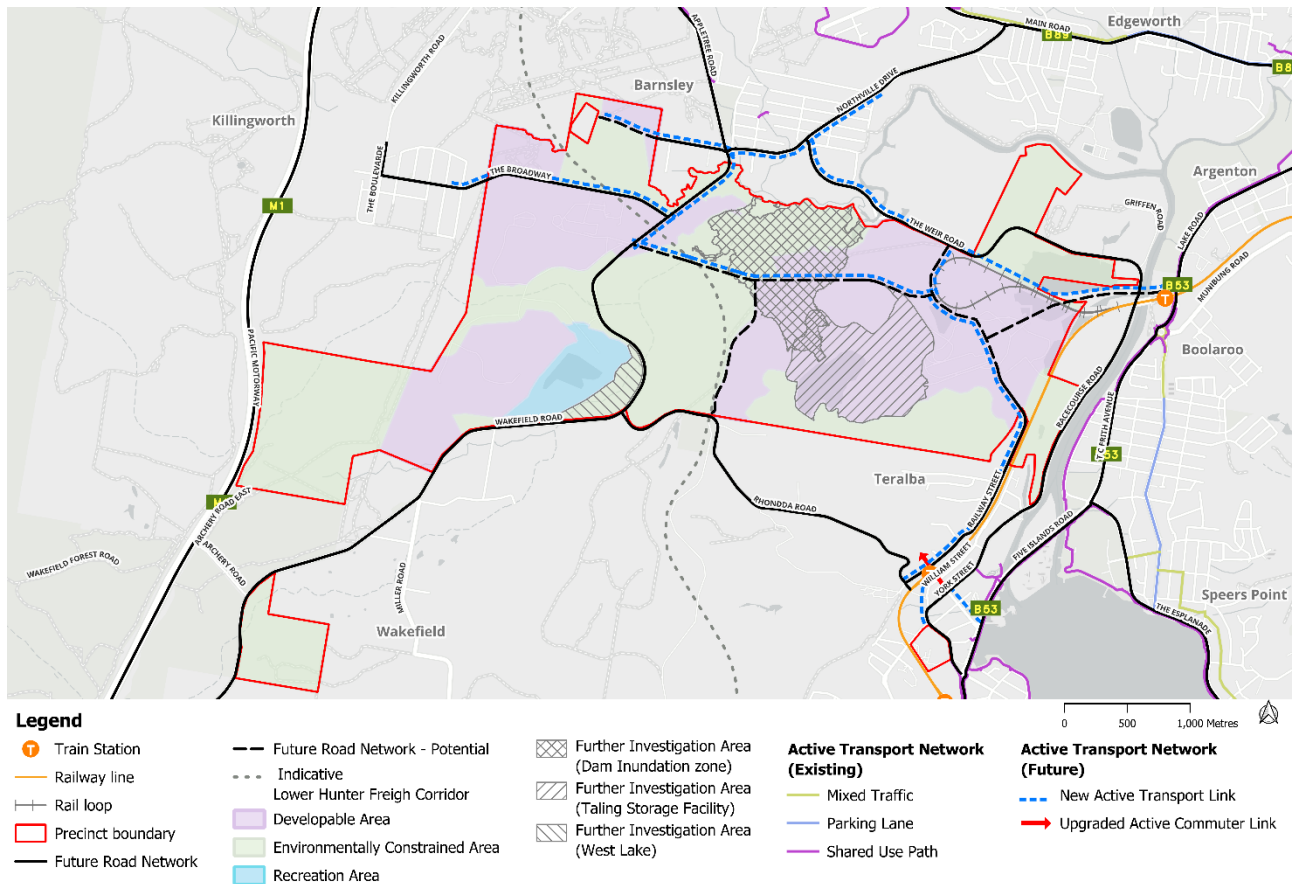


Figure 4-5: Existing and future active transport network

4.5.2 Public Transport

Road reserves and intersections are to be designed to accommodate bus movements, priority measures and accessible stops, with safe pedestrian connections to Teralba, Booragul and Cockle Creek stations. Bus services warranted by growth are to be advocated to TfNSW, including flexible or demand responsive coverage suited to the lower density Holmesville end. A new Cockle Creek crossing could provide an opportunity to provide a direct bus and/or active transport connection to Cockle Creek Station, which the committed station accessibility upgrade reinforces as the main public transport hub. An active transport crossing, in lieu of a full road crossing, could also be considered which would provide an opportunity to connect buses on Racecourse Road with Cockle Creek Station via a walk over the bridge.

Early provision of services is important rather than being retrofitted so that travel habits form around public transport. This would support development parking calibrated to the Lake Macquarie DCP reduced transit-oriented rates in the Teralba medium density area.

4.5.3 Road Network

The internal network comprises the Wakefield Road sub-arterial spine, collector connections at Rhondda Road and The Weir Road, and local streets to the development parcels, with cross sections per the Lake Macquarie DCP providing for walking and cycling on all streets. Internal roads must be located on developable land or existing alignments, must not be formed within the conservation areas or coastal wetland, and are to respond to steep and flood affected land and the heritage character of the Teralba and Holmesville centres. Road and intersection capacity is to be delivered in step with the development it serves, with preliminary potential delivery triggers and responsibilities set out in a staging plan.

The hierarchy is classified under the NSW Movement and Place Framework so that each street is designed to its intended balance of movement function and place intensity from the outset. The freight spine and the classified approaches are movement corridors where efficient and safe movement of people and goods is the priority and place activity is deliberately kept low, the employment core streets are industrial movement

streets with high goods movement and modest place demand at lot frontages, the residential streets are local streets where place and safety outweigh movement, and the Teralba centre streets are places for people whose movement function is deliberately reduced by the removal of the East Access from the long term RAV strategy.

4.5.4 Freight Access

A freight management plan is to be prepared to coordinate freight access, servicing and loading across the Precinct, embedding last mile strategies such as consolidated deliveries, designated servicing routes and hours, off street loading and provision for smaller and low emission delivery vehicles. Employment and business park access roads are to be designed as multi modal corridors consistent with the Place Strategy treatment of the Munibung Road corridor in the Cardiff Advanced Industry Precinct.

The forecast freight task is material but manageable. At full development the employment cores generate approximately 1,030 heavy vehicles per day two way, dominated by rigid trucks and articulated vehicles, with around 30 daily movements taken as 19 metre B-doubles. Peak hour demand is around 109 movements in the AM and 97 in the PM, roughly 10 per cent of total peak demand, following the *GTIA* warehousing profile of 62 per cent entry in the AM and 65.5 per cent exit in the PM.

On the busiest corridor, Wakefield Road, the heavy vehicle task of around 410 movements per day equates to roughly 28 per hour two way across the business day, of which B-doubles average around 12 per day. Flows of this order are well within the operating range of a sealed two-lane rural road, as such, the engineering challenge is one of geometry, structures and pavement rather than capacity. The modest B-double volume does not diminish the route requirements, as the design vehicle for the RAV network is set by the largest vehicle that must be accommodated, not by how often it arrives.

Three technical implications follow from the task:

- Pavement and structures– The Wakefield Road spine and internal freight collectors must be designed for the cumulative axle loading of around 260,000 heavy vehicle movements per year, driving a heavy duty pavement and whole of life maintenance approach rather than a standard local road treatment
- Design vehicle– All intersections and accesses serving the employment cores must accommodate the RAV 19 metre B-double to the Austroads swept path templates and should not preclude future considerations for higher order heavy vehicles
- Gap acceptance – Turn treatments at the M1 ramps and along Wakefield Road must account for laden vehicles over 50 tonnes, which need substantially longer gaps than the passenger car the warrants are calibrated to, with auxiliary lanes, protected turns and lighting where Austroads warrants indicate.

Network and route strategy

The highest order vehicle expected is the RAV 19 metre B-double over 50 tonnes. The approved 19 metre B-double network runs along the surrounding B89, Main Road and B53 corridors but does not extend into the interior where the future employment land is proposed, therefore, heavy vehicle access must be newly established rather than simply extended.

The strategy provides two RAV access routes in the short term, being the north access via Northville Drive from the B89 to Rhondda Road, or to The Weir Road which would require upgrade and formalising as an approved RAV route. The RAV routes during the short term are illustrated in Figure 4-6.

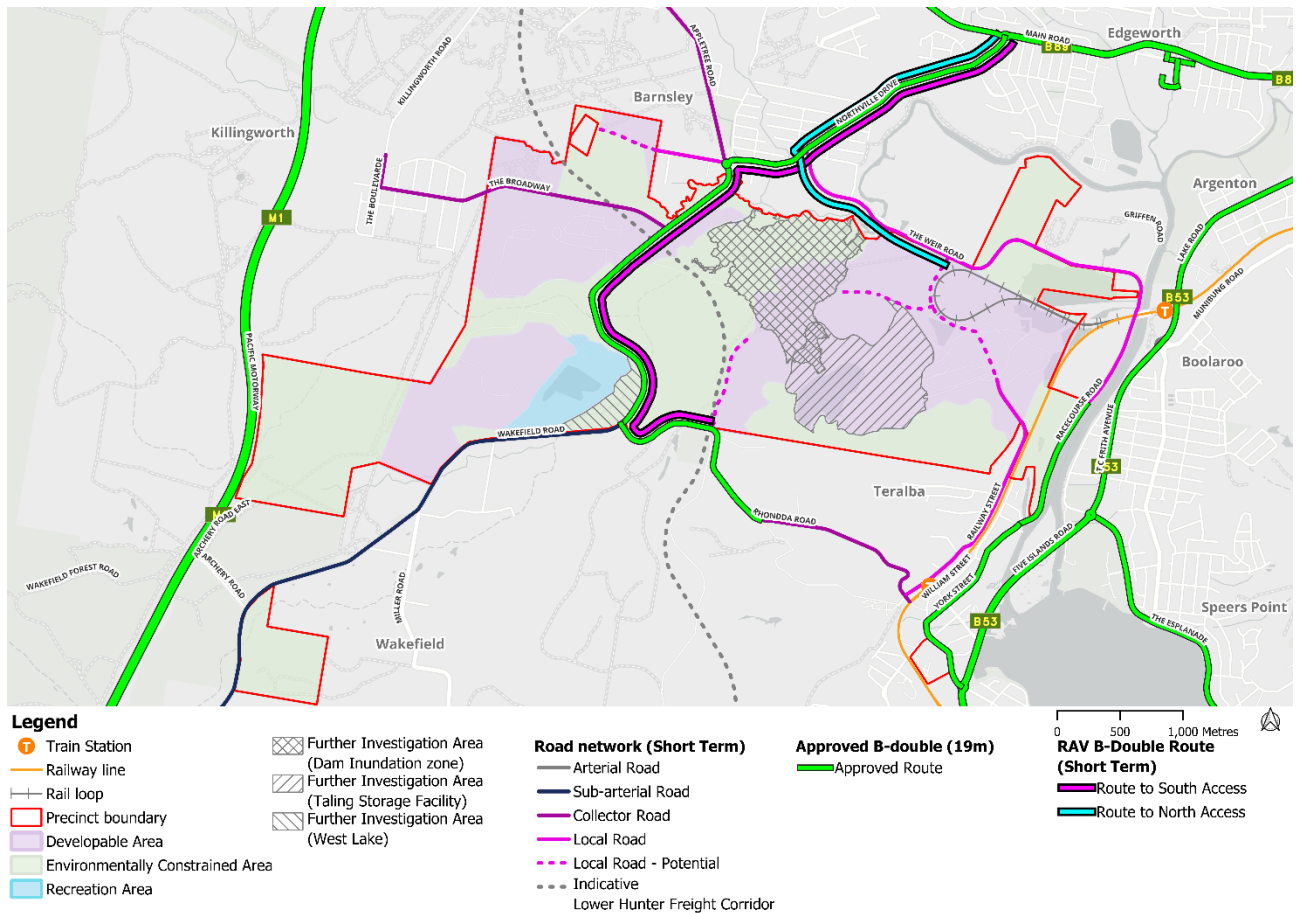


Figure 4-6: Future heavy vehicle network - short term

The medium term introduces three additional RAV routes, being access from the east through Teralba via the upgraded railway overpass to Railway Street, with improvements in the eastern connectivity and Wakefield Road south of Rhondda Road which introduces a new connection between the M1 Pacific Motorway and the Precinct. The RAV routes during the medium term which carry over to the long term are illustrated in Figure 4-7.

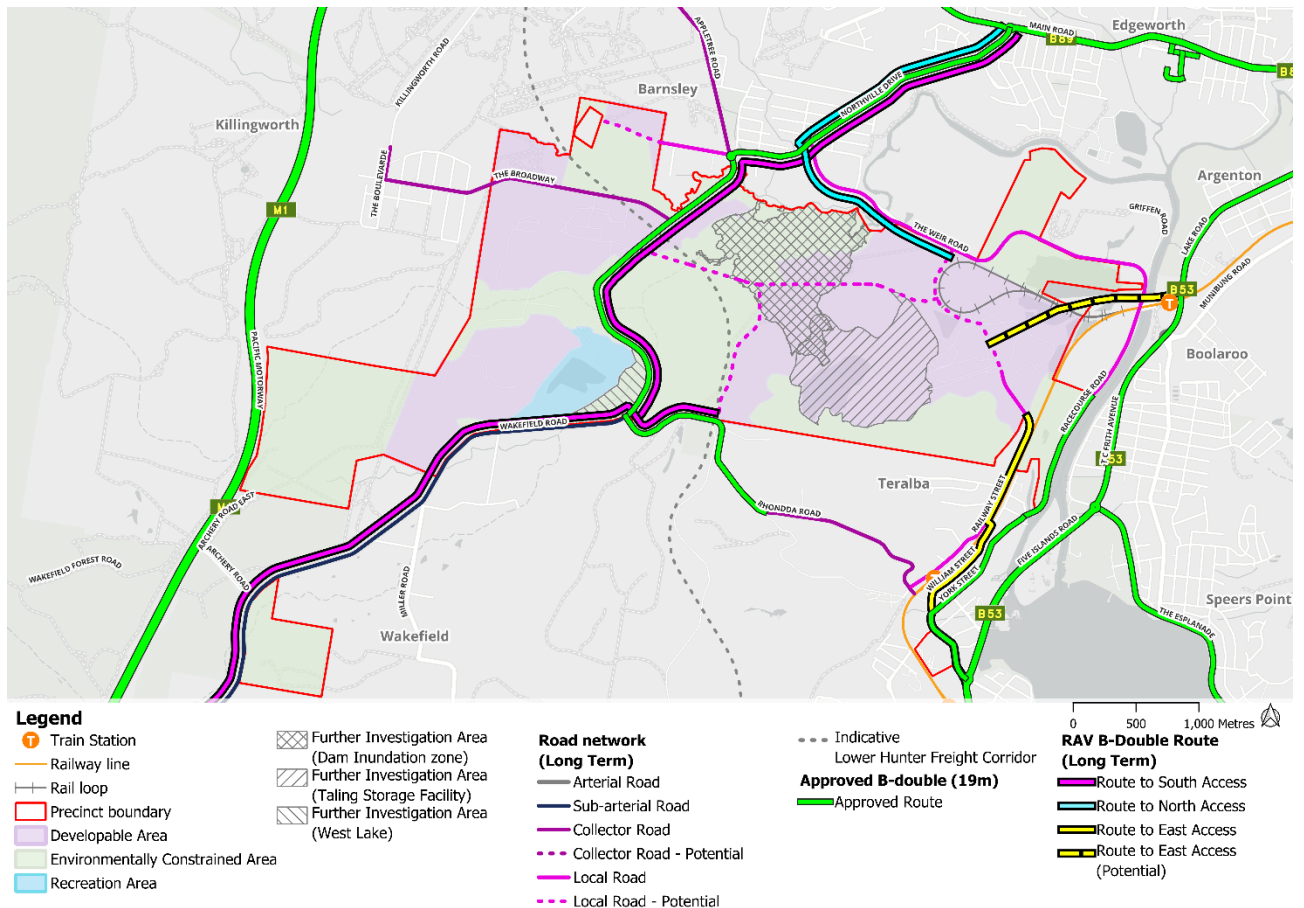


Figure 4-7: Future heavy vehicle network – medium and long term

Making Wakefield Road suitable for RAV 19 metre B-doubles

Wakefield Road is the preferred RAV spine as it connects directly to the M1 without passing residential catchments at Barnsley or Holmesville or the school zones on Northville Drive. Wakefield Road is partially approved (north of Rhondda Road) but the southern section to the M1 is not currently an approved RAV B-double route. Establishing the route requires a route assessment and approval under the Heavy Vehicle National Law through the NHVR, with Lake Macquarie City Council as the road manager and TfNSW for State Road connections.

The works program addresses six matters:

- **Geometry** – Swept path compliance for the 19 metre B-double at all intersections and accesses, curve widening on the winding alignment, sealed shoulders and clear zones, and auxiliary or overtaking provision on grades where speed differentials warrant
- **Structures** – Load capacity assessment of the bridges over Burkes Creek, Cockle Creek and Diega Creek and all culverts for vehicles over 50 tonnes gross, with strengthening or replacement where required
- **Pavement** – Structural assessment and strengthening for the forecast task of approximately 290 daily heavy vehicle movements, including B-doubles and articulated vehicles
- **Safety and amenity** – A Safe System review of the corridor, intersection sight distance, lighting, advance warning of the M1 ramps and noise assessment at the limited sensitive receivers near the southern section
- **Approval** – Submission of the route for gazettal on the NHVR approved network under the Heavy Vehicle National Law, with interim permit access during early stages if required.

Routes through Teralba are not currently suitable for RAV access to the Precinct. The Rhondda Road elevation profile shows approximately 90 m of climbing over 3.2 km at average grades of around 6 per cent with localised slopes approaching over 10 per cent, nearing the limit Austroads identifies as the maximum negotiable by heavy vehicles, and a laden B-double over 50 tonnes would crawl on the ascent and require

very low gear descent on the downgrade, with the William Street rail bridge and the Teralba centre adding width and amenity constraints on the approach. Heavy vehicle and freight movements are to be routed to avoid the Teralba local centre, heritage areas and residential streets, and freight generating land uses should be located with convenient access to the approved or proposed RAV network.

4.6 Integration of Transport Opportunities

4.6.1 Planned Transport Upgrades

Four committed or planned external upgrades shape the assessment:

- Speers Point Transport Improvements (TfNSW, committed). Address existing B53 congestion with travel time reductions of up to 50 per cent, assumed in the short term assessment
- Cockle Creek Station accessibility upgrade (TfNSW, committed). Strengthens the main public transport hub for the area
- M1 to Raymond Terrace extension (under delivery). Addresses motorway capacity north of the area and improves long haul reliability for the Port of Newcastle and Brisbane markets
- Place Strategy items. The Teralba and Booragul stations multimodal upgrades (S13 and S16) and the Pitt Street intersection upgrade (L19), adopted as performance criteria.

4.6.2 Strategic Transport Opportunities

The principal strategic opportunity is a new Cockle Creek crossing.

Depending on the form of this crossing this could provide a key sustainable transport link between the precinct and Cockle Creek Station and residential areas to the east of the creek, significantly shortening the travel distance and making active and public transport trips more viable.

While this connection has also been raised as a new general traffic connection, to relieve pressure on the northern corridor, similar results may be able to be achieved through other upgrades and improvements and will be subject to further modelling and investigation.

The Lower Hunter Freight Corridor between Fassifern and Hexham strengthens the case for freight-based land uses subject to delivery timing, as such land uses relying on the corridor are treated as a longer-term proposition while local freight planning remains consistent with it.

The Sydney to Newcastle High Speed Rail scheme includes a Lake Macquarie station at Morisset, and while a station within the Precinct is not part of the current scheme the alignment is still being finalised, so development must not prejudice the corridor, noting that even without a local stop the project would improve regional accessibility through interchange via the Northern line and could release capacity for additional local passenger and freight services. Finally, the West Wallsend Colliery rail loop offers either reactivation for rail served freight or adaptive reuse as a rail trail as illustrated in the potential active transport network in Section 4.5.1.

4.6.3 Sustainable Freight Opportunities

The Precinct presents an opportunity to embed sustainable freight principles from the outset rather than retrofitting them. The following measures are recommended to reduce the environmental footprint of the employment cores while maintaining freight efficiency and are to be developed through a freight management plan as the structure plan advances.

Low-emission freight and fleet transition

Supporting the transition to low and zero-emission freight vehicles is central to reducing the environmental footprint of the employment cores. The Precinct should actively enable the uptake of electric, hybrid and alternative-fuel freight vehicles. New developments should be future proofed for charging and refuelling

infrastructure so that capacity can be installed and expanded as demand grows. Providing charging at docking stations and loading bays allows vehicles to charge during dwell times, improving operational efficiency, and centralised or shared charging hubs may be appropriate for multi-tenant sites and logistics clusters within the Cockle Creek and Killingworth cores.

Outcomes:

- Reduced emissions
- Improved air quality and reduced noise impacts
- A future-ready precinct.

First and last mile freight solutions

Short-distance freight trips can generate a disproportionate level of congestion, noise and emissions, so sustainable first and last mile solutions should be explored. This includes promoting cargo bikes, electric vans and low-emission trucks for smaller deliveries, supported by the active transport and shared-path network proposed through the Precinct. To enable this shift, appropriate supporting infrastructure should be provided, including secure storage and parking, and charging facilities located near loading and service areas. Matching the vehicle type to the task allows the Precinct to reduce the impacts of freight activity while maintaining service levels.

Outcomes:

- Reduced impacts of short-distance freight trips
- Improved safety and amenity
- More efficient and sustainable local distribution.

Consolidated freight movements and trip generation

Reducing the number of freight vehicle trips is a key lever for improving sustainability while maintaining operational efficiency. This can be achieved through shared logistics and consolidation facilities, coordinated delivery schedules, and encouraging tenants to adopt consolidated servicing arrangements where feasible, so fewer vehicles are required to move the same volume of goods. Higher productivity freight vehicles, including the RAV 19 metre B-doubles the network is designed to accommodate, can carry greater volumes per trip and reduce the overall number of truck movements. Shifting freight activity away from peak commuter periods, consistent with the off-peak heavy vehicle profile adopted in this assessment, reduces conflicts with employee travel, improves travel time reliability for freight operators and makes better use of the Wakefield Road spine and the surrounding network.

Outcomes:

- Fewer freight vehicle trips
- Reduced congestion and emissions
- Improved freight efficiency and reliability.

Mode shift and rail freight integration

Mode shift is a critical opportunity for reducing road congestion and emissions, particularly for longer-distance freight, where rail can be explored as an alternative to long-haul road transport while road freight is retained for first and last mile distribution. The Precinct could leverage its proximity to the ARTC rail freight network, either through the reactivation of the West Wallsend Colliery rail loop with a south-facing junction at Cockle Creek, or through connection to the broader network and the future Lower Hunter Freight Corridor between Fassifern and Hexham. Integrating rail freight into the Precinct's logistics offering reduces heavy vehicle kilometres travelled, improves resilience in the freight network and enhances sustainability outcomes, while access to both rail and road freight modes make the employment cores more attractive to businesses seeking efficient and flexible supply chains.

Outcomes:

- Reduced reliance on long-haul road freight
- Reduced congestion and emissions
- Enhanced economic competitiveness of the employment cores.

Rail freight connection via the Lower Hunter Freight Corridor

A future rail freight connection could be provided by linking the Precinct to the ARTC network and the planned Lower Hunter Freight Corridor, giving the eastern employment land an intermodal interface with the regional and interstate freight task. Land uses relying on this corridor are treated as a longer-term proposition subject to delivery timing, and the structure plan should protect the alignment and rail loop options so that staging decisions do not foreclose a future rail-served capability.

Outcomes:

- A protected pathway to rail-served freight
- Reduced long haul road dependence over time
- A more diverse and resilient freight offering.

4.6.4 Potential Upgrades

The staged upgrades represent a potential program based on the scenario assessed. Short term works enable freight access and first stage development, the medium term could include the internal duplication associated with the full build out, and the long term would secure the strategic M1 Pacific Motorway and rail connections. These staging outcomes should be treated as indicative, and deviation from this program should be considered as the project progresses and trip volumes become clearer. The road upgrades attributed to each horizon are illustrated in Figure 4-8 summarised in Table 4-13.

A potential future Cockle Creek crossing is identified separately as an opportunity to improve eastern connectivity into the Precinct rather than a committed upgrades, and its need, timing and form would be subject to further investigation as the project progresses.

Horizon	Upgrade	Driven by	Trigger
	Upgrades to accommodate new RAV routes at Rhondda Road and Cessnock Road	Precinct	Formalising of Wakefield Road south of Rhondda Road as RAV route
	Road duplication where VCR exceeds 1.0 (Northville Drive corridor first)	Background traffic growth	Midblock VCR triggers in the background traffic long term horizon
	RAV route extension to internal employment roads and east access through Teralba via upgraded railway overpass north of Teralba Train Station	Precinct	Enable eastern sub-precinct access, relieving Northville Drive. Direct B-double access sub-precincts and an additional RAV approach from the east
Long term (15+ years)	Direct M1 Pacific Motorway connection, being a new or upgraded interchange via Wakefield Rd East and Archery Rd	Background traffic growth and Precinct	Full development freight task, reduced local network reliance
	Five Islands Rd / The Esplanade / T C Frith Ave further capacity works beyond the Speers Point project	Background traffic growth	Residual B53 growth with development
	West Wallsend Colliery rail loop reactivation with a south facing junction at Cockle Creek	Precinct	Road to rail mode shift of the freight task
	Improved connectivity to the M1 Pacific Motorway and integration with the Lower Hunter Freight Corridor and High Speed Rail	Background traffic growth	Regional gateway role and long term capacity
	Upgrade railway overpass on Racecourse Road	Precinct	Increase in Precinct freight intensity

4.6.5 Indicative Road Cross Sections

The Lake Macquarie Development Control Plan 2014

The road hierarchy and cross sections for the Precinct draw on two Council local guidelines and plans, which, while not directly relevant to the State-led rezoning, provide useful local context.

Lake Macquarie City Council's Engineering Design Guidelines (2016) set the overarching framework for planning, designing and constructing infrastructure across the local government area, defining a road network hierarchy in which road function, design speed and road design together determine the reserve width, carriageway width and verge treatment appropriate to each road type.

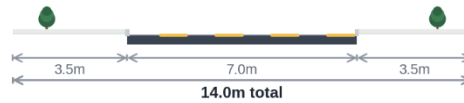
The Lake Macquarie Development Control Plan 2014 (Part 8, Subdivision Development) gives effect to these principles at the subdivision scale, setting out the criteria that describe each road type and the indicative cross sections with which subdivision design must comply. The road cross sections set out in the Lake Macquarie DCP are reproduced in Figure 4-9 and presented in Table 4-14.

These are local guides that establish the compliance baseline and are to be enhanced by the TfNSW Design of Roads and Streets guide (DORAS, 2024), to better consider movement and place and to provide increased recommendations for active transport across all levels of the road hierarchy, as discussed further below.

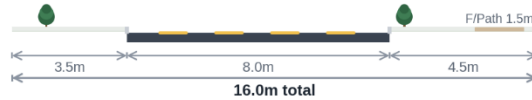
Access Lane
Minimal access handle



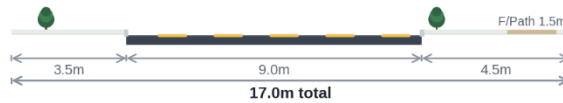
Cul-de-sacs / Access Streets
Low-volume residential



Local Street – Secondary
1.5 m footpath one side



Local Street – Primary
Bus shelter · 1.5 m footpath



Collector
2.5 m shared path + 1.5 m footpath



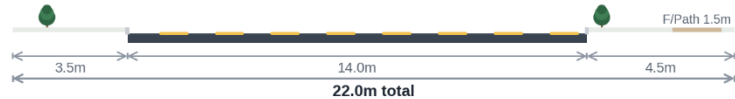
Sub-Arterial
2.5 m shared path + 1.5 m footpath



Commercial / Business Area
2.4 m footpaths both sides



Industrial
1.5 m footpath · heavy vehicle



Non-Urban Living
Rural / Environmental · 1.5 m shoulders

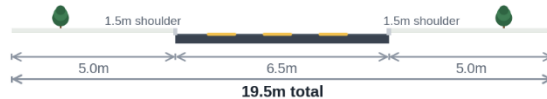


Figure 4-9: Indicative road cross sections (Lake Macquarie DCP 2014, Part 8 Subdivision Development, Figure 12)

Table 4-14: Road types and dimensions (Lake Macquarie DCP 2014, Part 8 Subdivision Development, Table 3)

Street type	Max lots	Max traffic volume (vpd)	Design speed ¹ (km/hr)	Road reserve width ² (m)	Carriage way width (m)	Footway verge width (m)	Footpath	Shared Pathway	Street trees	On-road cycling provision	Kerbing ³	Design considerations
Residential												
Access lane ⁴	N/A	2,000	10-30	8	6	1 & 1	No	No	No	No	Upright / flush	
Cul-de-sacs / Access streets	Up to 10	2,000	30	15	8	3.5 & 3.5	No ⁵	No	Yes	No	Upright	
Local Street - Secondary	10-100	2,000	30	16	8	4.5 & 3.5	1 x 1.5m	No	Yes	No	Upright	
Local Street - Primary	100-200	3,000	40	17 18	9 9	4.5 & 3.5 4.5 & 4.5	1 x 1.5m 2 x 1.5m ⁶	No	Yes	No	Upright	
Collector	200+	3,000 or more	50	24.6	14.6	4.5 & 5.5	1 x 1.5m	1 x 2.5 (min)	Yes	Yes, cycle lanes	Upright	Carriageway width based on 2 x 1.2 bicycle lanes plus 0.7m buffer x 2, 2 x 3.3 travel lanes, 2 x 2.1m parking lanes
Sub-Arterial	200+	3,000 or more	60 or more	24.8	14.8	4.5 & 5.5	1 x 1.5m	1 x 2.5 (min)	Yes	Yes, cycle lanes	Upright	
Other												

Street type	Max lots	Max traffic volume (vpd)	Design speed ¹ (km/hr)	Road reserve width ² (m)	Carriage way width (m)	Footway verge width (m)	Footpath	Shared Pathway	Street trees	On-road cycling provision	Kerbing ³	Design considerations
Commercial / Business area (excluding Business Park)	N/A	N/A	50	23.8 - 24.8	14.8 - 15.6	4.5 & 4.5	Yes, full width of verge both sides	No, see kerb separated cycle lanes	Yes	Yes, kerb separated main street	Upright	Carriageway width based on 2 x 3.3 travel lanes, 2 x 2.1m parking lanes, 1 x 4m two-way separated bicycle path OR 2 x 2.4m one-way separated bicycle paths
Industrial / Business Park	N/A	N/A	50	22	14	4.5 & 3.5	1 x 1.5m	Note	Yes	Yes, cycle lanes	Upright	
Non-Urban Living (Rural / Environmental Living)	N/A	N/A	90-110	19.5	6.5 with 2 x 1.5m shoulders	5	No	Note	Yes		WSUD	

Notes as per Lake Macquarie DCP 2014, Part 8 Subdivision Development, Table 3:

1. Design speed differs to sign-posted speed. Entry treatments including kerb extensions and continuous footpaths are preferred for local and access streets. Mid-block treatments such as roundabouts, chicanes, slow points and wombat crossings are appropriate for street blocks longer than 150m. Refer to Council's Walking, Cycling and Better Streets strategy and NSW Movement and Place design solutions.
2. In bushfire prone areas, the carriageway and road reserves may be required to be widened, to comply with the requirements of Planning for Bushfire Protection.
3. Where subdivision is designed using Water Sensitive Urban Design (WSUD), kerb and guttering may be replaced with table drains or roadside swales.
4. Access lanes should be provided as part of residential rear access (to accommodate shared paths on collector streets) and commercial subdivision for access and servicing.
5. Footpaths required on one side of the street within the Principal Pedestrian Network R3 medium density zoned areas.
6. Footpaths required on both sides of the road if located in the principal pedestrian network. Refer to the walking, cycling and better streets strategy and maps.

Design of Roads and Streets guide

While these Council instruments set an indicative baseline, they pre-date current practice and are oriented primarily towards vehicle movement. To deliver streets that better support walking, cycling and public transport across all levels of the hierarchy, the Precinct cross sections should also draw on the TfNSW Design of Roads and Streets guide (DORAS, 2024), which applies the NSW Movement and Place framework to street design.

DORAS does not assign a single fixed cross section to each road type. Instead, the cross section is derived from two things together, being the movement function of the road and the place context it sits within, so that the same nominal road type is treated differently in a residential street and an employment corridor. This document sets out the key principles, considerations and design aspects to carry into the Precinct road types, to complement and where appropriate exceed the minimum DCP cross sections.

DORAS sets out nine principles for NSW roads and streets. These principles frame every cross section and are summarised in Table 4-15, with the active transport implication for the Precinct noted for each.

Table 4-15: NSW roads and street principle (TfNSW Design of Roads and Streets (TS 00066:1.0), 2024))

Principle	Application to the Precinct
On Country	Roads and streets offer pathways for connecting to and building relationships with Country, integrated through early engagement and landscape design.
For people	Streets are welcoming, equitable, inclusive and accessible for people of all ages, genders and abilities, not designed for vehicles alone.
Safe	The Safe System approach is applied so that human or mechanical error does not result in death or serious injury, with lower speeds where vulnerable users and vehicles interact.
Healthy	Design enables walking and cycling for everyday trips and reduces congestion, urban heat, noise and emissions.
Public spaces	Streets are treated as part of the network of public spaces, supporting activity and social life rather than movement only.
Ecological network	Verges, street trees and water sensitive urban design form part of the green and blue grid within the cross section.
Carry people and goods efficiently	Efficiency is achieved by prioritising walking, cycling and public transport and by using appropriately sized vehicles for the context, including for freight and servicing, rather than by maximising vehicle throughput.
Create value	Streets underpin social interaction, local economies and land value.
Designed for context	Design responds to the natural, land use and cultural context of each part of the Precinct.

Movement and Place Design Parameters

The NSW Movement and Place framework, applied through the Transport for NSW Design of Roads and Streets guide (DORAS, 2024), classifies all roads and streets into four street environments based on movement function and place intensity. The consolidated Table 4-16 sets out the design parameter categories for each environment, with framework level characteristics and indicative ranges. The parameters are non-prescriptive guidance applied as ranges, complementing Austroads guides, Australian Standards and the Lake Macquarie DCP.

Table 4-16: Movement and Place design parameters by street environment (TfNSW Design of Roads and Streets (TS 00066:1.0), 2024))

Design parameter	Local streets	Main streets	Main roads	Civic spaces
Movement function	Lower. Local access and distribution rather than through movement.	High. Significant movement function, balanced against high place.	High. Strategic movement of people and goods, including freight and major public transport.	Low. Movement subordinate to place, through traffic strongly limited or excluded.
Place intensity	Lower to moderate, predominantly residential or local in character.	High. Significant place qualities, centres, retail and mixed use frontages with pedestrian activity.	Lower, though the corridor may hold local meaning.	High. Significant meaning, activity and built environment, people prioritised.
Indicative operating speed	Low speed, typically 30 to 40 km/h, 30 km/h where vulnerable users interact.	Lower speed, typically 40 to 50 km/h, 30 km/h where pedestrian activity is high.	Higher speed, typically 60 to 80 km/h and above, managed down at crossings.	Very low speed, typically 10 to 30 km/h, including shared zone speeds.
Through traffic and freight	Minimised, servicing and local deliveries only, through movement discouraged.	Accommodated but not prioritised over place, servicing managed by time and design.	Primary function, high volumes and heavy vehicles including design freight vehicle.	Minimised, access limited to servicing, timed deliveries and emergency vehicles.
Carriageway / lanes	Narrow carriageways, two clear lanes on neighbourhood streets, with traffic calming.	Constrained lane widths for lower speeds, space reallocated to people and place.	Multiple through lanes with auxiliary and turn lanes, sized for movement priority.	Minimal or shared surface, carriageway de-emphasised in favour of pedestrian space.
Intersections	Safe walking and cycling crossings on all legs, raised crossings reinforce pedestrian priority.	Manage complex vehicle and people interactions, prioritise walking, minimise approach speeds.	Grade separated or signalised, efficient through movement while reducing crossing delay and risk.	Main roads should not directly intersect, turning vehicles into civic spaces limited.
Pedestrians	Continuous footpaths connecting homes to parks, schools, stops and the network.	High priority, wide continuous footpaths and frequent direct crossings on desire lines.	Provided at controlled crossing points, footpaths set back from the carriageway.	Highest priority, generous continuous pedestrian space with priority over vehicles.
Cyclists	On-street or quiet-street cycling supported by low speeds, separated on busier links.	Separated cycleways or low-speed mixed conditions integrated into the network.	Principal bicycle network as physically separated cycleways or paths.	Low-speed shared or integrated cycling consistent with pedestrian priority.
Public transport	Generally local bus routes and stops where provided.	Stops, shelters and priority within an active frontage environment.	Major route function, bus priority, stops and interchange provided.	Accommodated where required at low speed, integrated with the public space.
Kerbside / parking	On-street parking common, reallocated to build-outs, trees or place where not required.	Managed for short-stay parking, loading, dining and bus stops, not continuous parking.	Limited or none, kerbside reserved for movement, transit and turn lanes.	Minimal, kerbside given to people, dwelling, planting and activity.
Verge / landscape	Street trees, verge planting and WSUD, planting narrows the corridor and calms traffic.	Street trees, planting, seating and high-quality paving supporting place and canopy.	Setbacks and landscaped verges for separation, canopy and WSUD, manage severance.	High-quality public domain, paving, trees, furniture and lighting define the space.
Place / frontage	Local residential or neighbourhood frontage, amenity and safety prioritised.	High active frontage, the street is a destination with space for dwelling.	Lower frontage activity, access managed to limit conflict with through movement.	Highly active frontages with space for dwelling, gathering and public life.

5 Conclusions

The assessment demonstrates that the Precinct can be supported by a feasible and appropriately staged access strategy. At full theoretical development of the Structure Plan land use schedule, being approximately 211,100 m² GFA of large format warehousing, a mixed use commercial core of around 18,100 m² gross leasable area at West Lake and around 10 dwellings at Barnsley West, the Precinct is anticipated to generate in the order of 12,470 daily trips of which 1,030 trips are expected be heavy vehicles and 30 RAV 19 metre B-doubles. Peak hour volumes are expected to be in the order of 1,400 vehicles in the AM peak and 1,300 in the PM peak.

Comparing the baseline with the Precinct impact shows that the constraints on the network are driven primarily by background growth rather than by the development.

- In the 2026 base year the network has substantial spare capacity. The Wakefield Road spine, Rhondda Road, The Broadway and the western local roads operate at LoS A to B, and only the Northville Drive corridor toward the B89 is close to desirable its limit, at LoS C to D before any Precinct traffic.
- Background growth alone, without the Precinct, pushes the Northville Drive corridor to LoS E by the long term (Northville Drive north of The Weir Road reaching a volume to capacity ratio of around 0.97), confirming the constraint exists independent of the Precinct.
- The southern and western network carries the Precinct comfortably in every scenario, with Rhondda Road at the southern access remaining at LoS A throughout, as such, the impact is concentrated on the northern corridors rather than spread across the network.

Cockle Creek is a major line of severance for access to the Precinct resulting in an already congested access route linking the Precinct to the east, and limited capacity for both background growth and precinct traffic to the north. Reducing the impact of this line of severance through traffic management, mode shift, infrastructure upgrades and potential new infrastructure will be appropriately incorporated within the onward planning for the precinct.

Potential upgrades as identified at this stage have been staged so that each is triggered by an indicative outcome of this assessment, and the indicative program distinguishes capability and access works from background driven capacity works to which the development contributes proportionately.

Development is to be staged consistent with the assessed capacity of each access point, land uses generating RAV movements are to be located on the approved or proposed RAV network, and detailed SIDRA modelling is to confirm intersection geometry, control type and treatment at the development application or subdivision stage. Subject to the staged upgrade program identified in this report, there are no presently known transport issues that would make the proposed rezoning inherently inappropriate.

The assessment provides a high-level basis for rezoning and infrastructure planning. Further investigations will be required during subsequent planning and design stages to confirm network performance, infrastructure requirements, and opportunities to support sustainable and efficient transport outcomes.

- Undertake wider network assessment and detailed traffic modelling, informed by updated STFM demand forecasts and land-use assumptions, to confirm cumulative precinct impacts, validate corridor and intersection performance, and identify any required network upgrades.
- Future investigations should consider rail access opportunities for freight, including a potential rail loop or siding, and public transport integration with nearby stations. Road reserves and intersections should also support future bus movements, priority measures, accessible stops, and safe pedestrian links, with ongoing TfNSW engagement to confirm service improvements. A future Cockle Creek crossing may further improve bus and active transport access to Cockle Creek Station and support transit-oriented parking outcomes.

Appendices

Appendix A

Historical Background Growth and Existing Trip Patterns Review

Historical traffic growth trends were assessed using permanent traffic count stations located on the M1 Pacific Motorway, B89 Main Road and B53 Lake Road corridors surrounding the Precinct. These permanent classifiers provide Annual Average Daily Traffic (AADT) data, which represents the average number of vehicles travelling past a location on a typical day over a calendar year. Three permanent classifiers with the most comprehensive datasets were selected for review and are summarised in Table A- 1.

Table A- 1: Historical traffic count locations

Permanent Classifier ID	Road	Direction	Data period
05095	M1 Pacific Motorway	Northbound/Southbound	2007–2016
05208	B53 Lake Road	Northbound/Southbound	2008–2023
05571	B89 Main Road	Eastbound/Westbound	2009–2020

Figure A- 1 presents AADT trends for the B53 Lake Road between 2008 and 2023. Traffic volumes on this corridor have been broadly stable, growing modestly from an average of approximately 12,800 vpd in 2008 to a pre-COVID peak of 13,440 vpd in 2019 an increase of around 5% over eleven years, averaging less than 0.5% per year. Volumes declined by approximately 6% in 2020, consistent with COVID-19 related travel restrictions, before recovering strongly to a record high of 14,040 vpd in 2022. By 2023, volumes had moderated slightly to 13,720 vpd, remaining above pre-COVID levels and indicating sustained demand on the corridor.

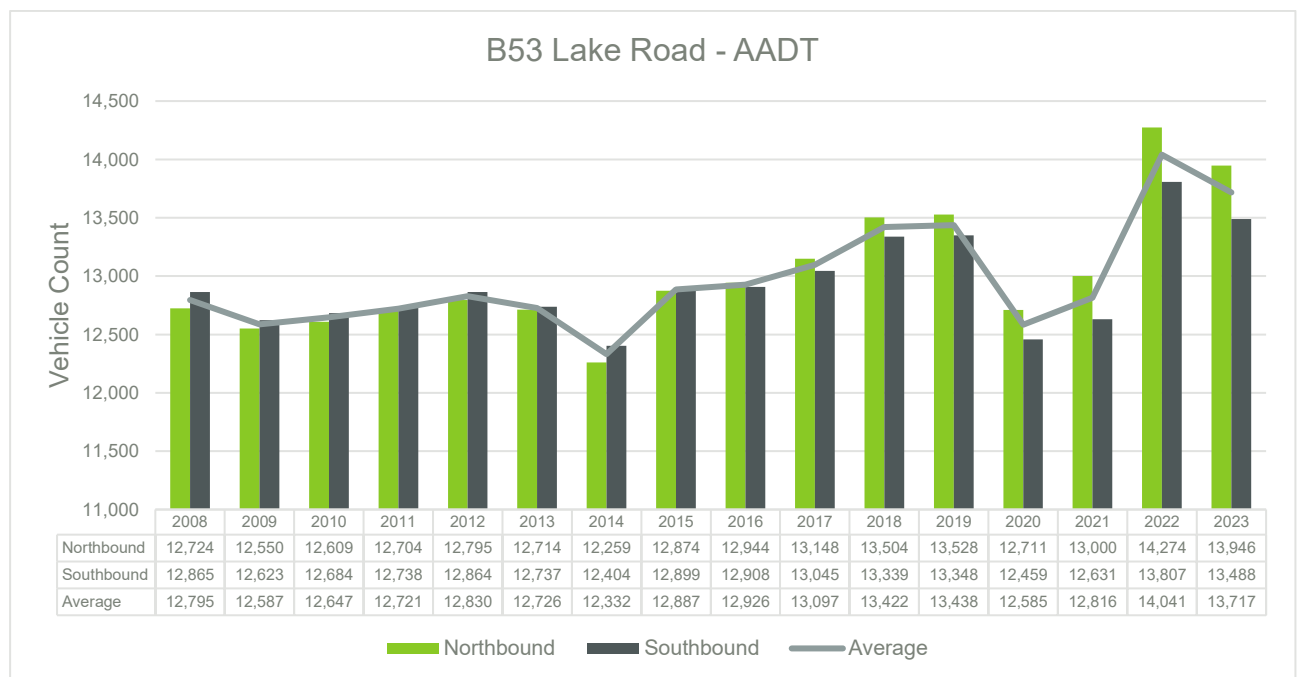


Figure A- 1: B53 Lake Road AADT, 2008-2023 (Traffic counter ID: 05208)

Figure A- 2 presents AADT trends for the B89 Main Road between 2009 and 2020. This corridor exhibited stronger underlying growth than the B53, with average daily volumes increasing from approximately 11,190 vpd in 2009 to a peak of 12,625 vpd in 2019, an increase of around 13 per cent over ten years,

averaging approximately 1 per cent per year. As with the B53, a noticeable decline occurred in 2020, with volumes falling approximately 8 per cent to around 11,670 vpd.

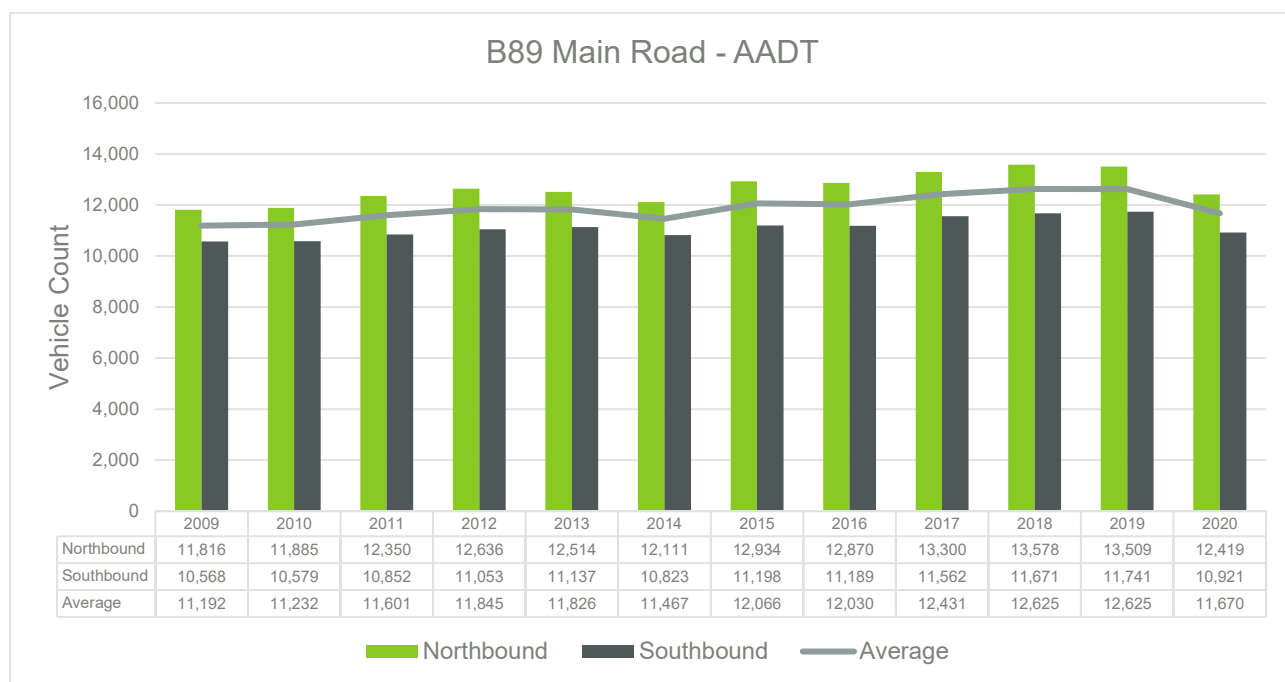


Figure A- 2: B53 Lake Road AADT, 2008-2023 (Traffic counter ID: 05208)

Overall, the classifier record indicates modest long term background growth on the corridors framing the study area, with temporary reductions during the pandemic period. While this classifier data is available, the permanent counters sit on the higher order corridors surrounding the study area (the M1, the B53 and the B89) and therefore do not provide an indication of the growth trends experienced on the lower order local roads through the Precinct, which carry the movements most relevant to this assessment.

Vehicle ownership

The Bolton Point-Teralba SA2 exhibits a strongly car dependent household profile. Vehicle ownership data from the 2016 and 2021 ABS Census indicates growing private vehicle dependency, with multi-vehicle households increasing their share and remaining well above NSW averages across both periods.

Table A- 2: Vehicle Ownership by Dwelling, Bolton Point–Teralba SA2, ABS 2016–2021

Category	2016	2021	Change (pp)
Average vehicles per dwelling	1.8	1.9	+0.1
No motor vehicles	6.6%	4.9%	-1.7%
1 motor vehicle	38.3%	34.9%	-3.3%
2 motor vehicles	34.9%	38.5%	+3.6%
3 motor vehicles	13.0%	14.2%	+1.2%
4 or more motor vehicles	7.2%	7.4%	+0.3%

As shown in Table A- 2, car-free households account for just 4.9% of dwellings in 2021, roughly half the NSW average of 9.2% while 60.1% of households owned two or more vehicles, compared with 52.4% across NSW. Vehicle-free households, most likely to rely on public and active transport for all trip purposes, represent a markedly smaller share of the local population than across NSW generally.

Figure A- 3 illustrates the contrast in vehicle ownership between Bolton Point–Teralba and NSW across all ownership categories.

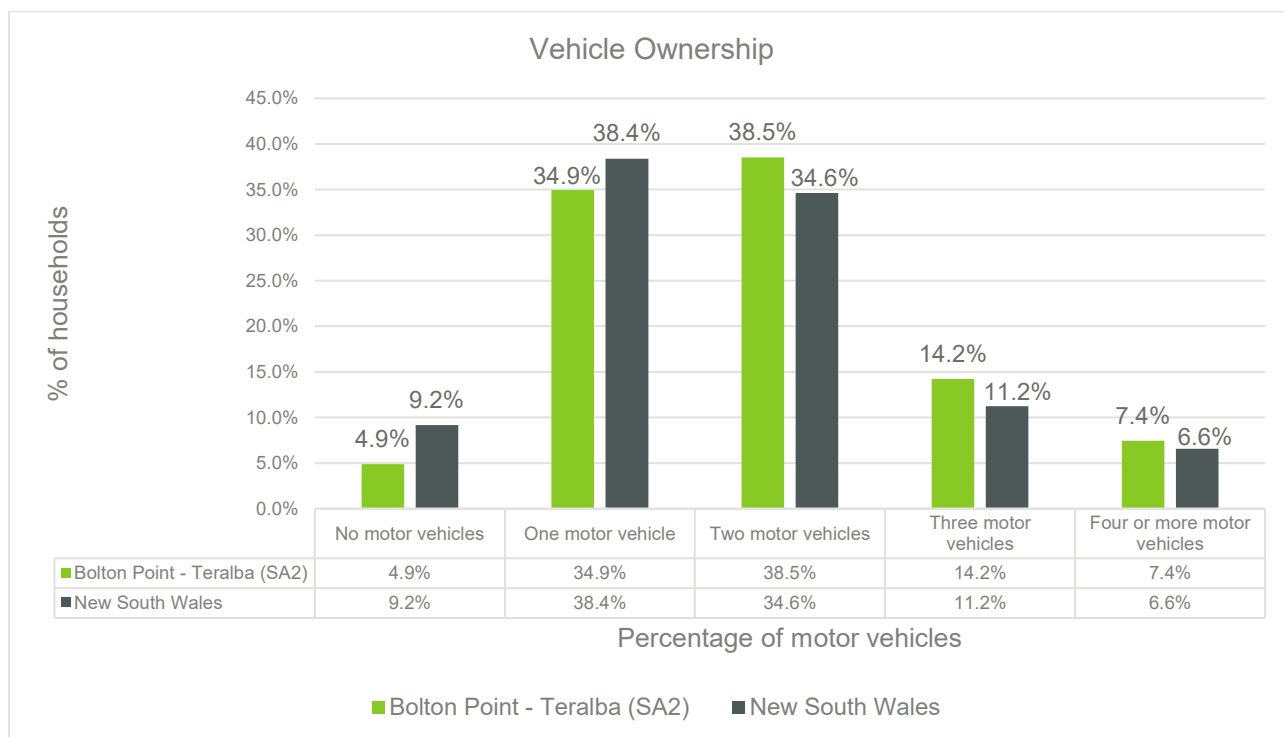


Figure A- 3: Comparison of motor vehicle ownership in 2021 between Bolton Point – Teralba (SA2) and NSW

Journey to work

The following travel characteristics relating to modes of transport used to travel to work by Lake Macquarie - West residents (refer to Figure A- 4), workers travelling to Lake Macquarie - West, and NSW were identified:

- Private vehicle travel is the dominant mode for both residents and workers, accounting for approximately 63 to 66% of journeys - significantly higher than the NSW average (48%).
- Public transport usage is very low at 0-1%, well below the NSW average of 3%, reflecting the limited availability of higher-frequency services in the area.
- Active transport modes account for a small proportion of journeys (1 to 2%), slightly below the NSW average (3%).
- A higher proportion of 'other' modes, including working from home, is recorded for residents (35%) than for incoming workers (31%), with both below the NSW average (46%).
- The major trip attractors within Lake Macquarie - West are health care and social assistance facilities, construction, education and training, and retail trade, which together account for approximately 58 per cent of all workers travelling to the area. Mining also represents a notable attractor, reflecting proximity to Hunter Valley coal operations.

Overall, the data indicates a strong reliance on private vehicles within Lake Macquarie - West, likely reflecting the regional context, lower density land use, and public and active transport connectivity.

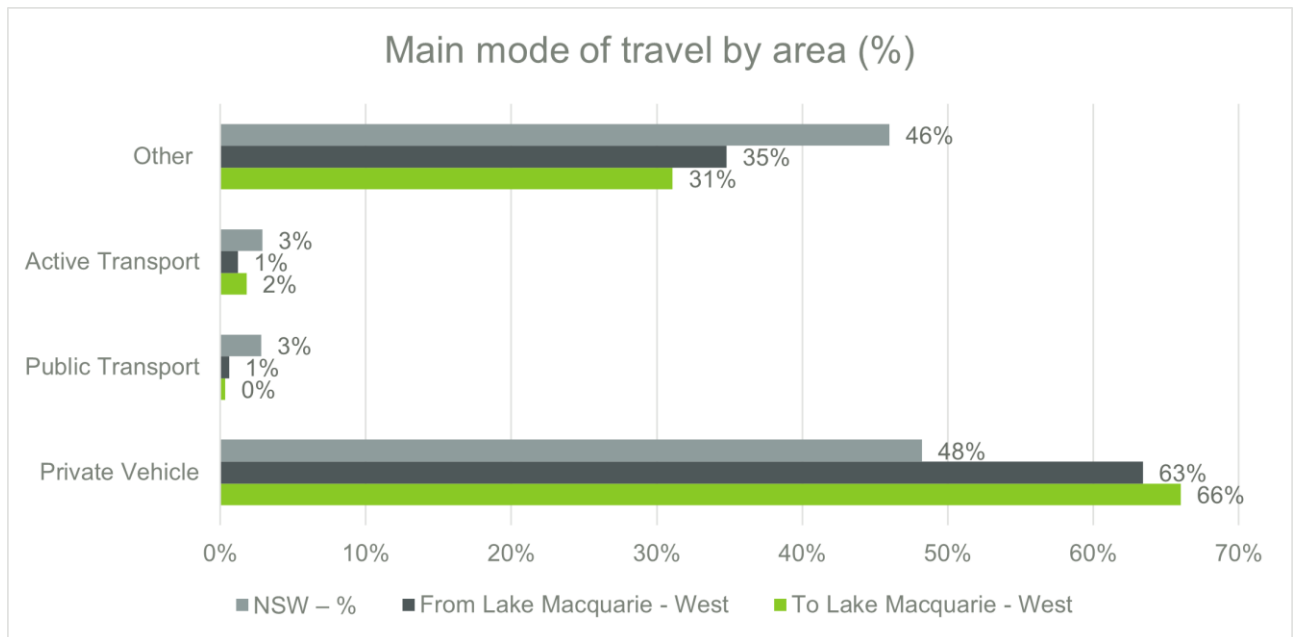


Figure A- 4: Journey to work data (2021) – Mode of transport used by residents and workers of Lake Macquarie (West)

Appendix B

Detailed Midblock Assessment Results

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